

CSE4227 Digital Image Processing

Chapter 9 – Morphological Image Processing- (Part-I)

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Today's Contents

❑ Morphological image processing: pixel shape based analysis

❑ Structuring Element

❑ basic morphology operations

- Dilation - grow image regions ✓
- Erosion - shrink image regions ✓
- ✓ ▪ Opening - structured removal of image region boundary pixels
- ✓ ▪ Closing - structured filling in of image region boundary pixels

■ Chapter 9 from R.C. Gonzalez and R.E. Woods, Digital Image Processing (3rd Edition), Prentice Hall, 2008 [**Section 9.1, 9.2, 9.3**]

■ <https://www.cs.auckland.ac.nz/courses/compsci773s1c/lectures/ImageProcessing-html/topic4.htm>

Morphological Image Processing

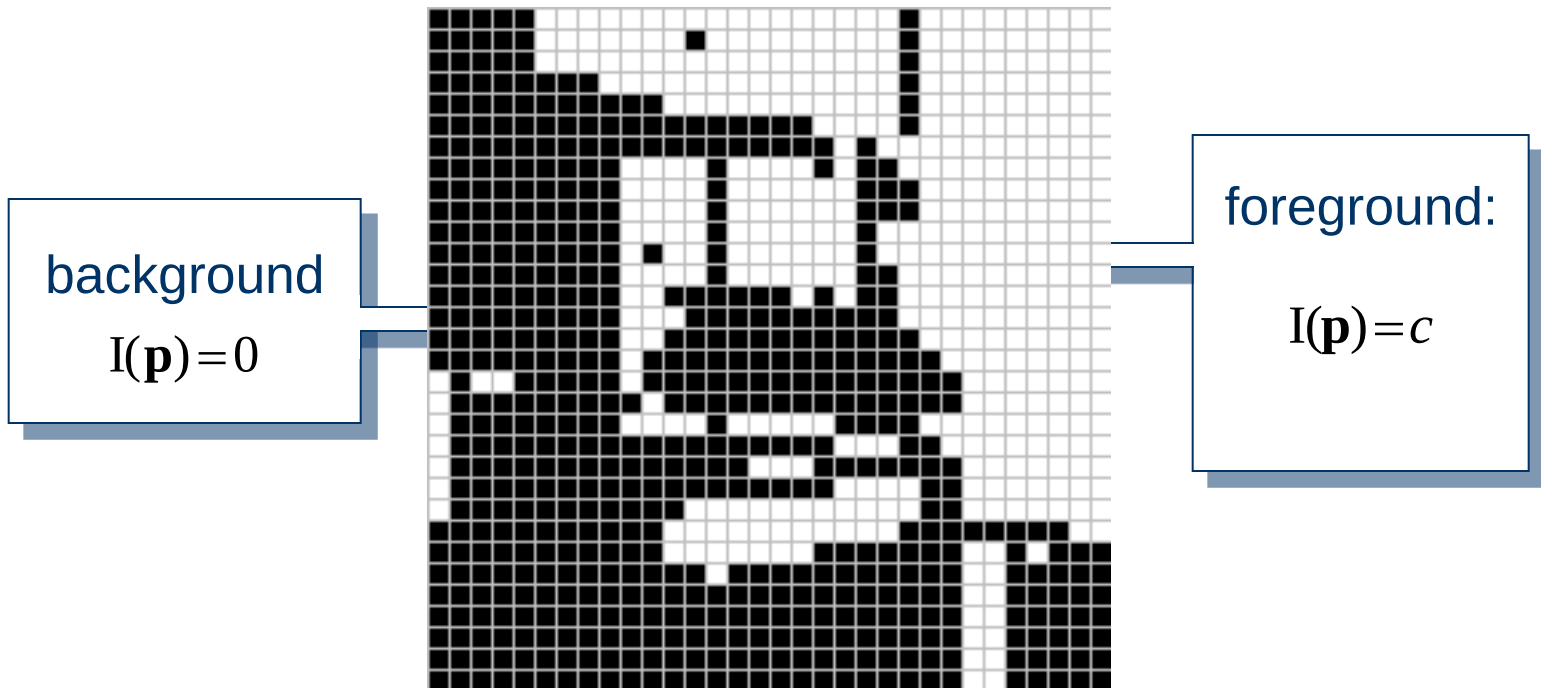
- A tool for **extracting image components** that are useful in the representation and description of image.
- It deals with the **shape** (or morphology) of objects/features in an image
- Rely only on the **relative ordering of pixel values**, not on their numerical values
- Probe an image with a small shape or template called a **structuring element**.
- The language of mathematical morphology of an image is **Set Theory**

Morphological Image Processing

- ❑ Morphology **applies** certain **simple rules and shapes** (such as squares, circles, diamonds, cubes and spheres) to process images.
- ❑ The objective is usually **to identify features** of interest in images
 - as a prelude to performing high-level inspection, or
 - machining, functions.
- ❑ Two kinds :
 - Binary Morphology
 - Grey Level Morphology
- ❑ Four basic morphology operations for **binary images**:
 - **Erosion**
 - **Dilation**
 - **Opening and**
 - **Closing**

Binary Image

- ❑ Representation of individual pixels as 0 or 1, convention:
 - foreground, object = 1 (white)
 - background = 0 (black)



This represents a digital image. Each square is one pixel.

Structuring Element

A small image/template that helps to produce new image from the old one i.e. a small binary array.

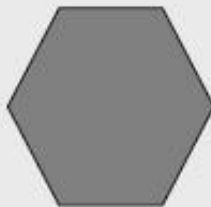
A **structuring element** is a shape mask used in the basic morphological operations.

They can be any shape and size that is digitally representable, and each has an **origin**.



box

`box(length,width)`

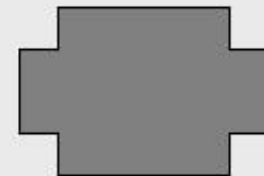


hexagon



disk

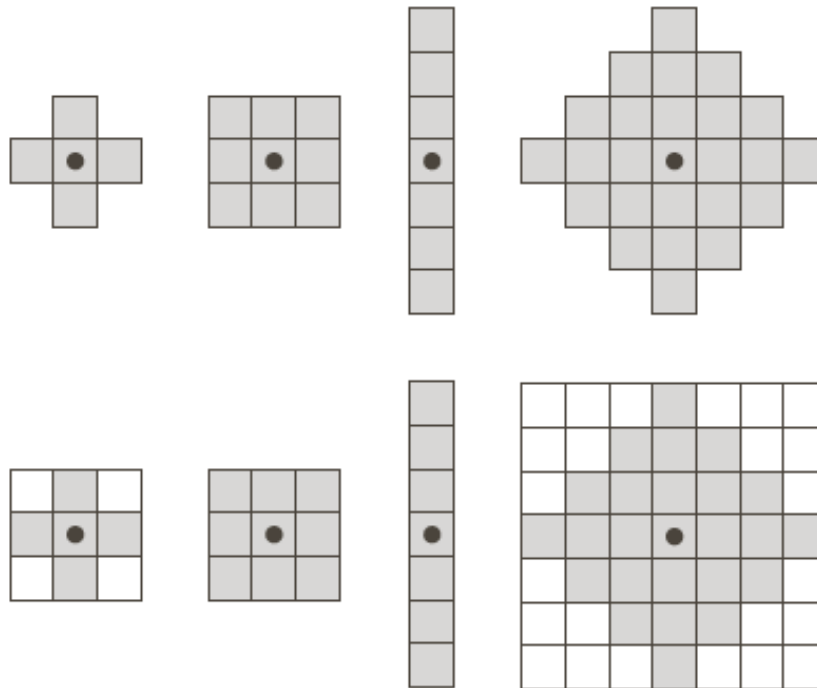
`disk(diameter)`



any shape

Structuring Element

A structuring element is a small image – used as a moving window



**Example
Structuring
Elements**

**Structuring
Elements
converted to
rectangular
arrays**

Structuring Element

- For simplicity we will use rectangular structuring elements **with their origin point.**
- Origin point can be at the middle pixel or at any position.

1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1

Square 5x5 element

0	0	1	0	0
0	1	1	1	0
1	1	1	1	1
0	1	1	1	0
0	0	1	0	0

Diamond-shaped 5x5 element

0	0	1	0	0
0	0	1	0	0
1	1	1	1	1
0	0	1	0	0
0	0	1	0	0

Cross-shaped 5x5 element

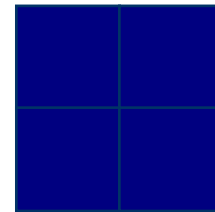
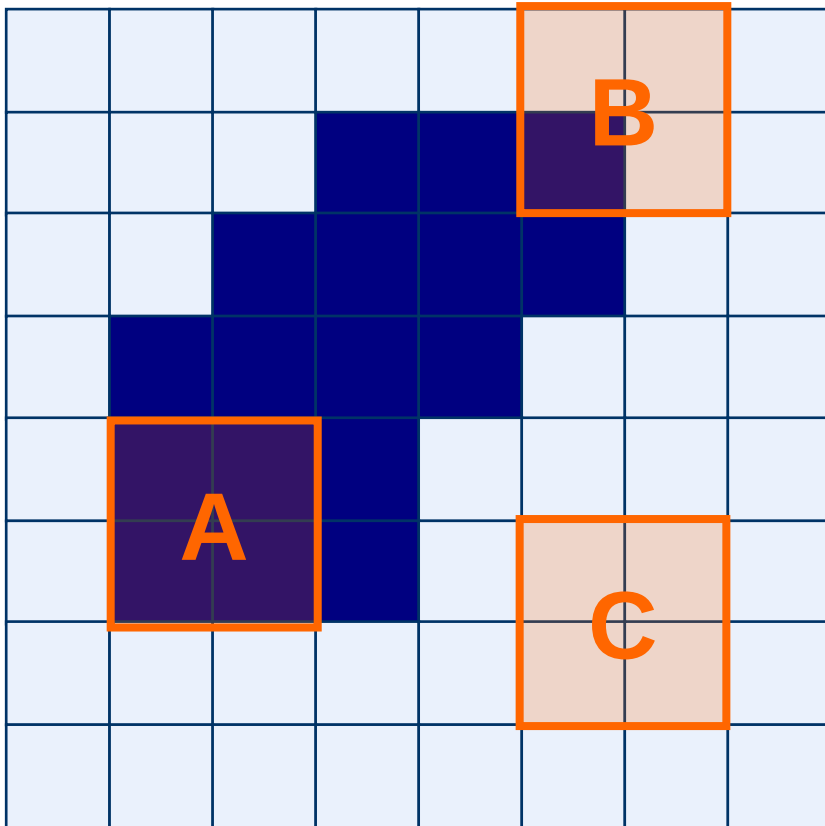
■ ← Origin

1	1	1
1	1	1
1	1	1

Square 3x3 element

❖ By default origin point is at the center

Structuring Elements: Hits & Fits



Structuring Element

Fit: All “**ON**” *pixels* in the structuring element cover “**ON**” *pixels* in the image

Hit: Any “**ON**” *pixel* in the structuring element covers an “**ON**” *pixel* in the image

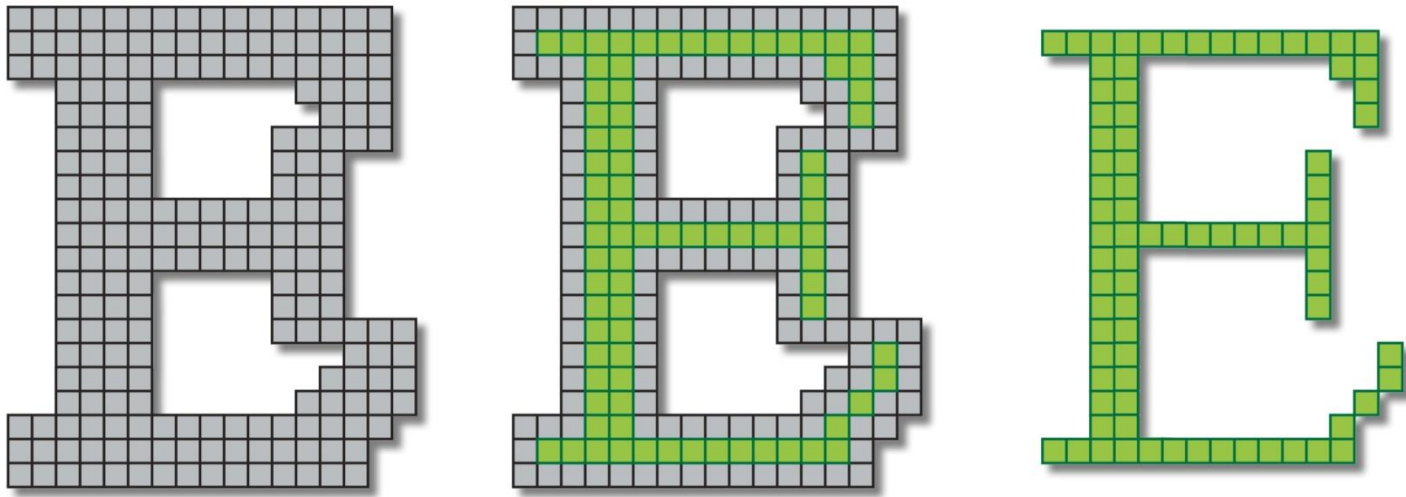
All morphological processing operations are based on these simple ideas

Fundamental Operations

- ♦ Fundamentally morphological image processing is very like spatial filtering
- ♦ The structuring element is moved across every pixel in the original image to give a pixel in a new processed image
- ♦ The value of this new pixel depends on the operation performed

There are two basic morphological operations: **Erosion** and **Dilation**

Erosion

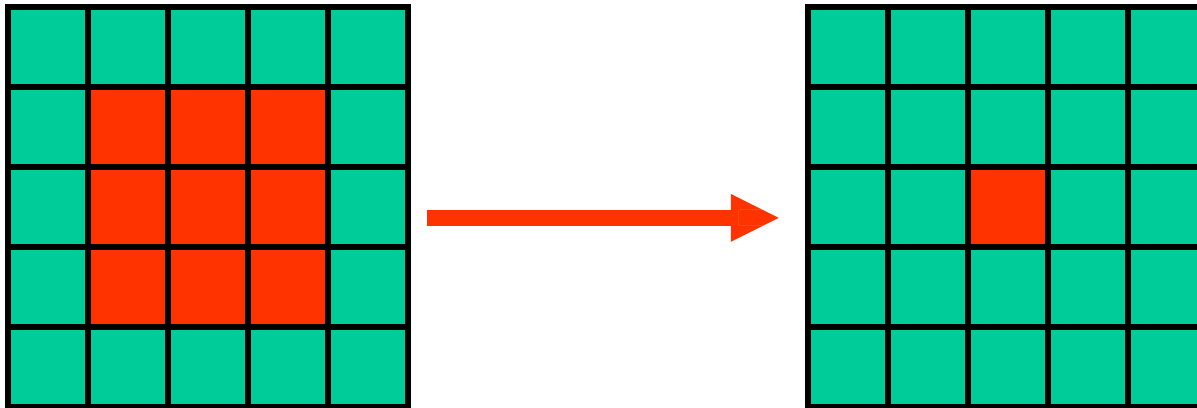


Shrink the object

Shrinks foreground, enlarges Background

Erosion

- Erosion is used for shrinking of element A by using element B.
- One of the simplest uses of erosion is for eliminating irrelevant details from a binary image.



Original Image

Eroded Image

Erosion

Definition 1:

Fit: All “**ON**” pixels in the structuring element cover “**ON**” pixels in the image

□ Does the structuring element **fit the set**?

$$A \ominus B = \{z \mid (B)_z \subseteq A\}$$

- Erosion of a set A by structuring element B :
- Set of all points z , such that B translated by z is contained in A .

Erosion

Definition 2:

Does the structuring element **fit the set?**

Erosion of image f by structuring element s is given by $f \ominus s$

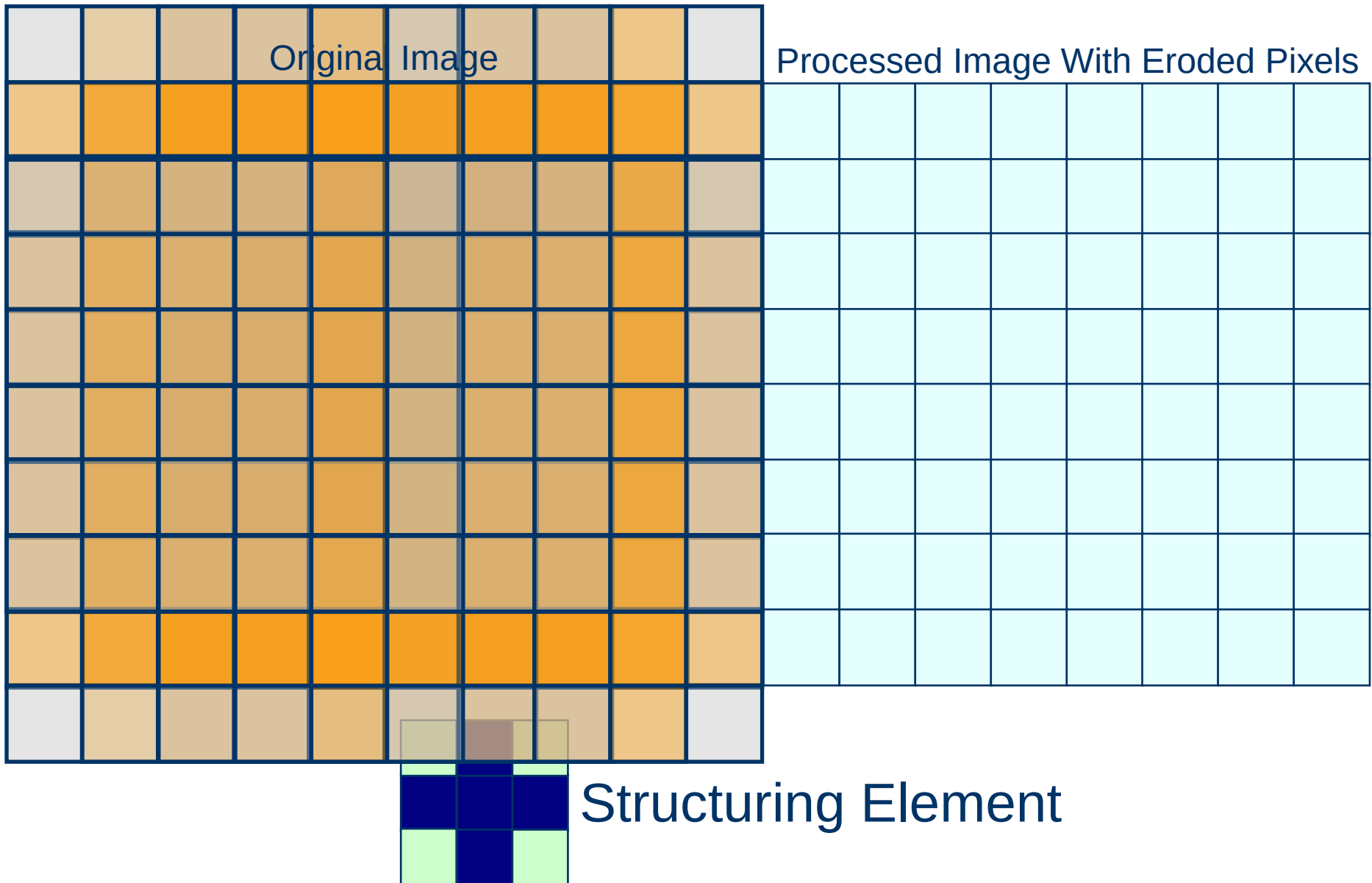
The structuring element s is positioned with its origin at (x, y) and the new pixel value is determined using the rule:

$$g(x, y) = \begin{cases} 1 & \text{if } s \text{ fits } f \\ 0 & \text{otherwise} \end{cases}$$

Erosion – How to compute

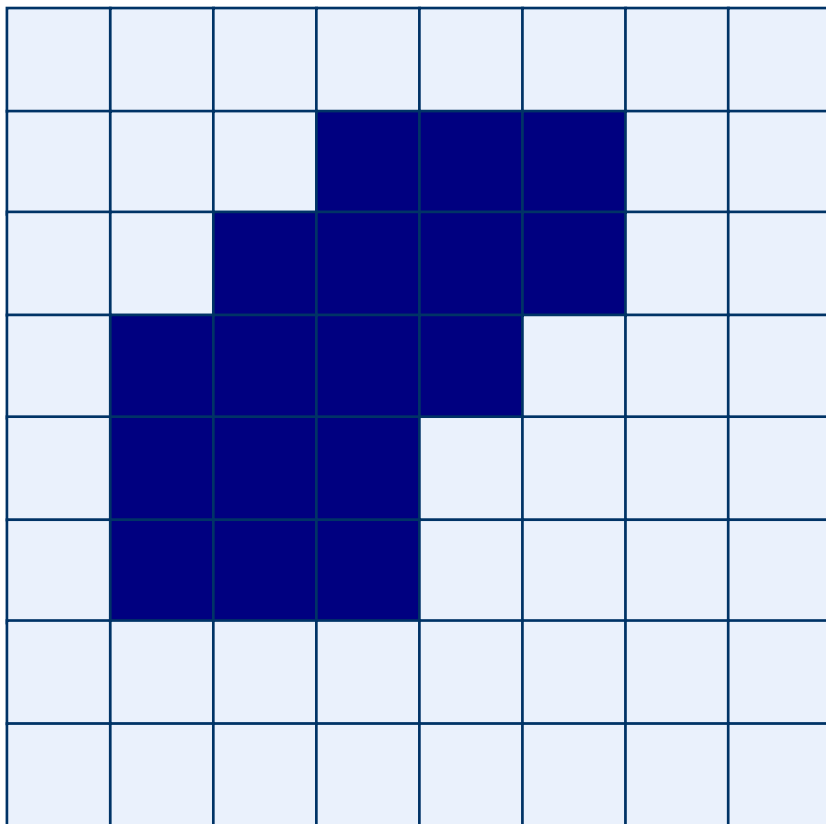
- ◆ For each foreground pixel (which we will call the *input pixel*)
 - Superimpose the structuring element on top of the input image so that the origin of the structuring element coincides with the input pixel position.
 - If *for every* pixel in the structuring element, the corresponding pixel in the image underneath is a foreground pixel, then the input pixel is left as it is.
 - If any of the corresponding pixels in the image are background, however, the input pixel is also set to background value.

Example for 2D Erosion

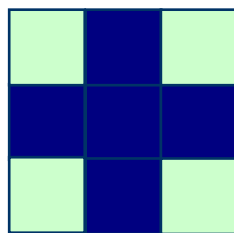
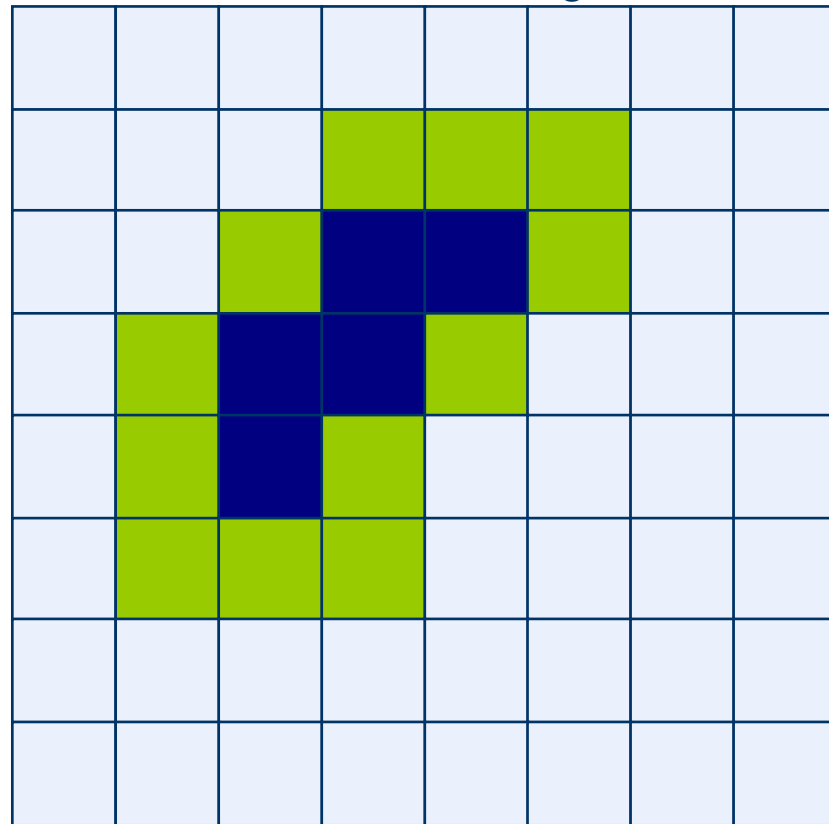


Erosion: Example 1

Original Image

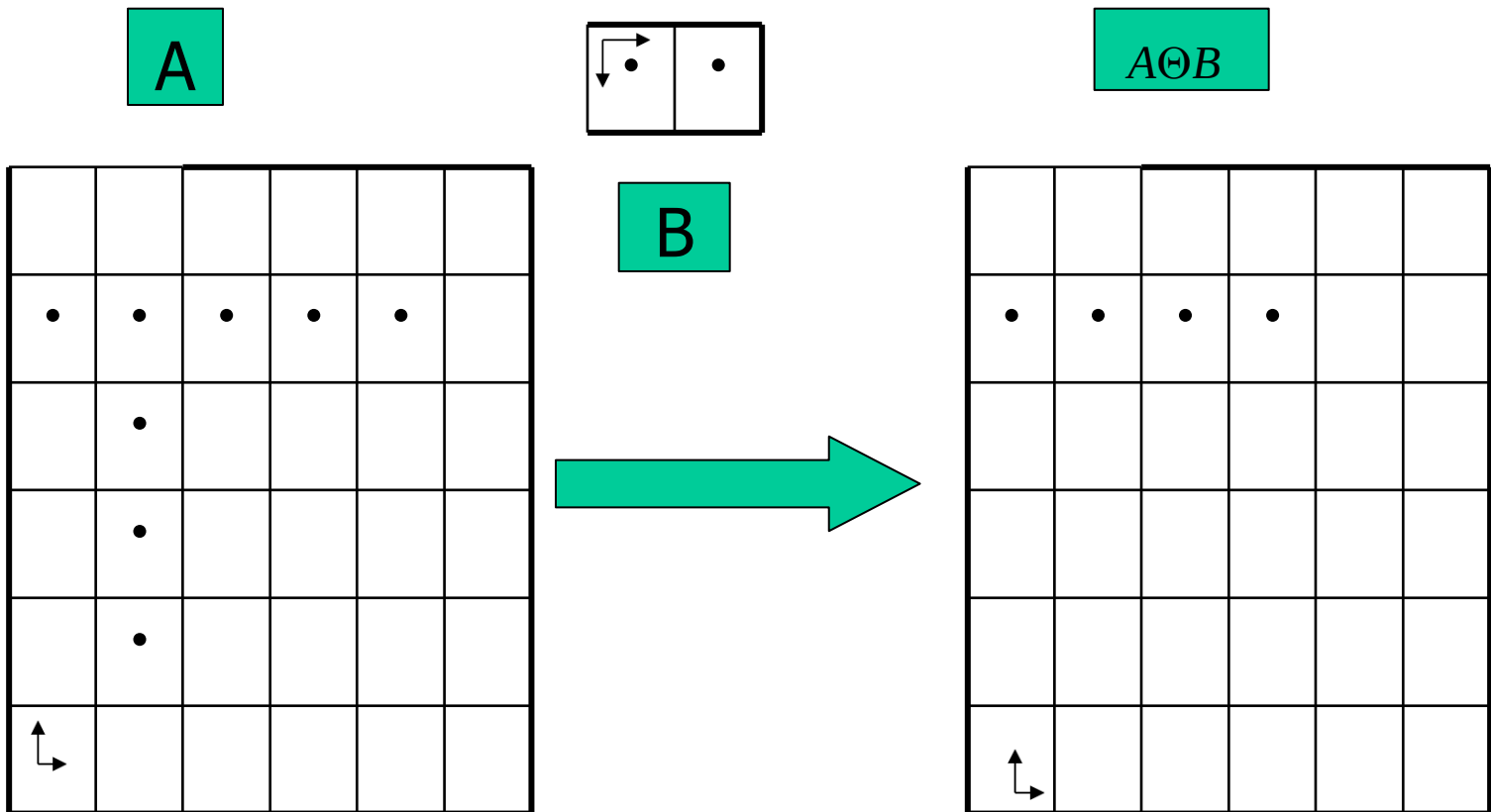


Processed Image

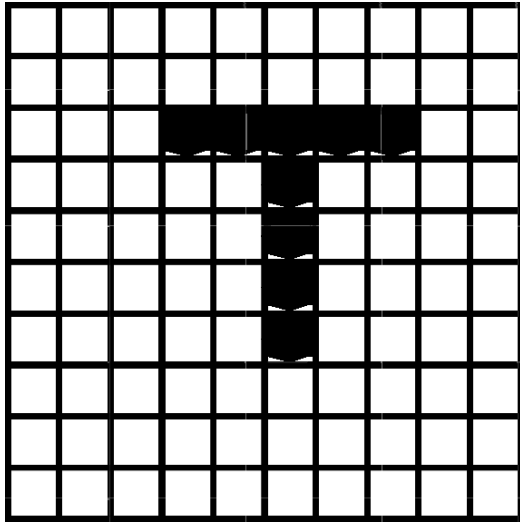


Structuring Element

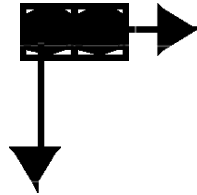
Erosion explained pixel by pixel



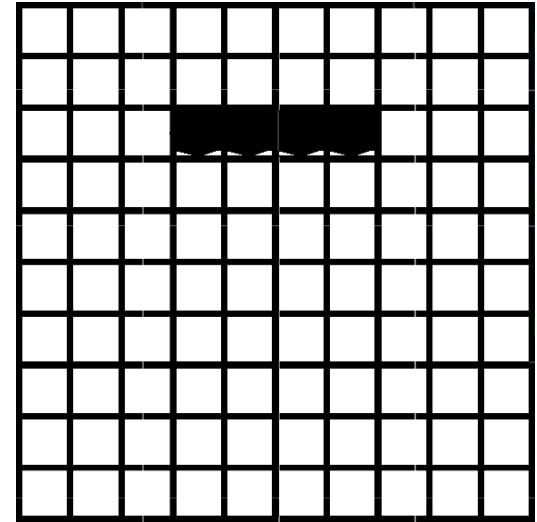
Structuring Element in Erosion Example



Image

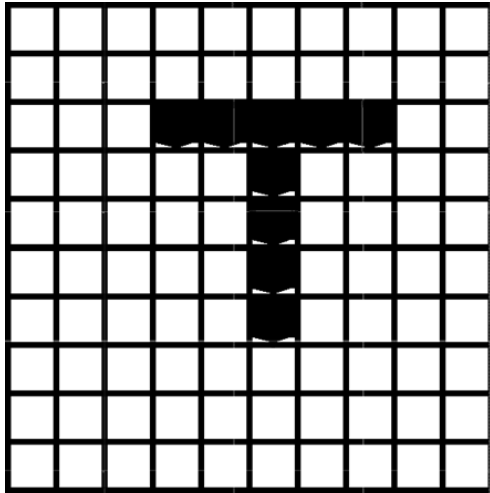


Structuring Element

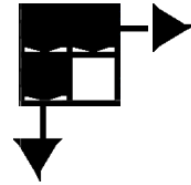


Result

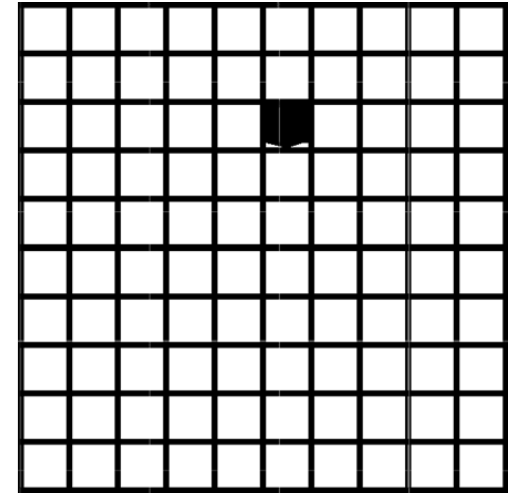
Structuring Element in Erosion Example



Image

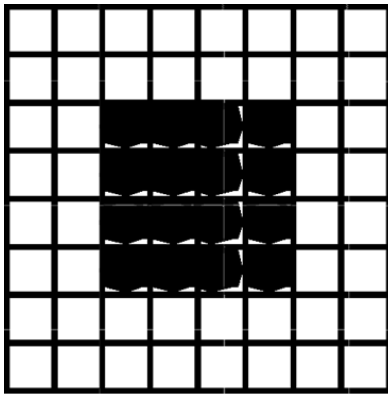


Structuring Element

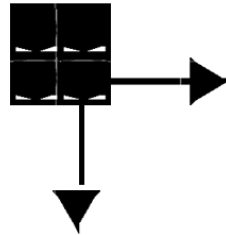


Result

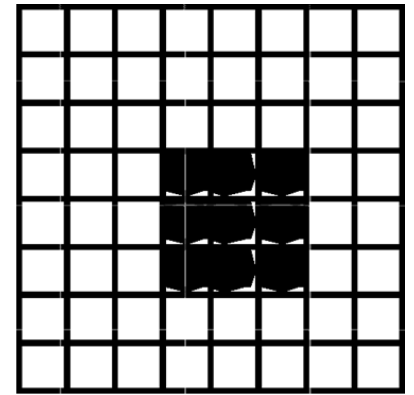
Structuring Element in Erosion Example



Image

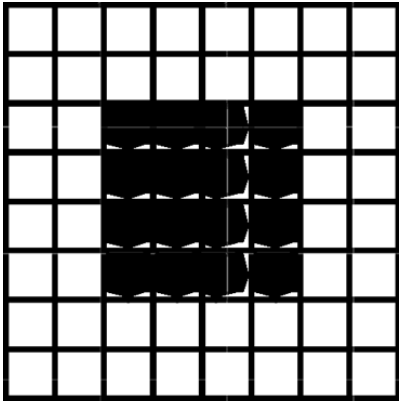


Structuring Element

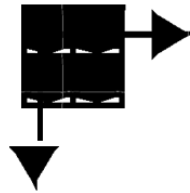


Result

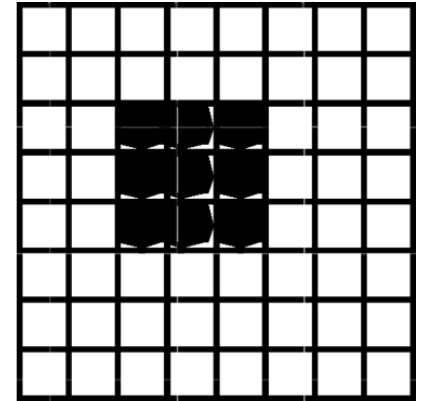
Structuring Element in Erosion Example



Image

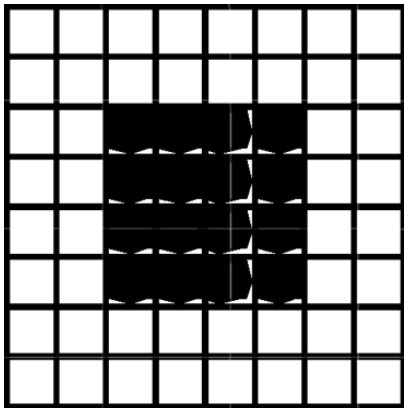


Structuring Element

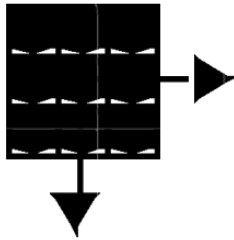


Result

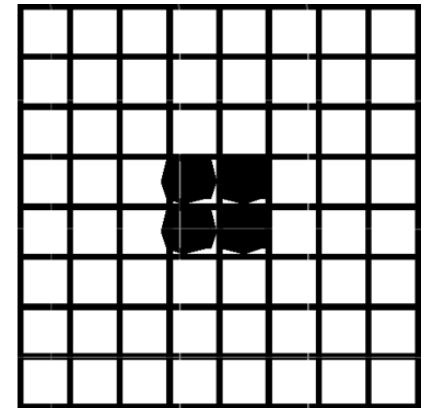
Structuring Element in Erosion Example



Image

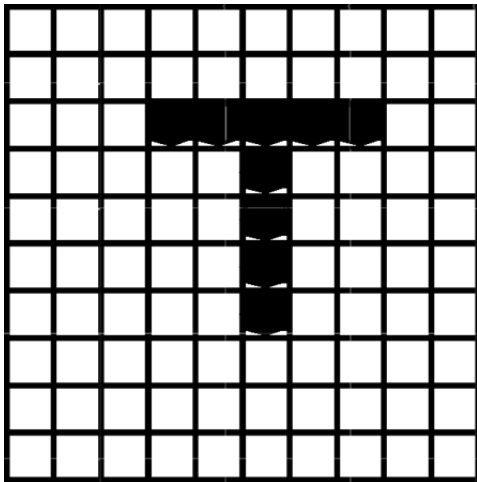


Structuring Element

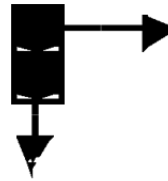


Result

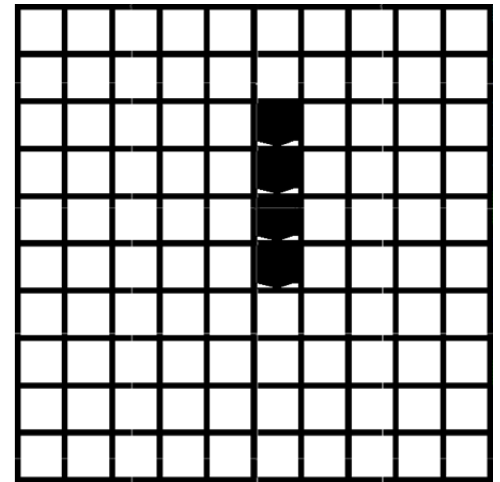
Structuring Element in Erosion Example



Image

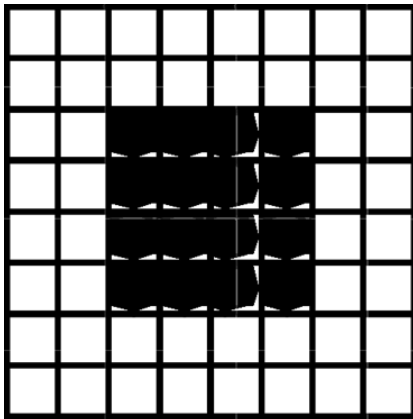


Structuring Element

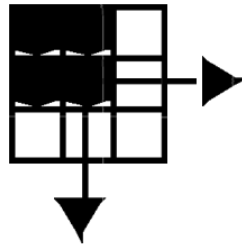


Result

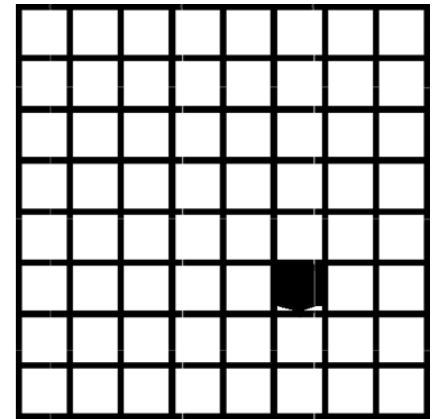
Structuring Element in Erosion Example



Image

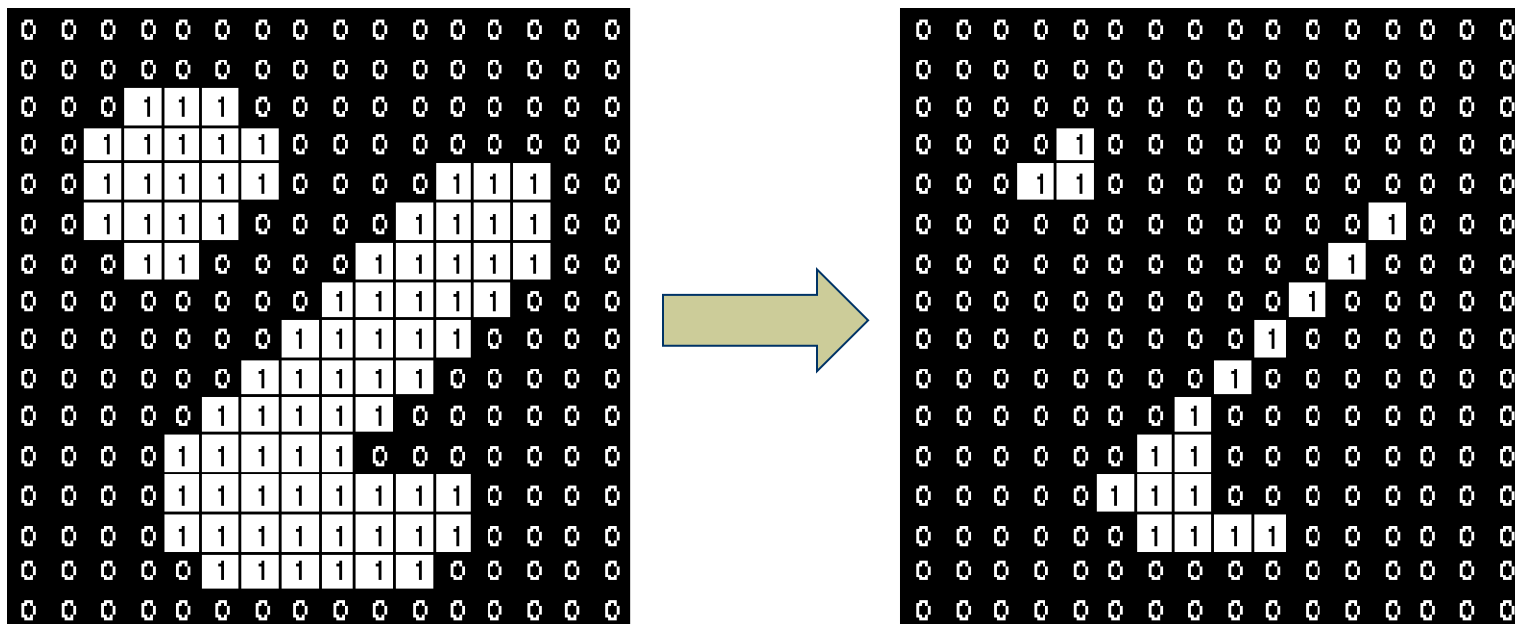


Structuring Element



Result

Erosion: Example 2



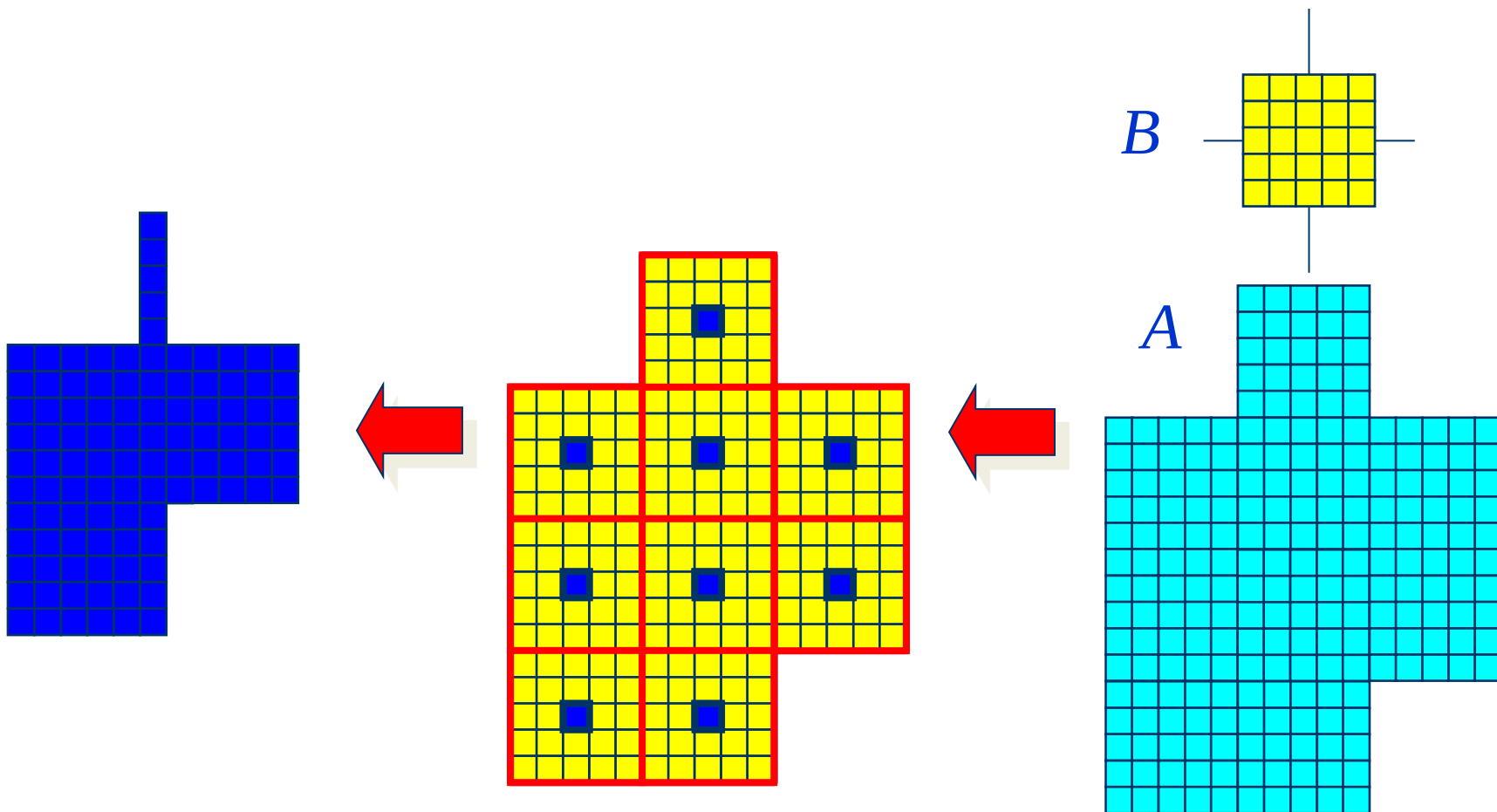
Erosion with a structuring element of size 3x3

1	1	1
1	1	1
1	1	1

Set of coordinate points =

{ (-1, -1), (0, -1), (1, -1),
 (-1, 0), (0, 0), (1, 0),
 (-1, 1), (0, 1), (1, 1) }

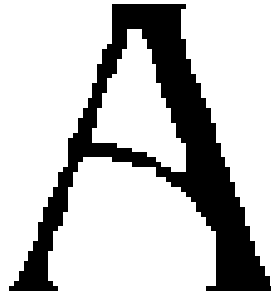
Erosion: Example 3



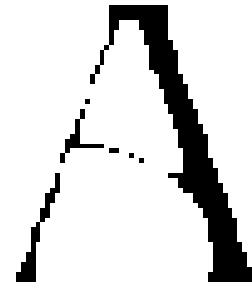
Erosion: Example 4



Original image



Erosion by 3*3
square structuring
element

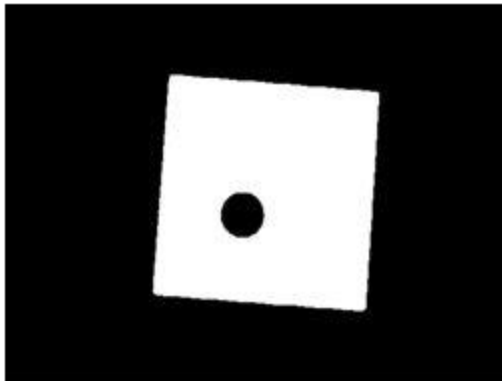


Erosion by 5*5
square structuring
element

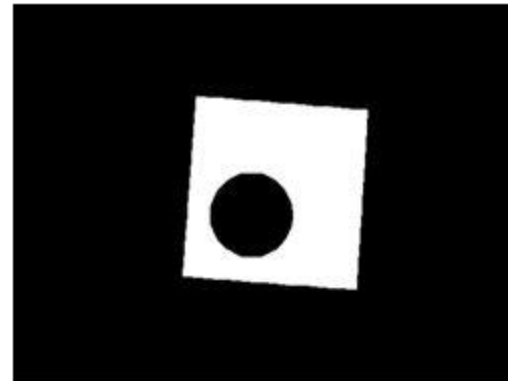
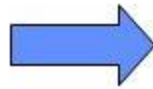
Note: In these examples a 1 refers to a black pixel!

Erosion: Example 5

⇒ Example: Binary erosion



Original thresholded image



Result of erosion four times with a disk shaped structuring element of 11 pixels in diameter

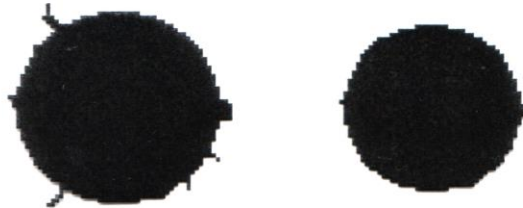
- ➔ It shows that the hole in the middle of the image increases in size as the border shrinks.
- ➔ Erosion using a disk shaped structuring element will tend to round concave boundaries, but will preserve the shape of convex boundaries.

Erosion: Example 6

Erosion can split apart joined objects



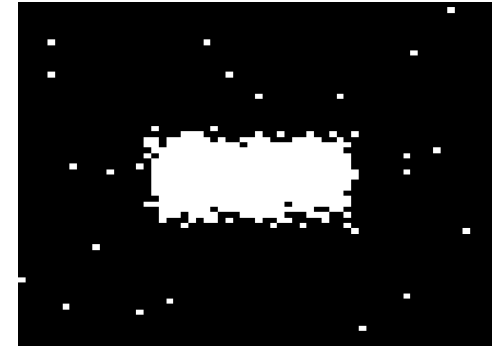
Erosion can strip away extrusions



Watch out: Erosion shrinks objects

Typical use of Erosion

- Removes isolated noisy pixels.
- Smooths object boundary (removes spiky edges).
- Removes the outer layer of object pixels:
 - Object becomes slightly smaller.
 - Sets contour pixels of object to background value



Erosion

◆ Effects

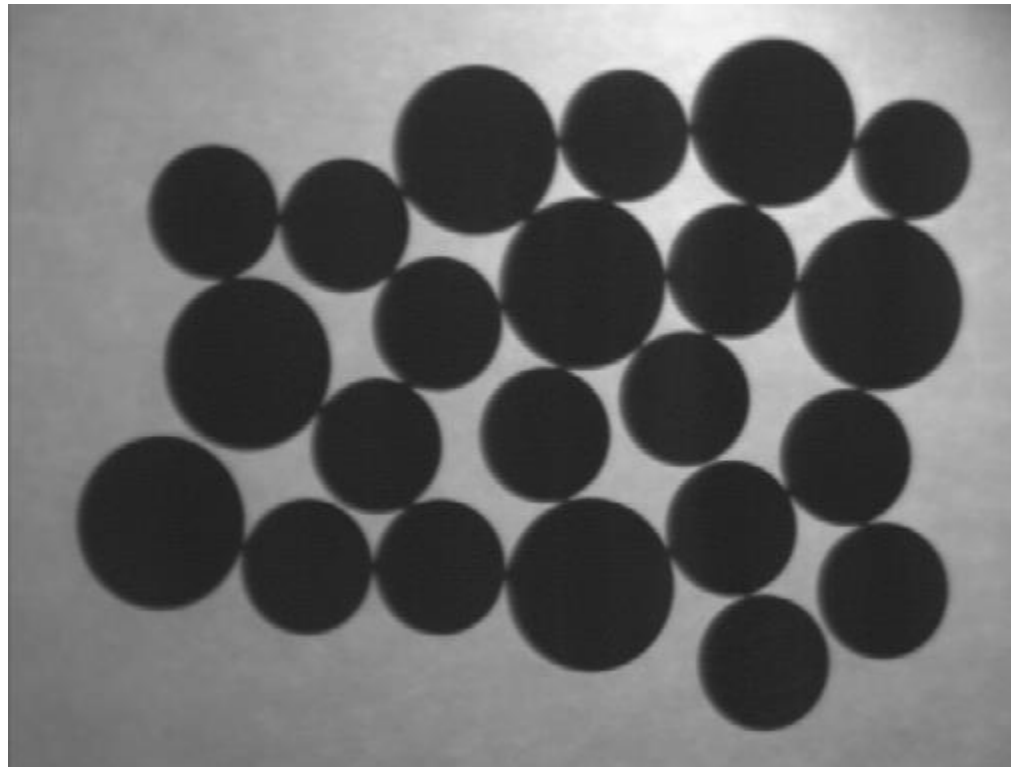
- Shrinks the size of foreground (1-valued) objects
- Smooths object boundaries
- Removes small objects

◆ Rule for Erosion

In a binary image, if any of the pixel (in the neighborhood defined by structuring element) is 0, then output is 0

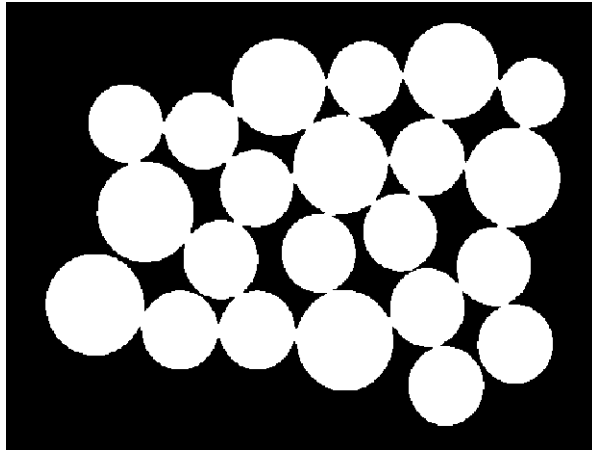
Exercise

- ❖ Count the number of coins in the given image performing Erosion operation.

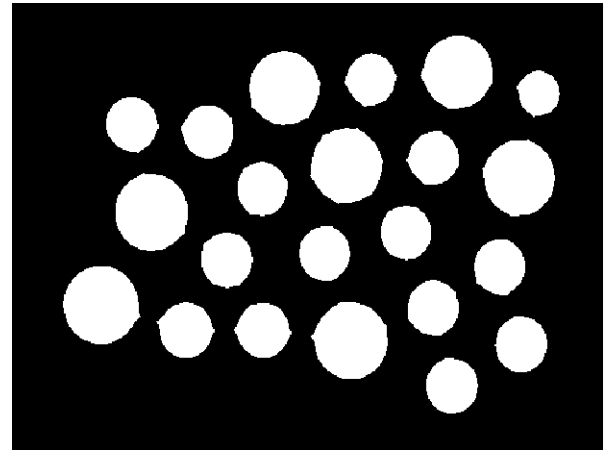


Exercise: Solution

Binarize the image

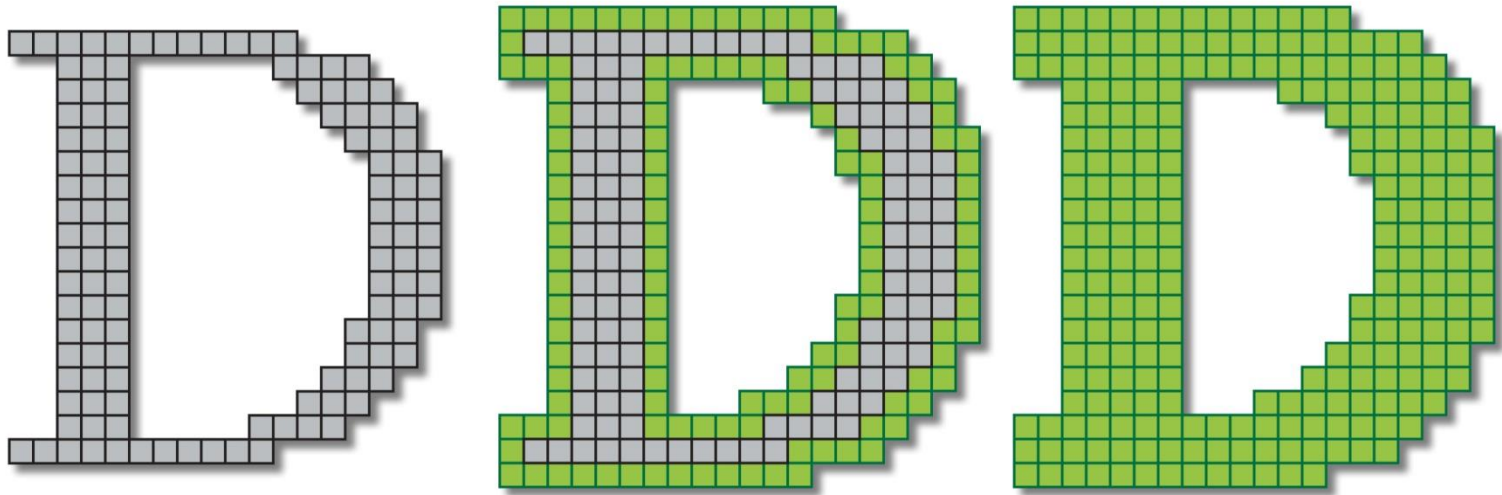


Perform Erosion



Use connected component labeling to count the number of coins

Dilation

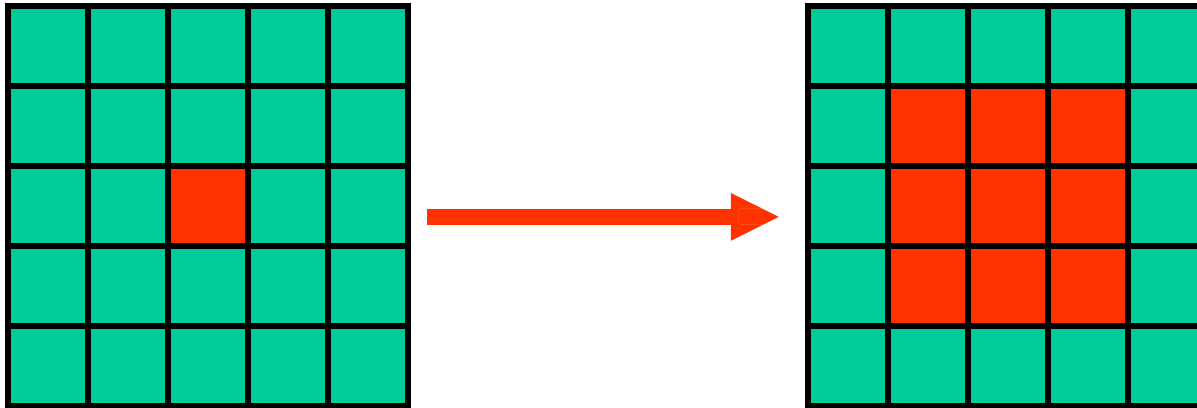


Grows the object

Enlarges foreground, shrinks background

Dilation

- Dilation is used for expanding an element A by using structuring element B.
- The dilation operator takes two pieces of data as input
 1. A binary image, which is to be dilated
 2. A structuring element (or kernel), which determines the behavior of the morphological operation



Original Image

Dilated Image

Dilation

Definition 1:

Hit: Any “**ON**” pixel in the structuring element covers an “**ON**” pixel in the image

- Does the structuring element **hit the set**?

$$A \oplus B = \left\{ z \mid (\hat{B})_z \cap A \neq \Phi \right\}$$

- Dilation of a set A by structuring element B:
 - all z in A such that B hits A when origin of B=z
 - such that overlap A by at least one element

Dilation

Definition 2:

Dilation of image f by structuring element s is given by $f \oplus s$

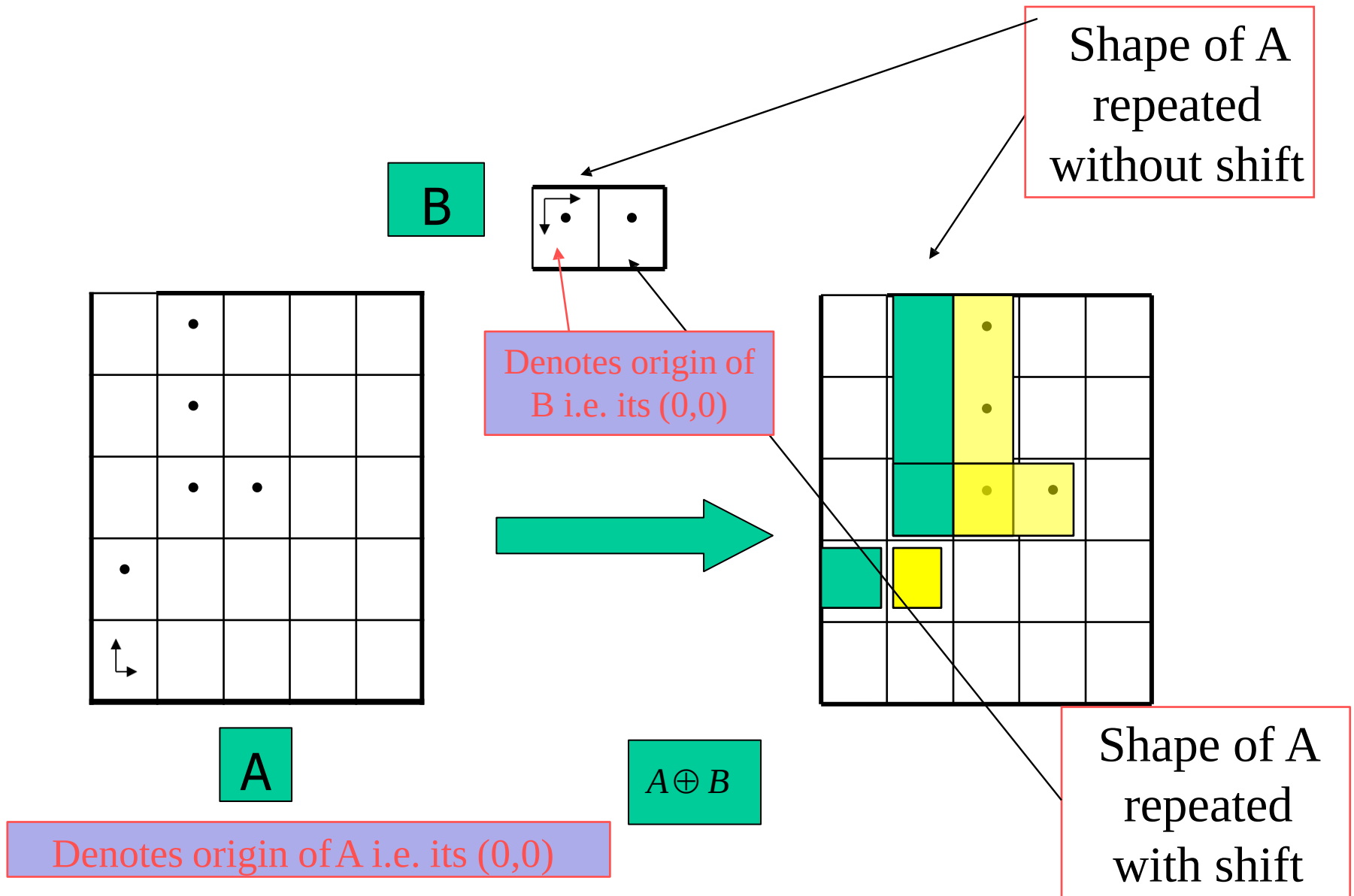
The structuring element s is positioned with its origin at (x, y) and the new pixel value is determined using the rule:

$$g(x, y) = \begin{cases} 1 & \text{if } s \text{ hits } f \\ 0 & \text{otherwise} \end{cases}$$

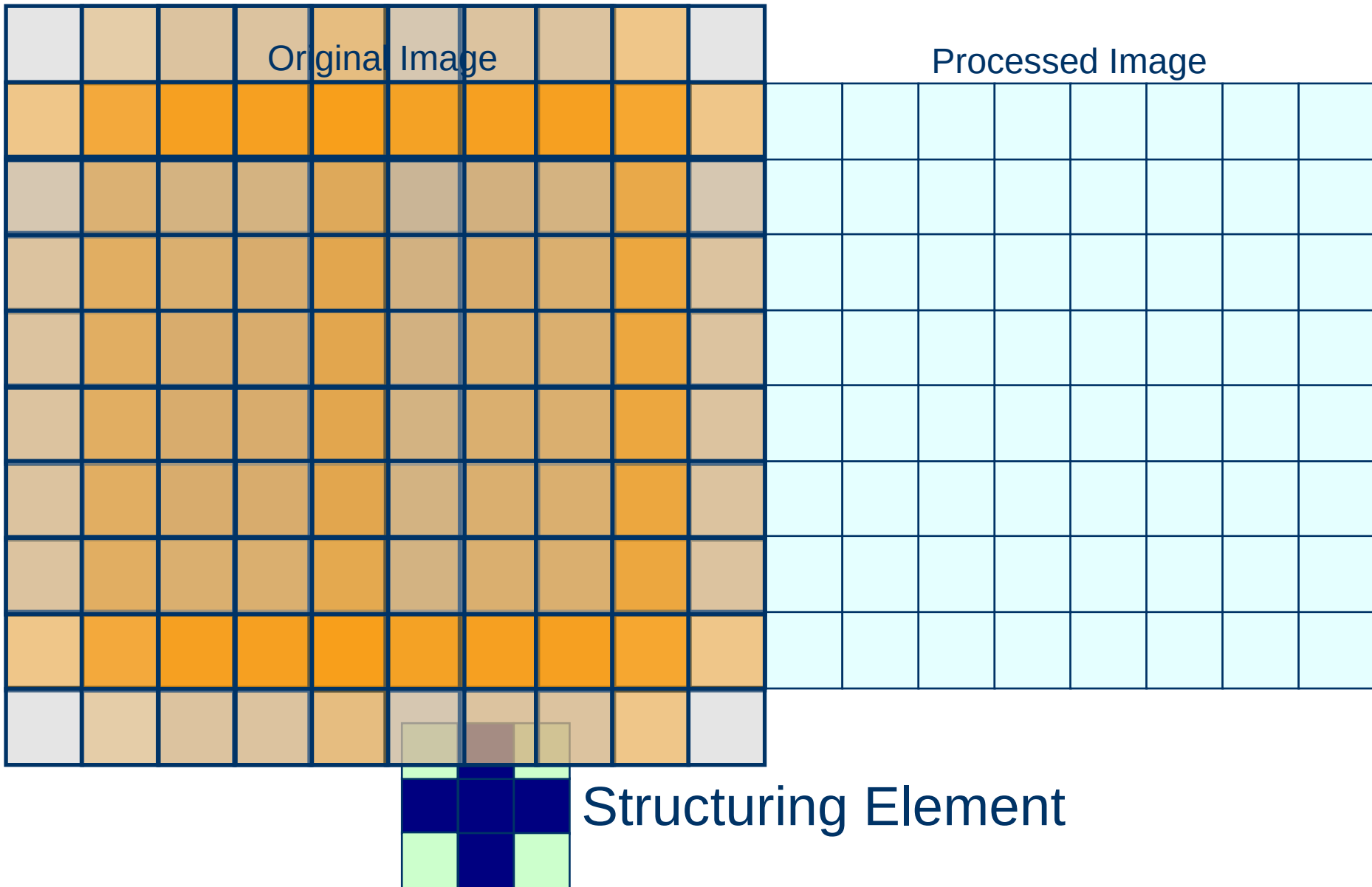
Dilation – How to compute

- ◆ For each background pixel (which we will call the *input pixel*)
 - Superimpose the structuring element on top of the input image so that the origin of the structuring element coincides with the input pixel position
 - If *at least one* pixel in the structuring element coincides with a foreground pixel in the image underneath, then the input pixel is set to the foreground value
 - If all the corresponding pixels in the image are background, however, the input pixel is left at the background value

Dilation explained by shape of A

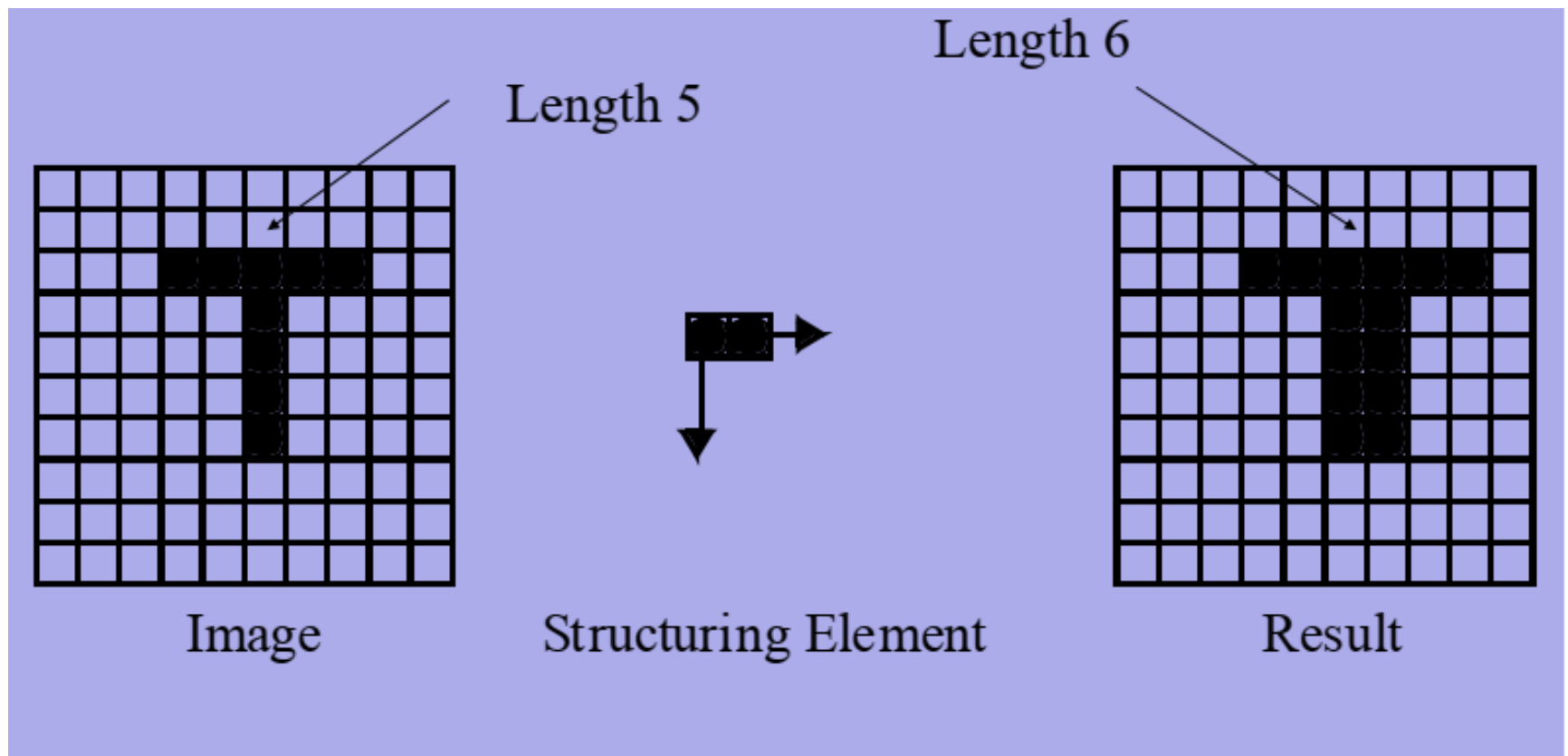


Dilation: Example



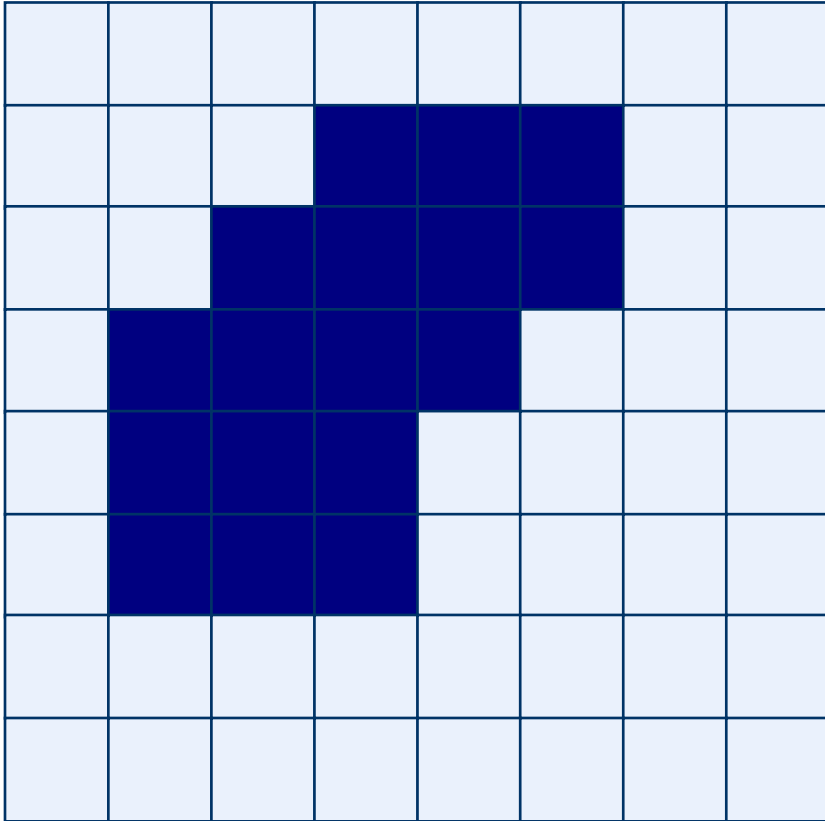
Structuring Element in Dilation

Example

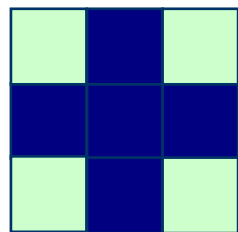
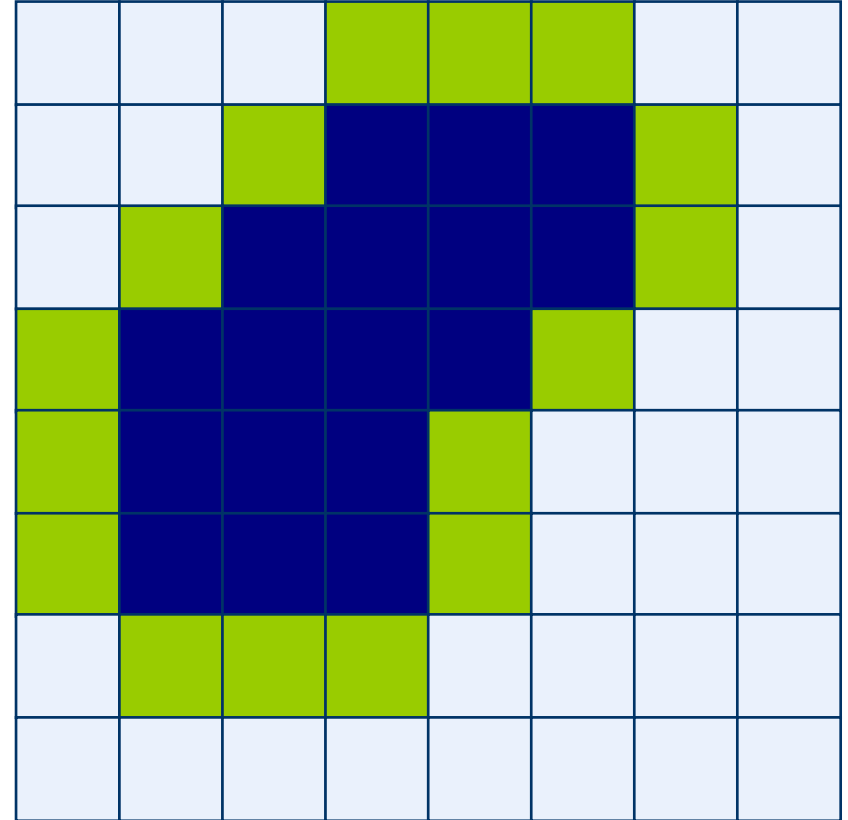


Dilation: Example 1

Original Image



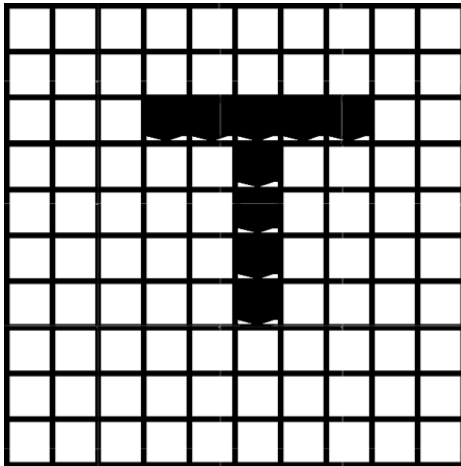
Processed Image With Dilated Pixels



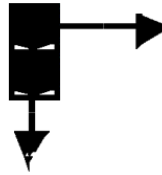
Structuring Element

Structuring Element in Dilation

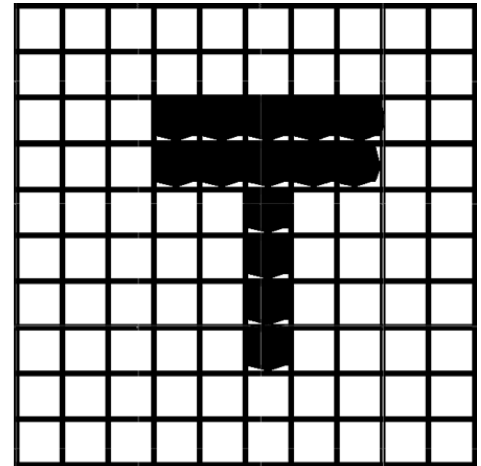
Example



Image



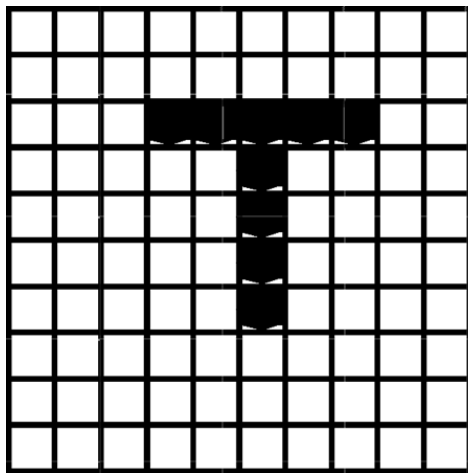
Structuring Element



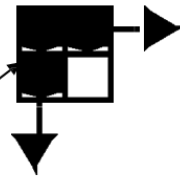
Result

Structuring Element in Dilation

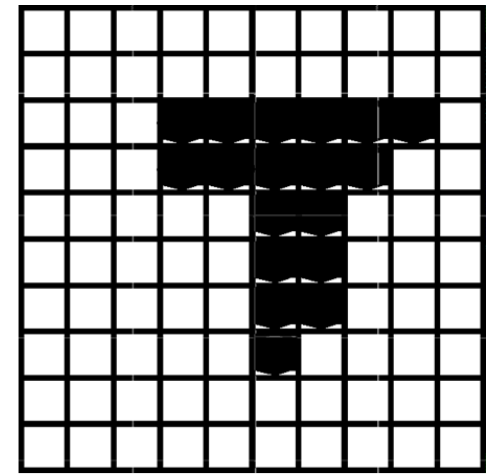
Example



Image



Structuring Element

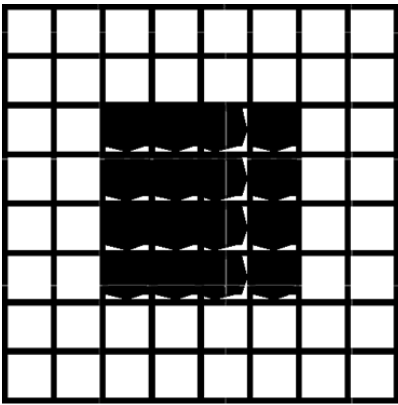


Result

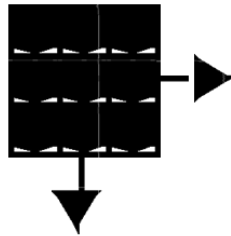
Single point in Image replaced with
this in the Result

Structuring Element in Dilation

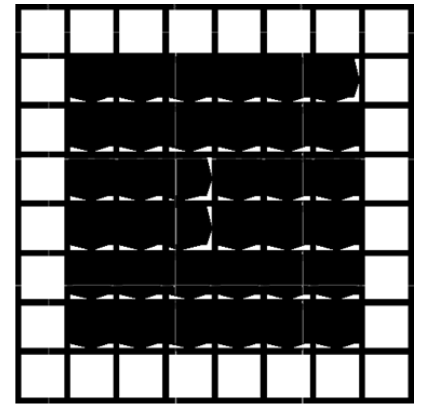
Example



Image



Structuring Element



Result

Dilation

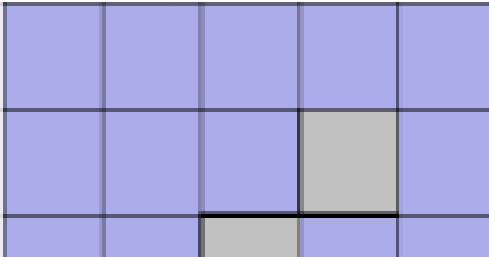
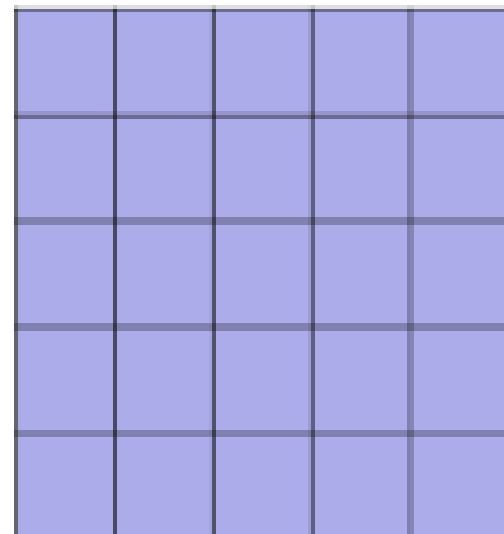
Question: Suppose that the structuring element is a 3x3 square with the origin at its center evaluate the

$$B = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

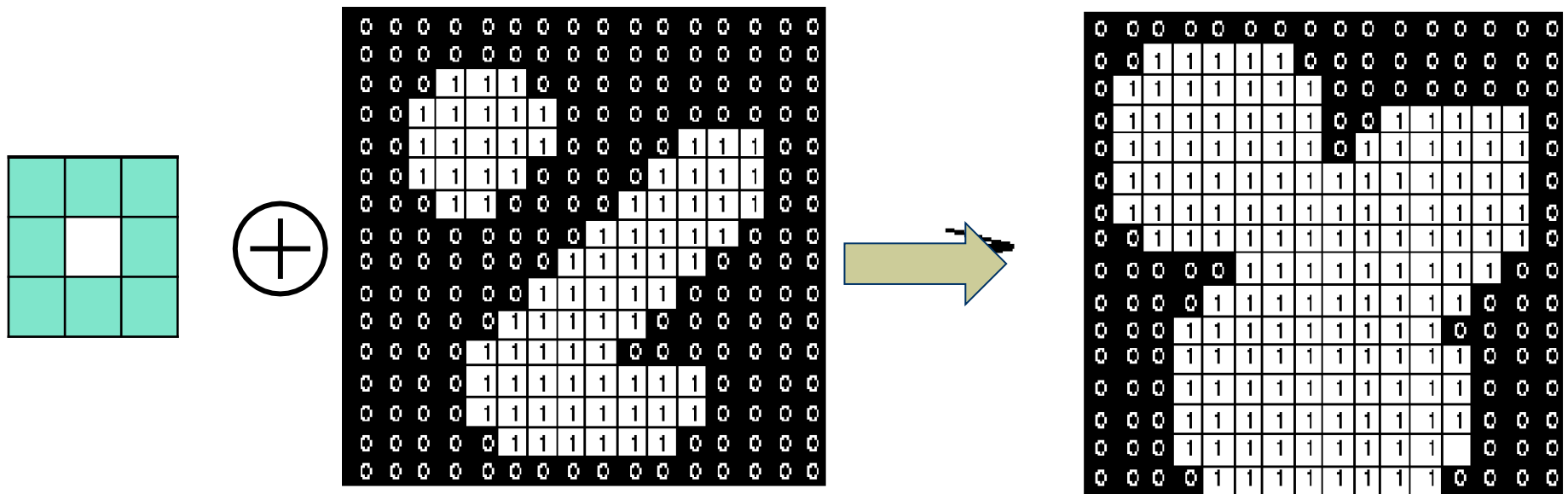
new image

$$\{ \begin{array}{lll} (-1, -1), & (0, -1), & (1, -1), \\ (-1, 0), & (0, 0), & (1, 0), \\ (-1, 1), & (0, 1), & (1, 1) \end{array} \}$$

$A =$

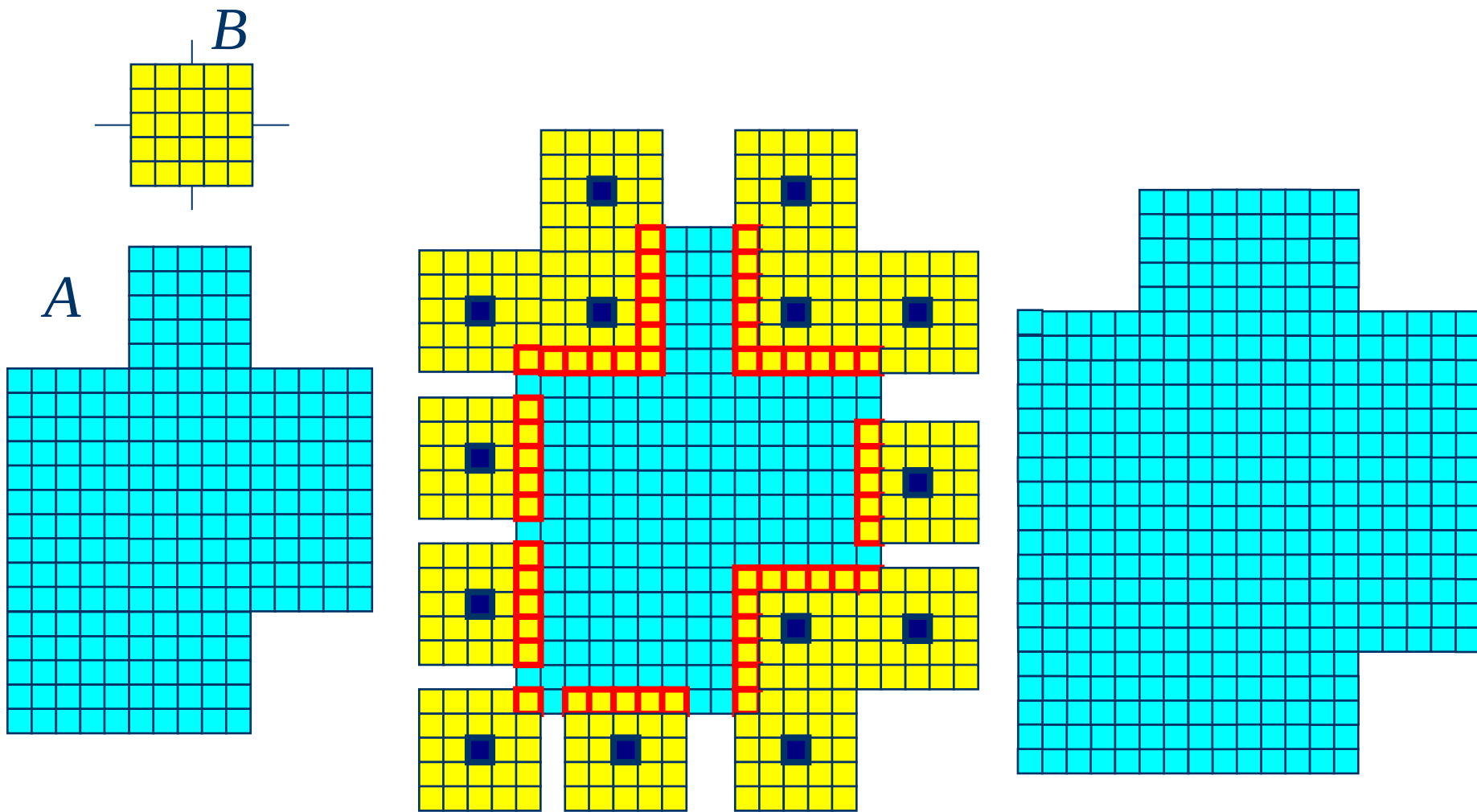

$$A \oplus B$$


Dilation: Example 2



Effect of dilation using a 3×3 square structuring element

Dilation: Example 3



Dilation: Example 4



Original image



Dilation by 3*3
square structuring
element



Dilation by 5*5
square structuring
element

Note: In these examples a 1 refers to a black pixel!

Dilation : Bridging gaps

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.



Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.



0	1	0
1	1	1
0	1	0

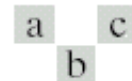


FIGURE 9.5
(a) Sample text of poor resolution with broken characters (magnified view).
(b) Structuring element.
(c) Dilation of (a) by (b). Broken segments were joined.

Dilation: Example 5

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.

Dilation: Example 6

Dilation can repair breaks



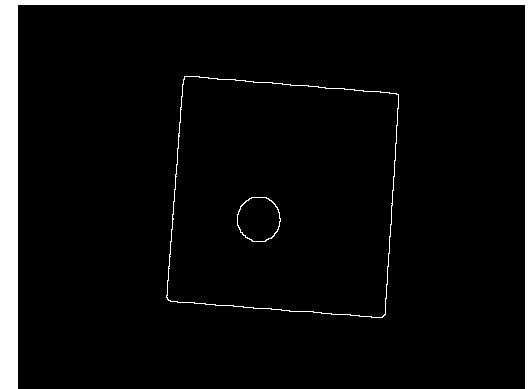
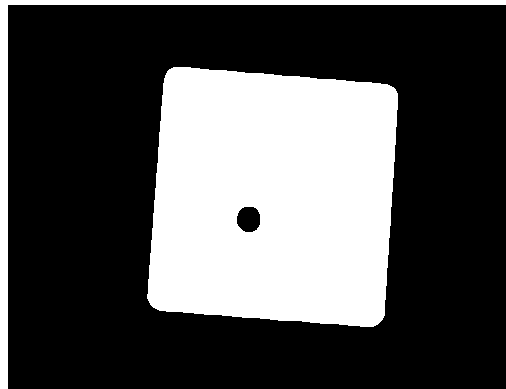
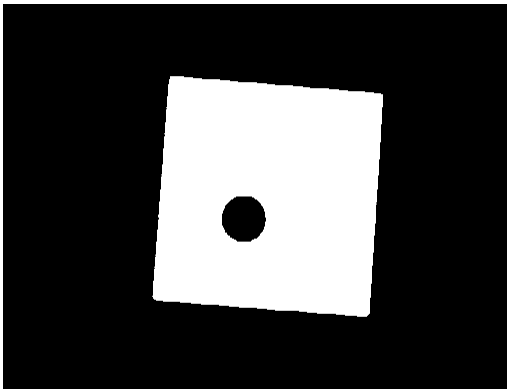
Dilation can repair intrusions



Watch out: Dilation enlarges objects

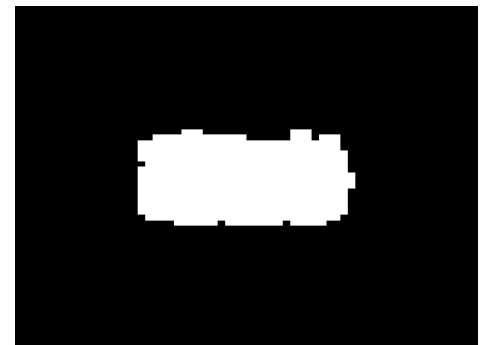
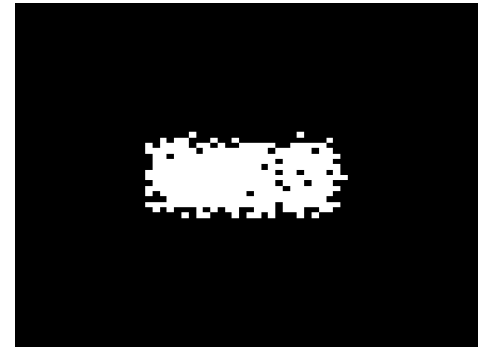
Dilation: Example 7

- Edge Detection
 1. Dilate input image
 2. Subtract input image from dilated image
 3. Edges remain!



Typical use of Dilation

- Fills in holes.
- Smoothens object boundaries.
- Adds an extra outer ring of pixels onto object boundary, ie, object becomes slightly larger.



Dilation

◆ Effects

- Expands the size of foreground(1-valued) objects
- Smooths object boundaries
- Closes holes and gaps

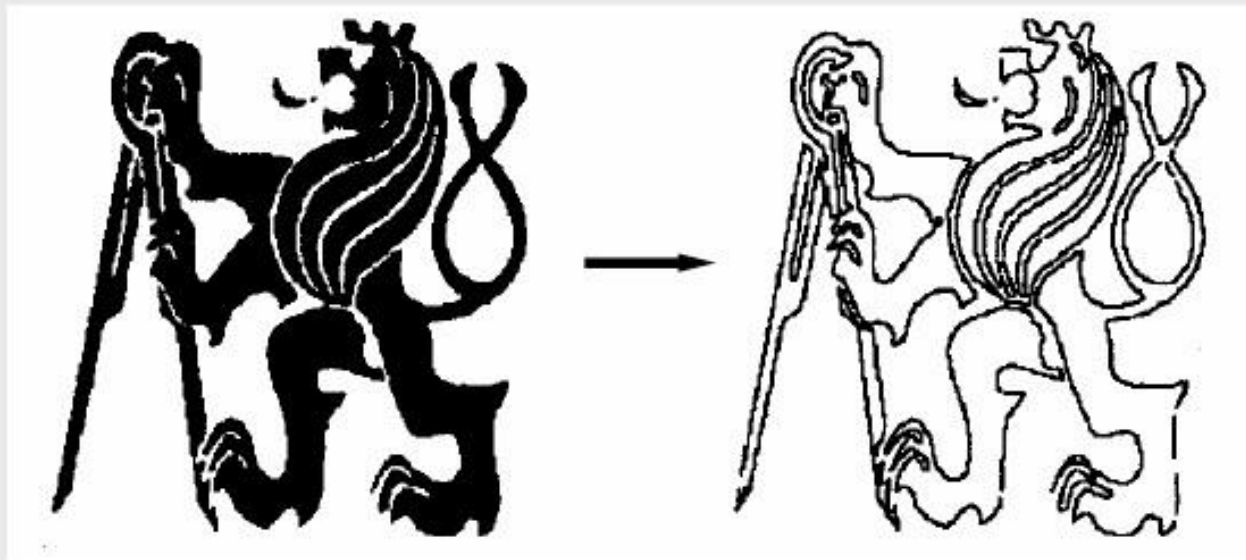
◆ Rule for Dilation

In a binary image, if any of the pixel (in the neighborhood defined by structuring element) is 1, then output is 1

More Applications of Erosion and Dilation

Boundary Extraction

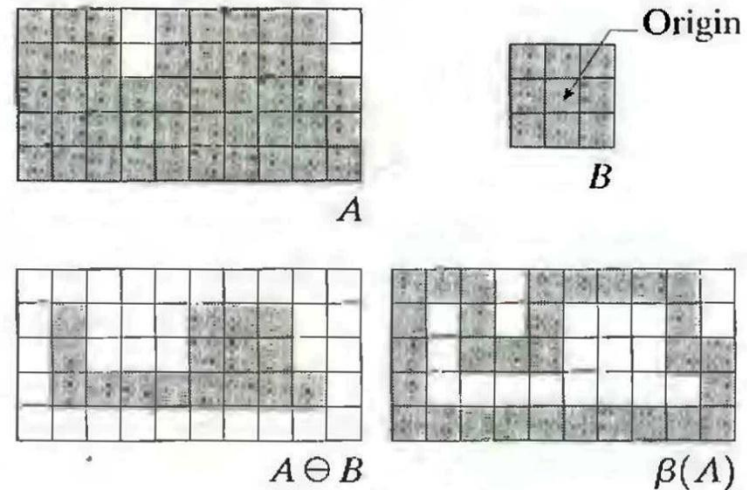
Contours can be extracted by **subtraction** of the **eroded** image from the original.



$$\beta(A) = A - (A \ominus B)$$

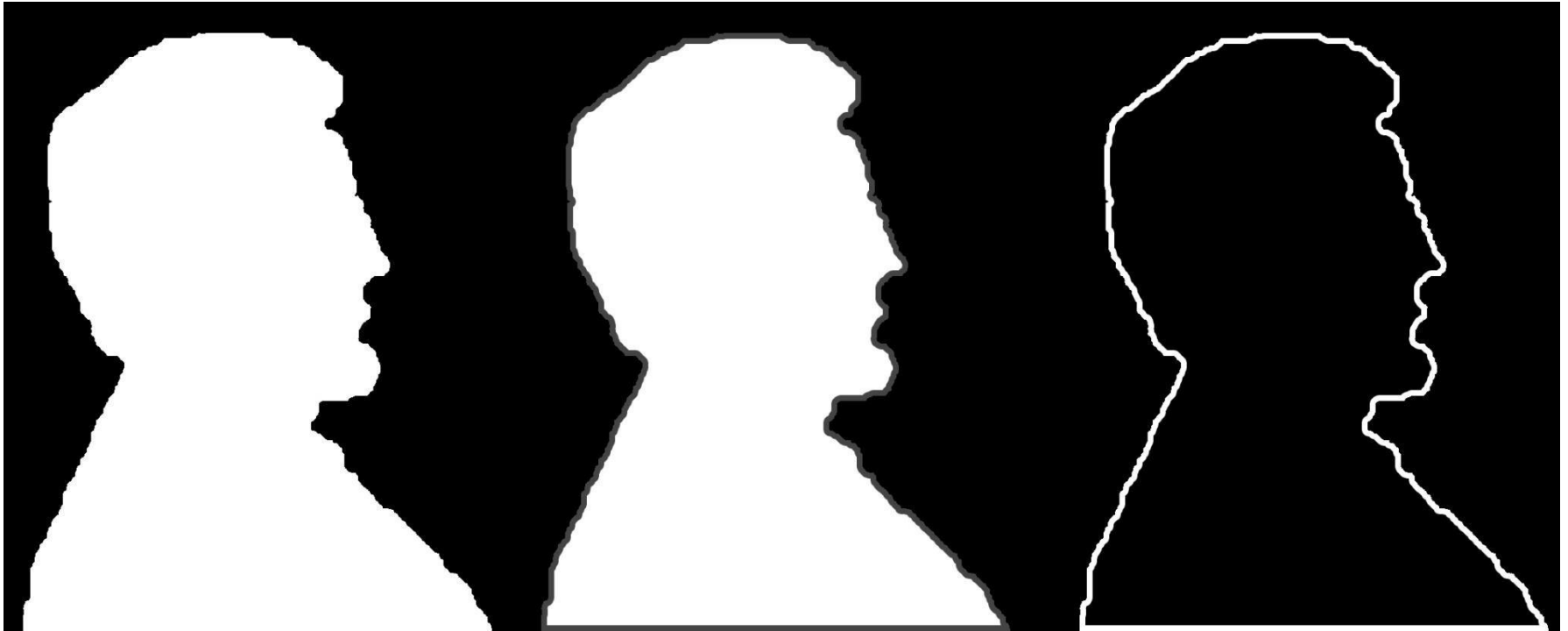
Boundary Extraction

- First, erode A by B , then make set difference between A and the erosion
- The thickness of the contour depends on the size of constructing object – B



$$\beta(A) = A - (A \ominus B)$$

Boundary Extraction



$$\beta(A) = A - (A \ominus B)$$

Edge detection

original



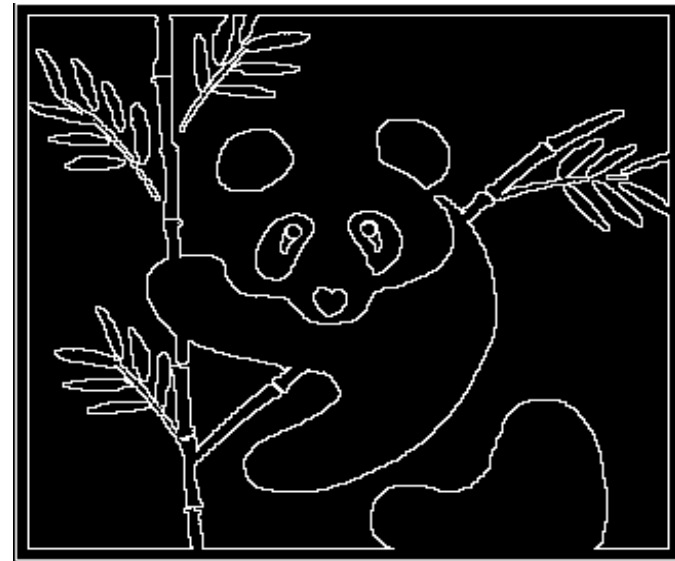
A

Dilate



$A \oplus B$

Dilate - original



$A \oplus B - A$

Duality relationship between Dilation and Erosion

- When one operation is the dual of the other, **it means that one can be written in terms of the other**. This does not, however, mean that they are opposites.
- ♦ Dilation and erosion are duals of each other:

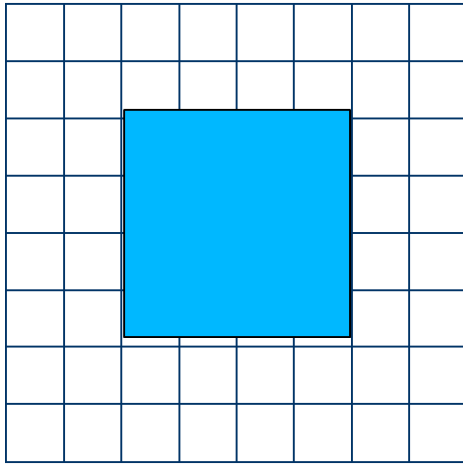
$$(A \ominus B)^c = A^c \oplus \hat{B}$$

- ♦ For a symmetric structuring element:

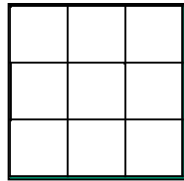
$$(A \oplus B)^c = A^c \ominus \hat{B}$$

It means that we can obtain erosion of an image A by B simply by dilating its background (i.e. A^c) with the same structuring element and complementing the result.

Duality of Dilation and Erosion Example



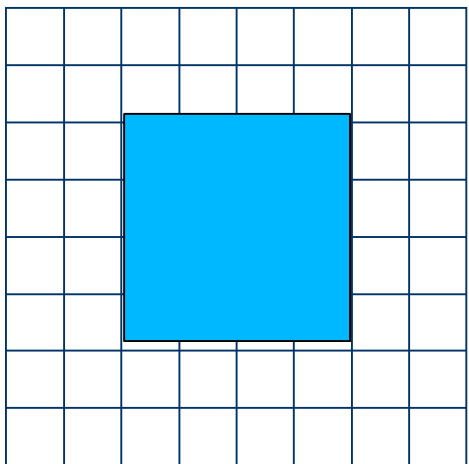
A



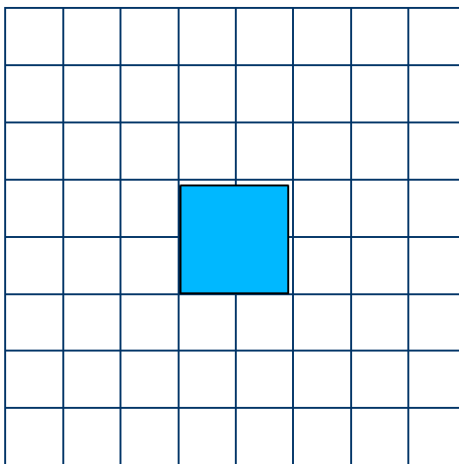
B

$$(A \ominus B)^c = A^c \oplus \hat{B}$$

Duality of Dilation and Erosion Example



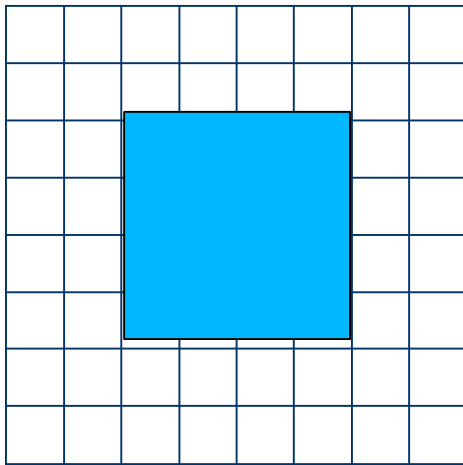
A



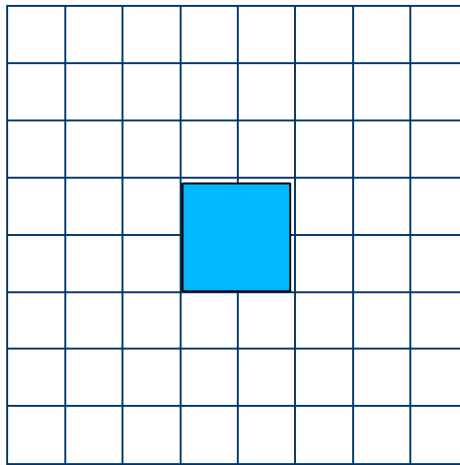
$(A \ominus B)$

$$(A \ominus B)^c = A^c \oplus \hat{B}$$

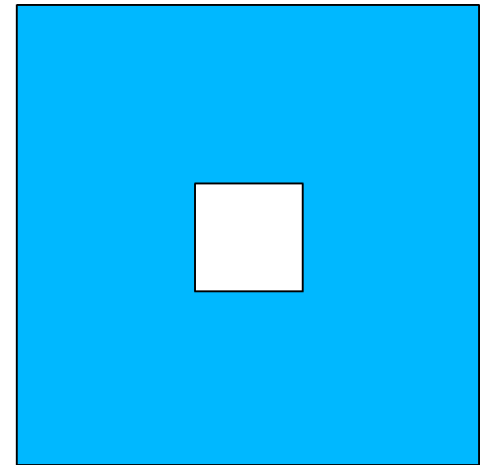
Duality of Dilation and Erosion Example



A



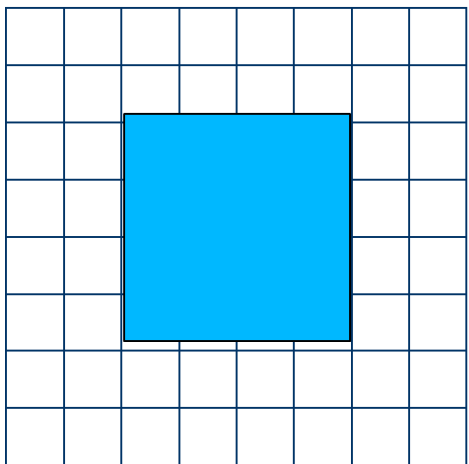
$(A \ominus B)$



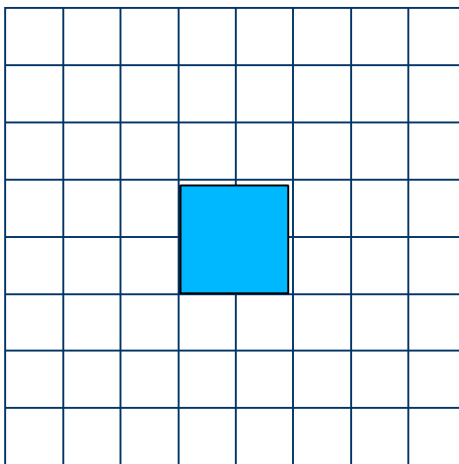
$(A \ominus B)^c$

$$(A \ominus B)^c = A^c \oplus \hat{B}$$

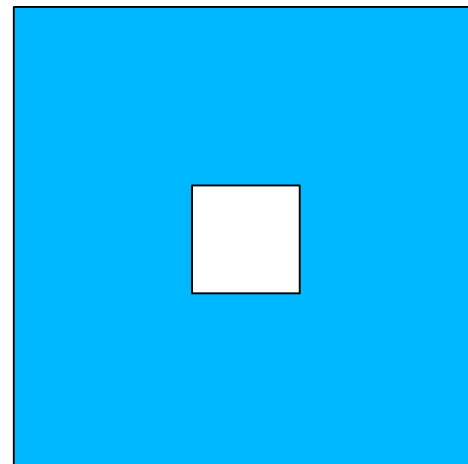
Duality of Dilation and Erosion Example



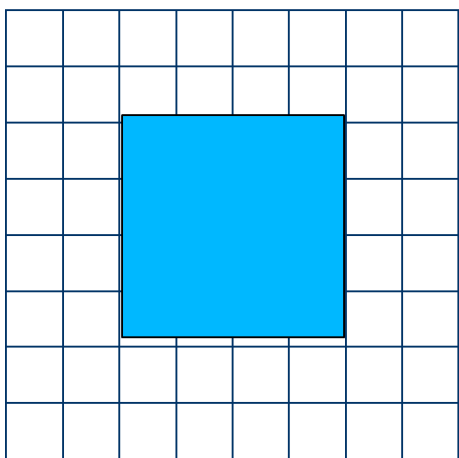
A



$(A \ominus B)$



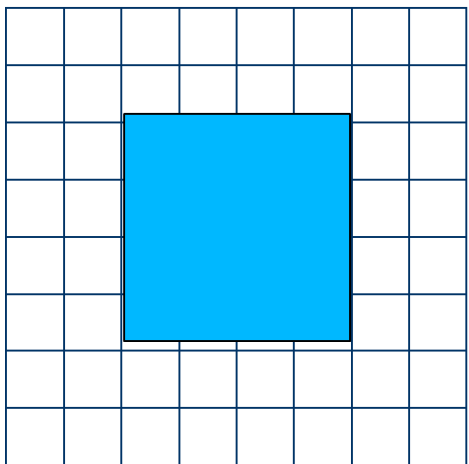
$(A \ominus B)^c$



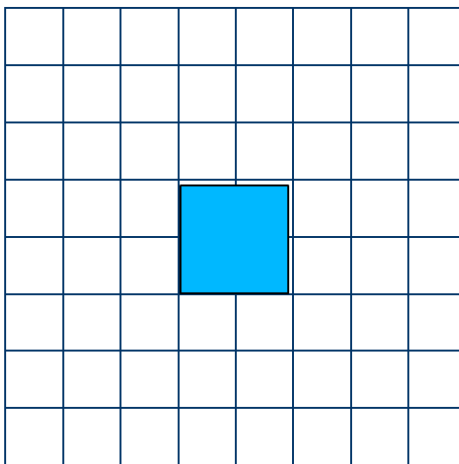
A

$$(A \ominus B)^c = A^c \oplus \hat{B}$$

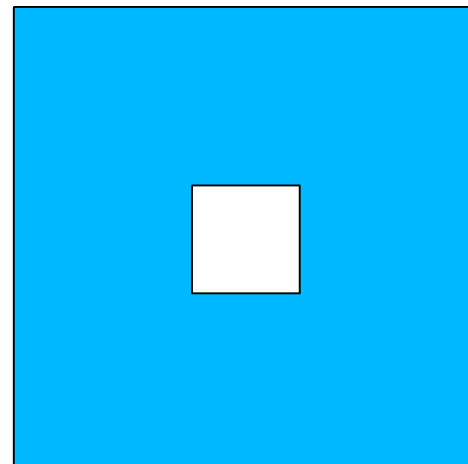
Duality of Dilation and Erosion Example



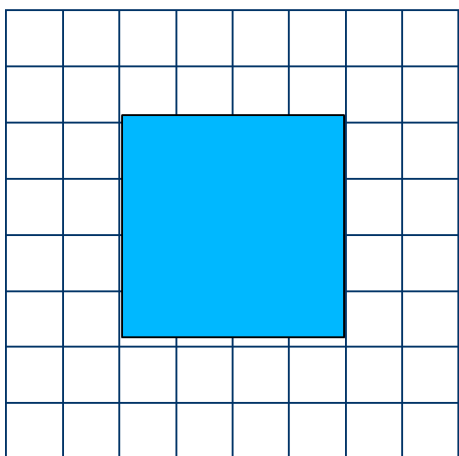
A



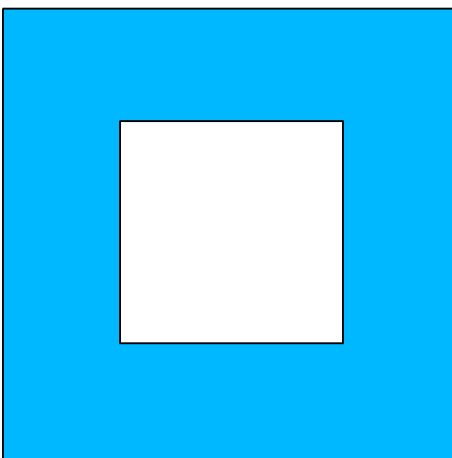
$(A \ominus B)$



$(A \ominus B)^c$

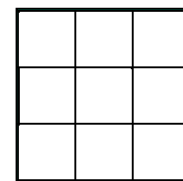


A



A^c

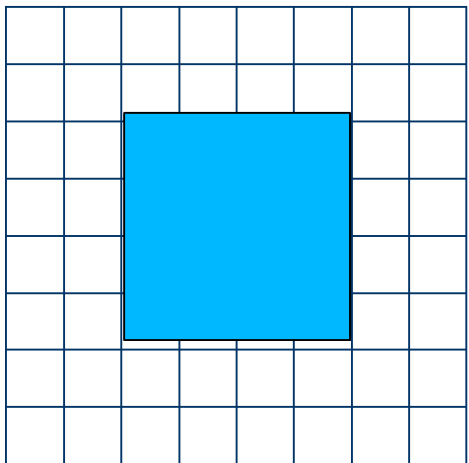
$$(A \ominus B)^c = A^c \oplus \hat{B}$$



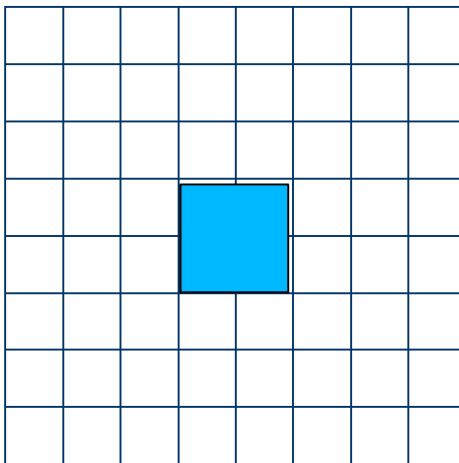
B

Duality of Dilation and Erosion Example

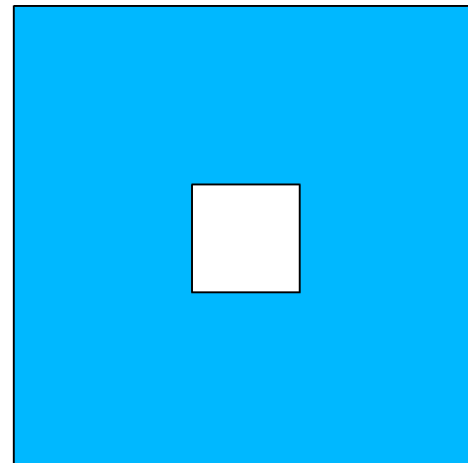
$$(A \ominus B)^c = A^c \oplus \hat{B}$$



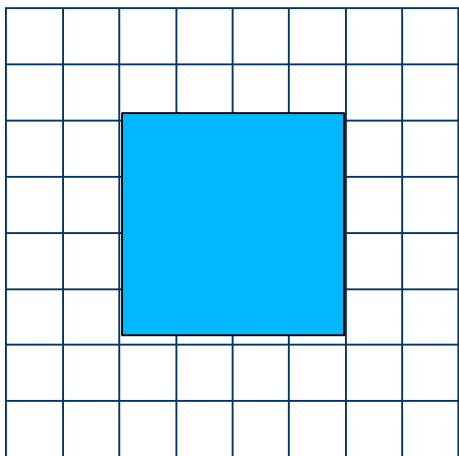
A



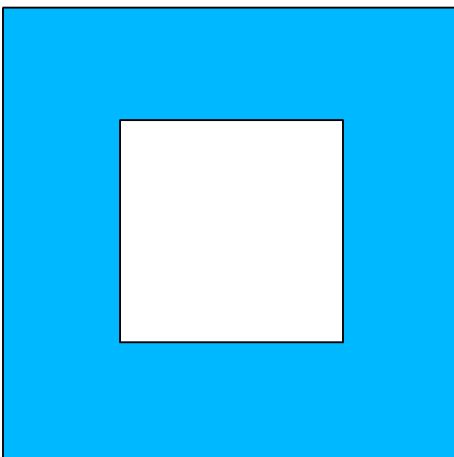
$(A \ominus B)$



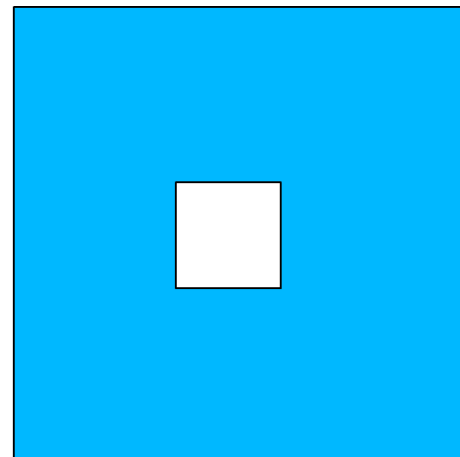
$(A \ominus B)^c$



A



A^c

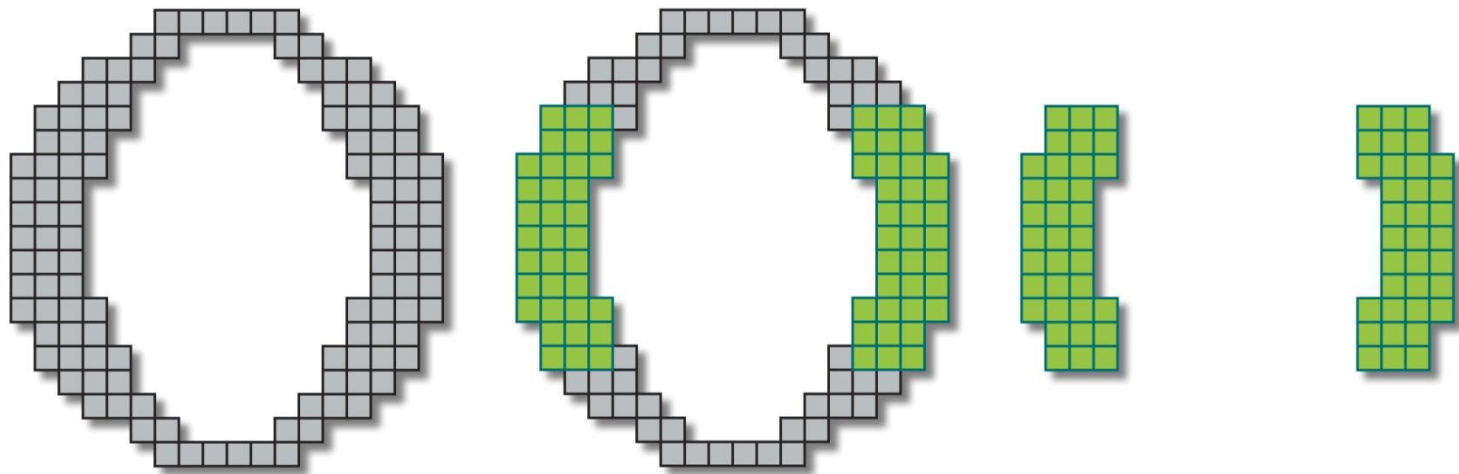


$A^c \oplus \hat{B}$

Compound Operations

- ❑ More interesting morphological operations can be performed by performing combinations of erosions and dilations
- ❑ The most widely used of these *compound operations* are:
 - Opening
 - Closing

Opening

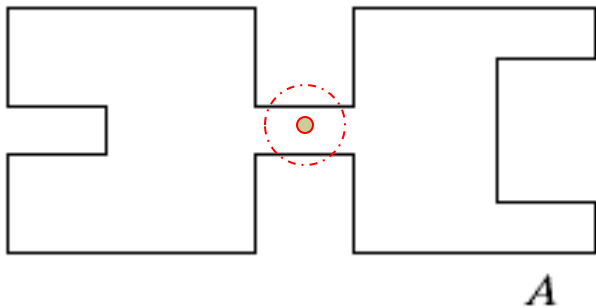


Erosion followed by a dilation

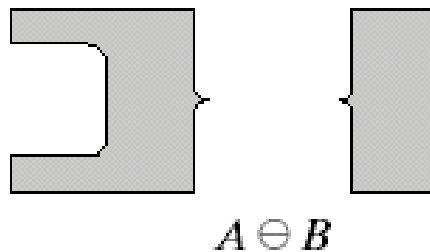
Opening

The opening of image A by structuring element B , denoted by $A \circ B$ is simply an erosion followed by a dilation

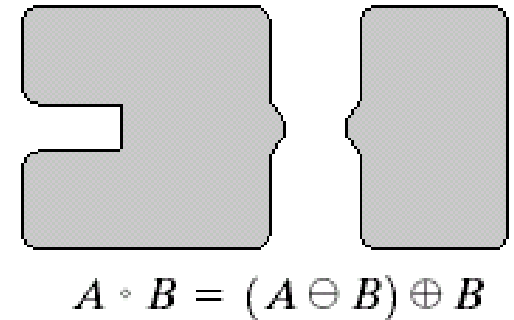
$$A \circ B = (A \ominus B) \oplus B$$



Original shape



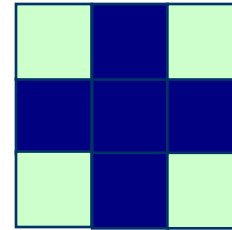
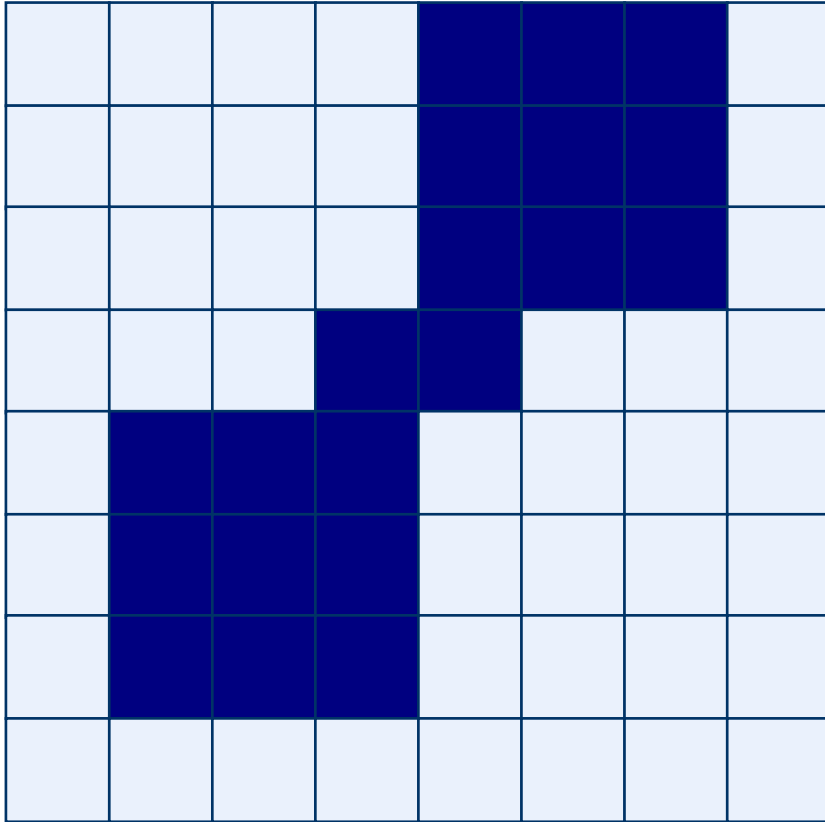
After erosion



After dilation
(opening)

Opening: Example

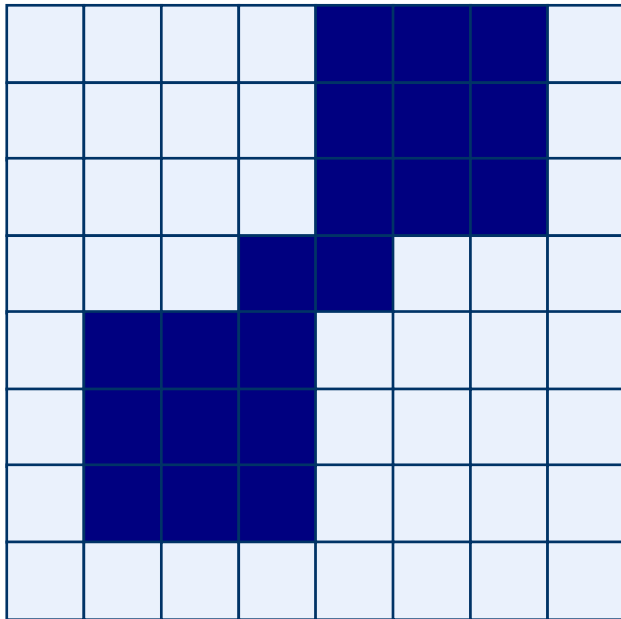
Original Image



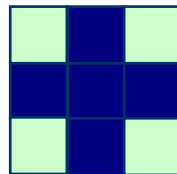
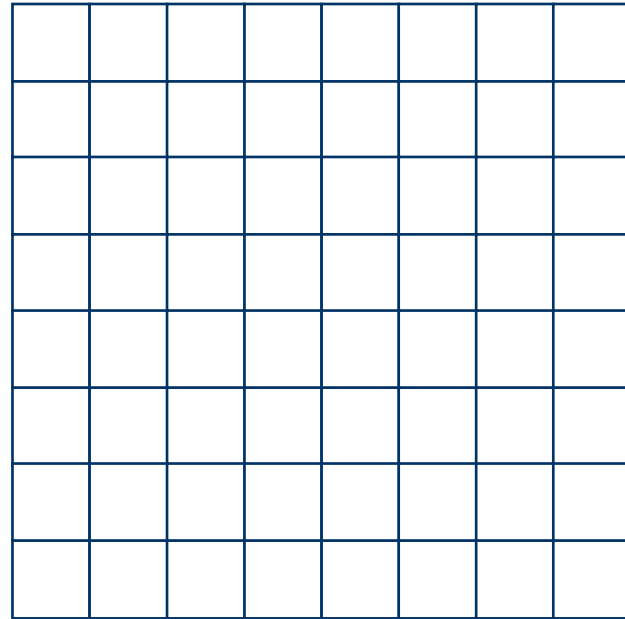
Structuring Element

Opening: Example

Original Image

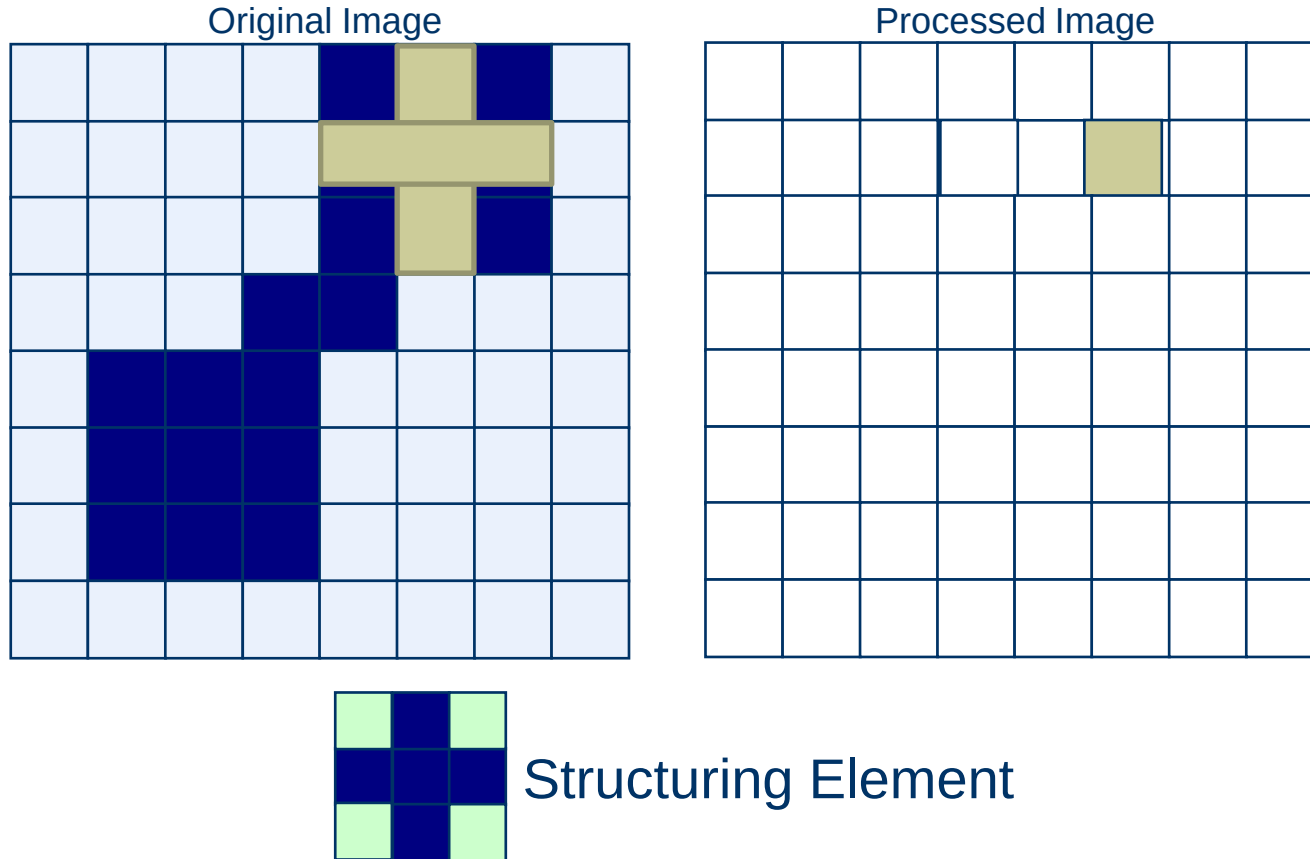


Processed Image

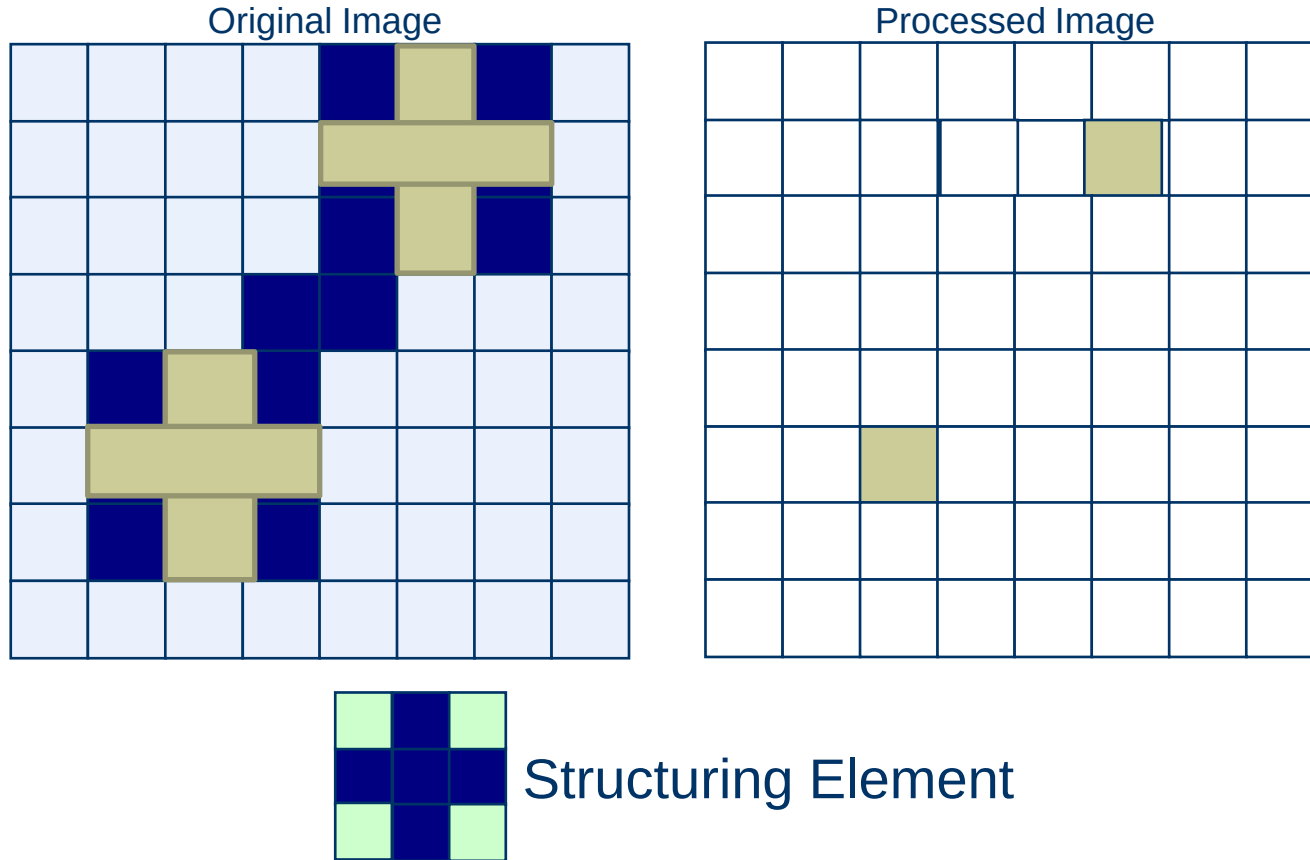


Structuring Element

Opening: Example (performing erosion)

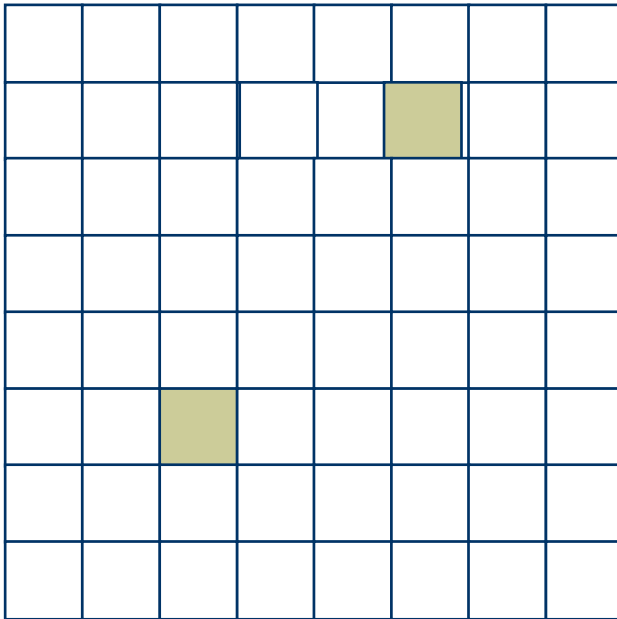


Opening: Example (performing erosion)

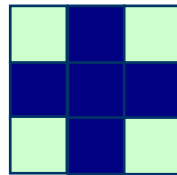
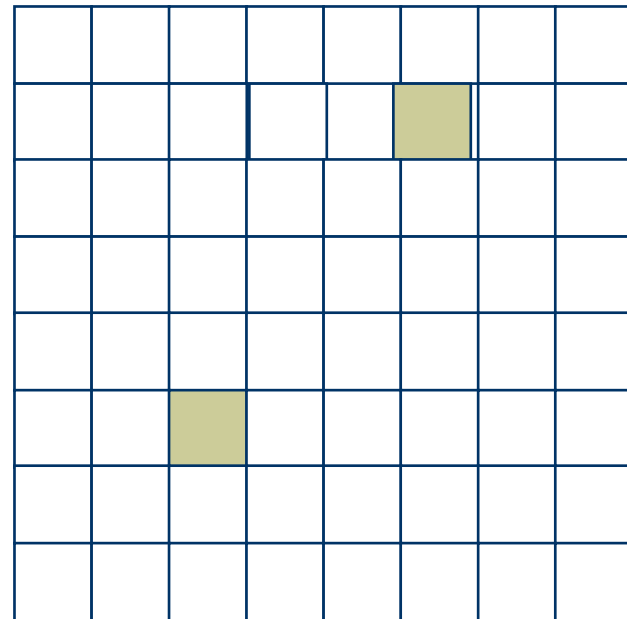


Opening: Example (performing dilation to errored image)

Original Image

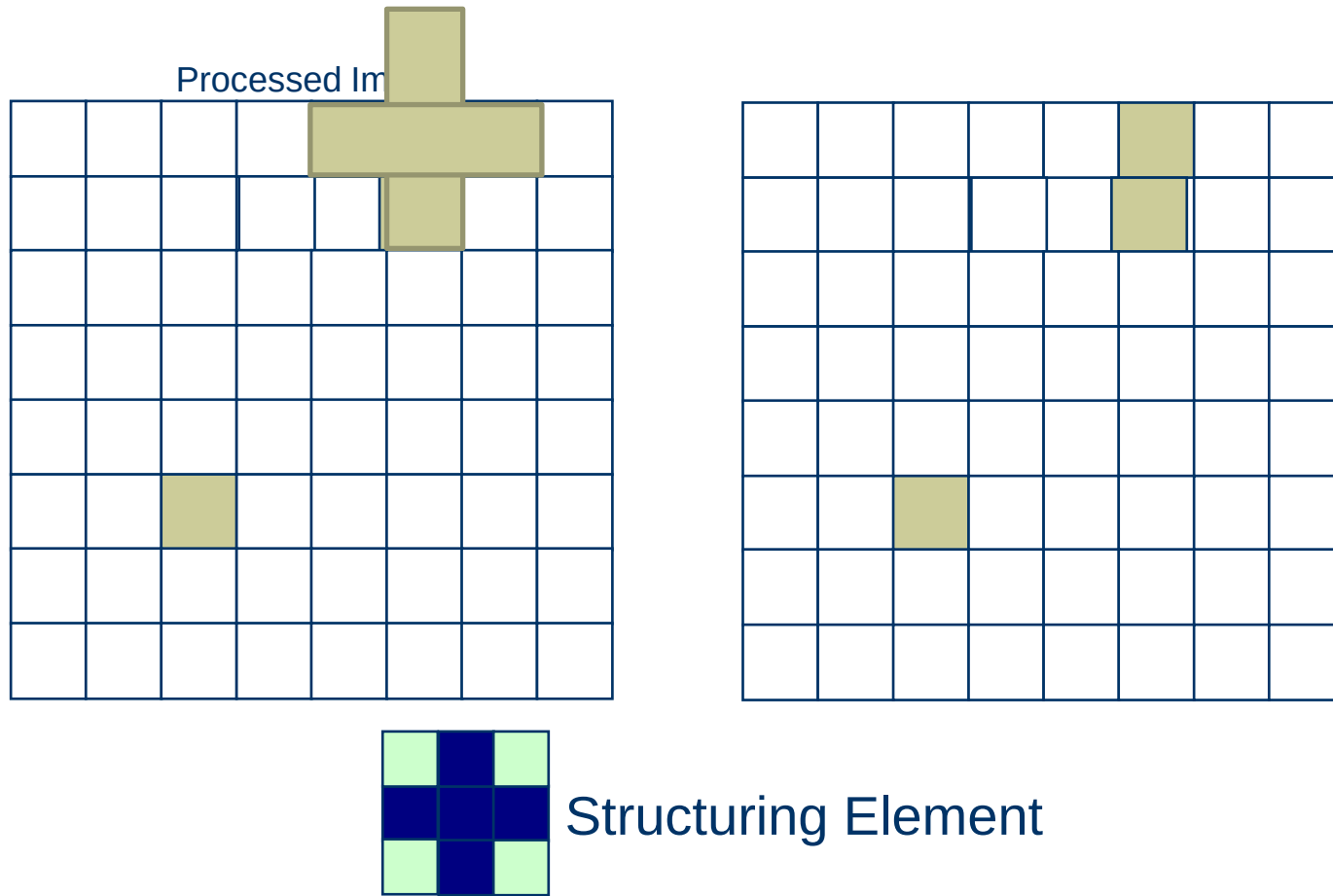


Processed Image

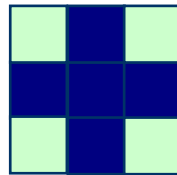
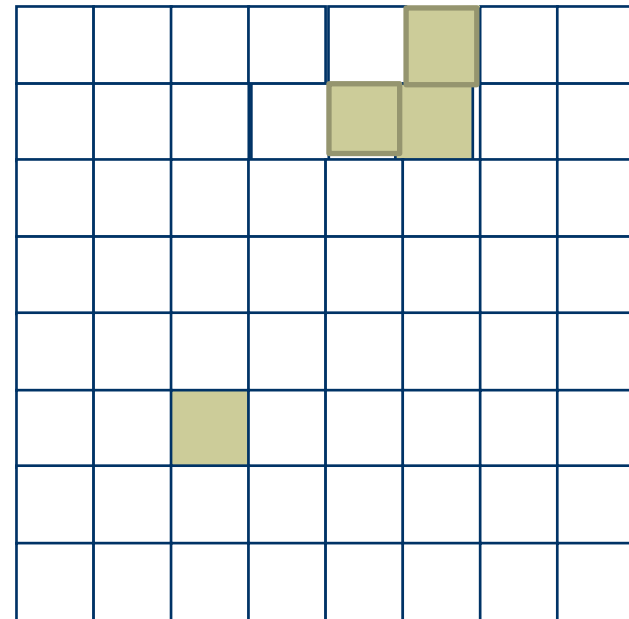
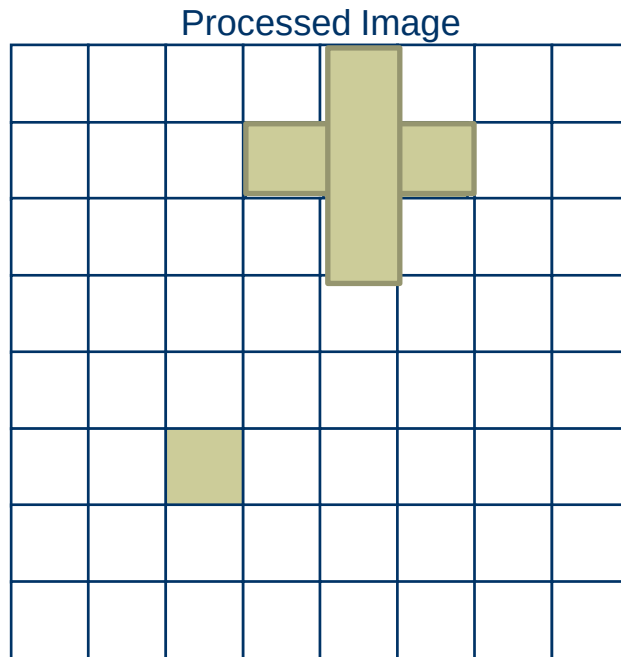


Structuring Element

Opening: Example (performing dilation to errored image)

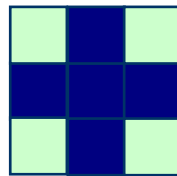
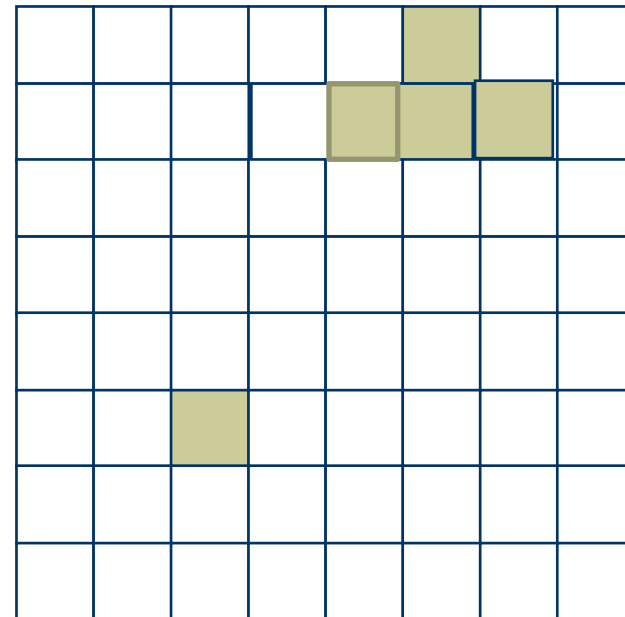
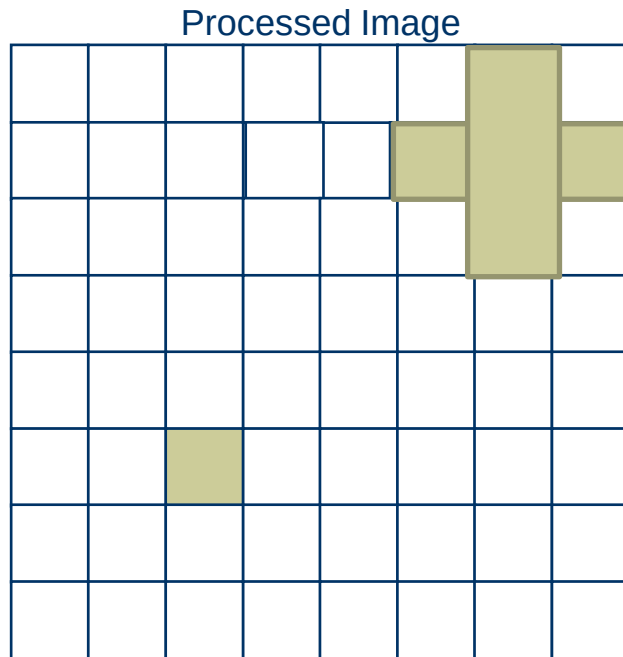


Opening: Example (performing dilation to errored image)



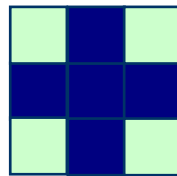
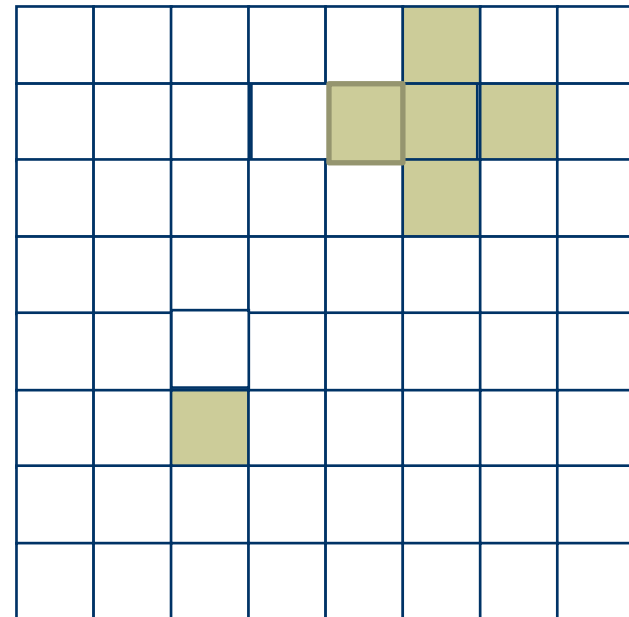
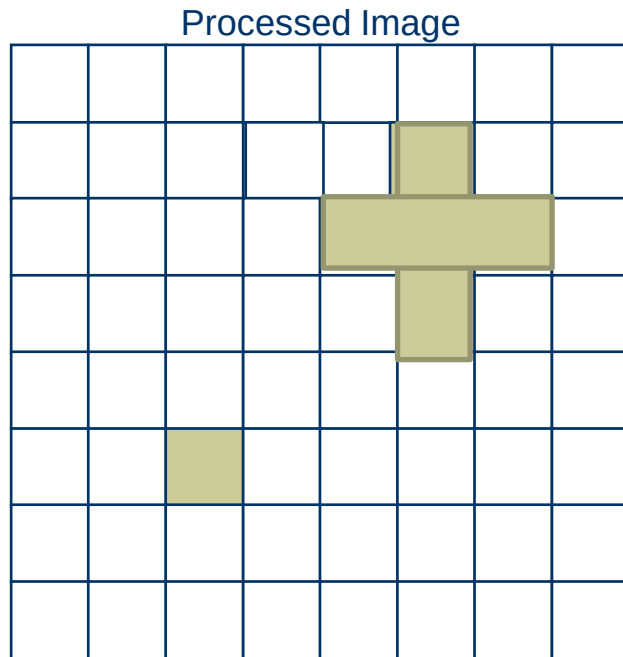
Structuring Element

Opening: Example (performing dilation to errored image)



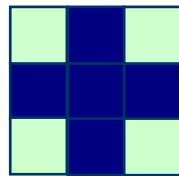
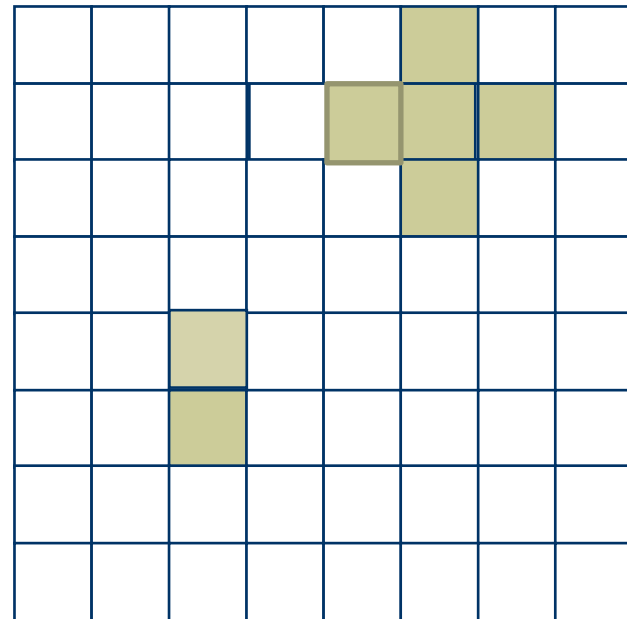
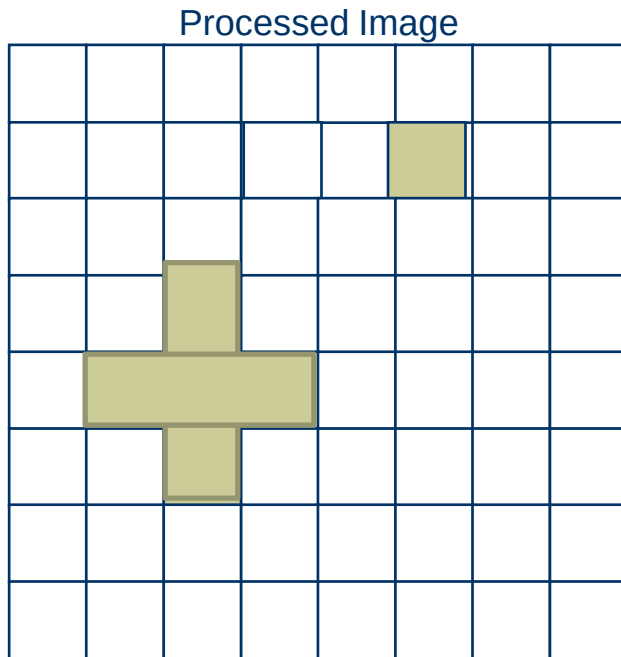
Structuring Element

Opening: Example (performing dilation to errored image)



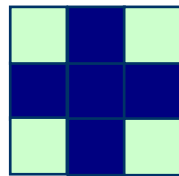
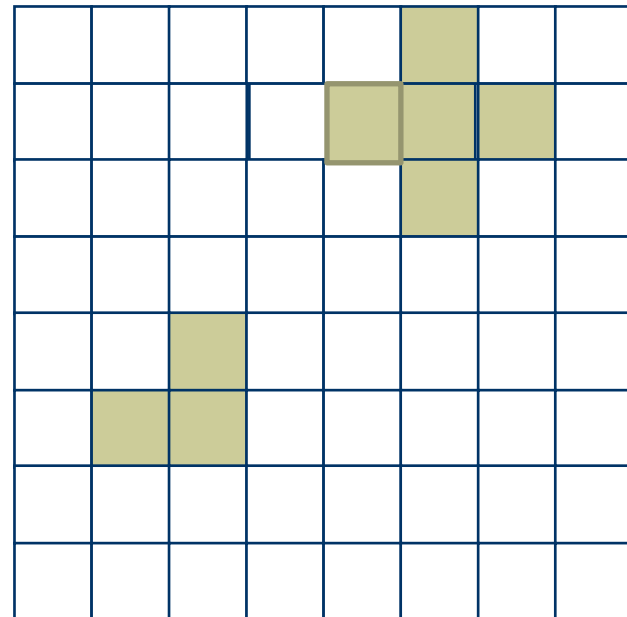
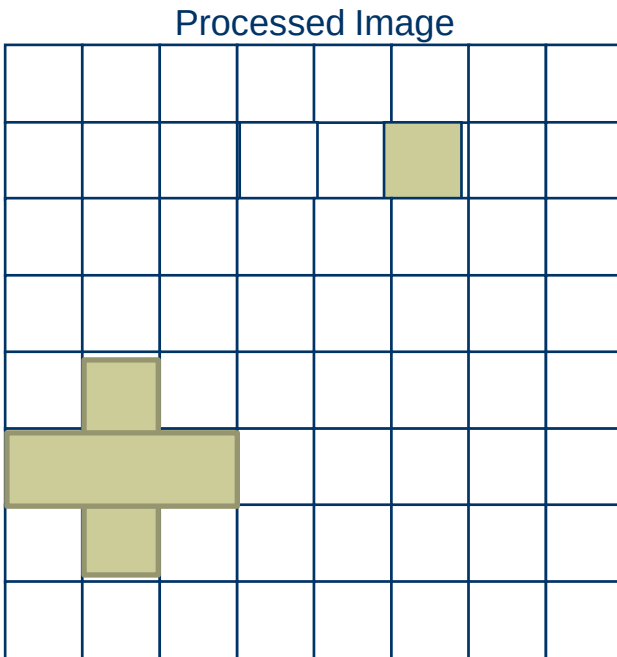
Structuring Element

Opening: Example (performing dilation to errored image)



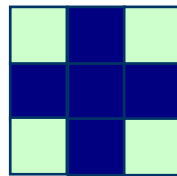
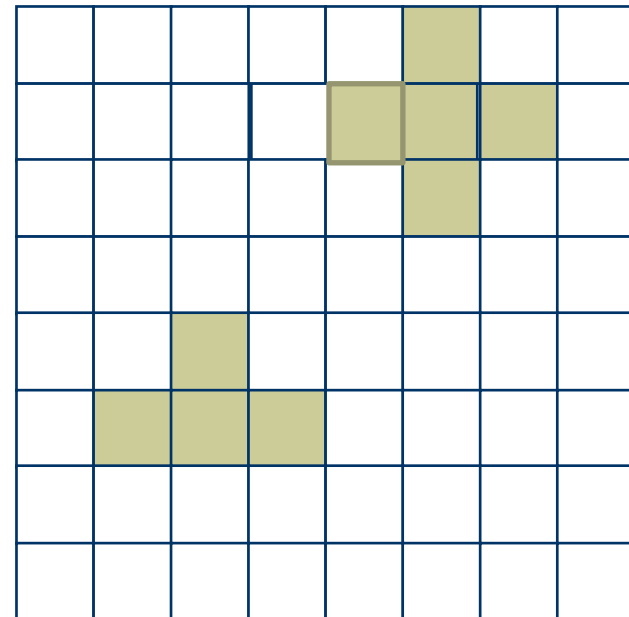
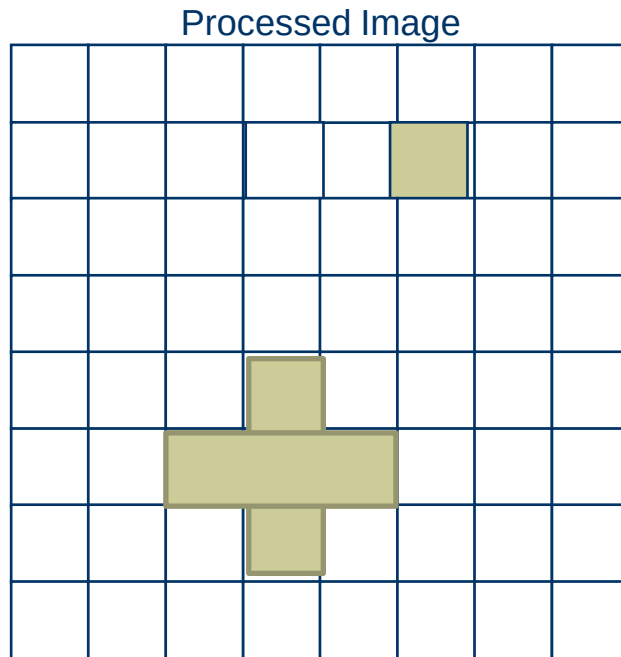
Structuring Element

Opening: Example (performing dilation to errored image)



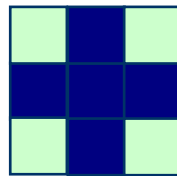
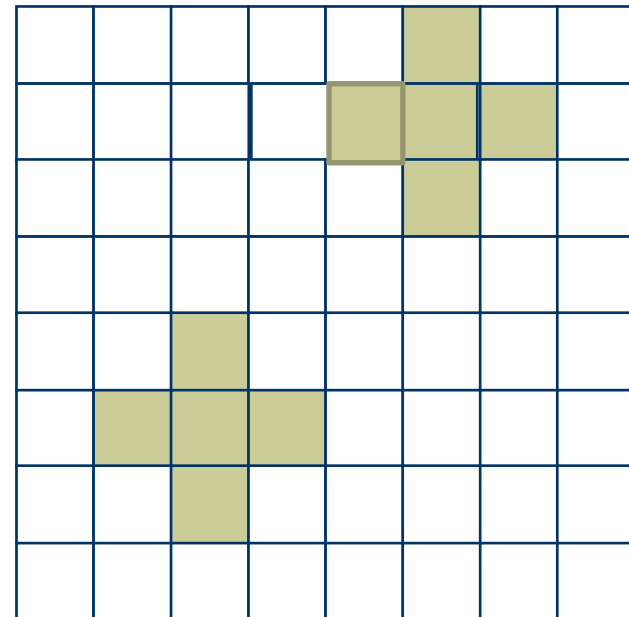
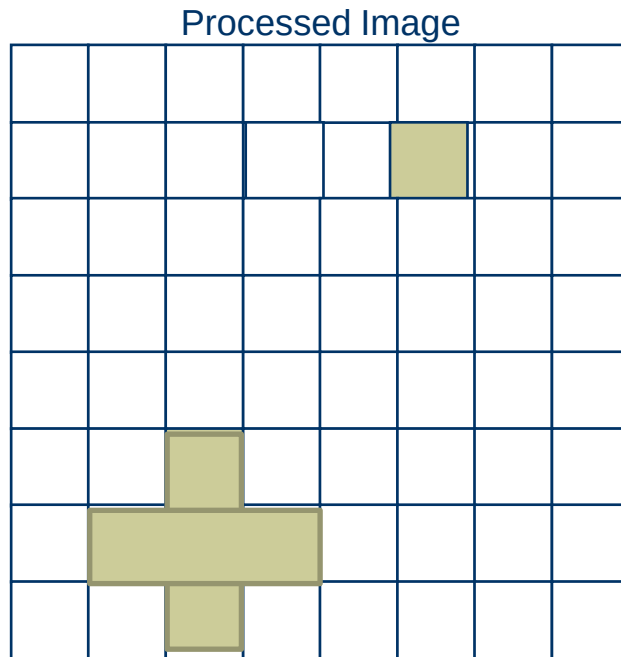
Structuring Element

Opening: Example (performing dilation to errored image)



Structuring Element

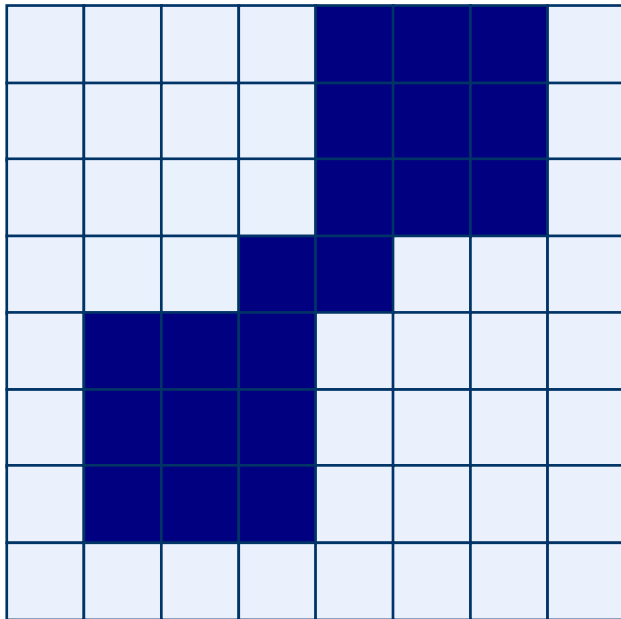
Opening: Example (performing dilation to errored image)



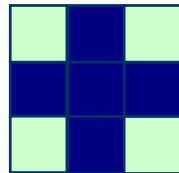
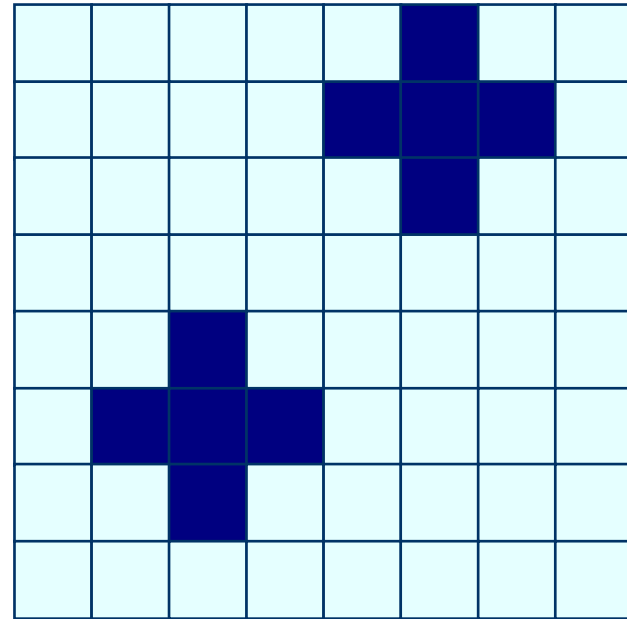
Structuring Element

After Opening:

Original Image



Processed Image

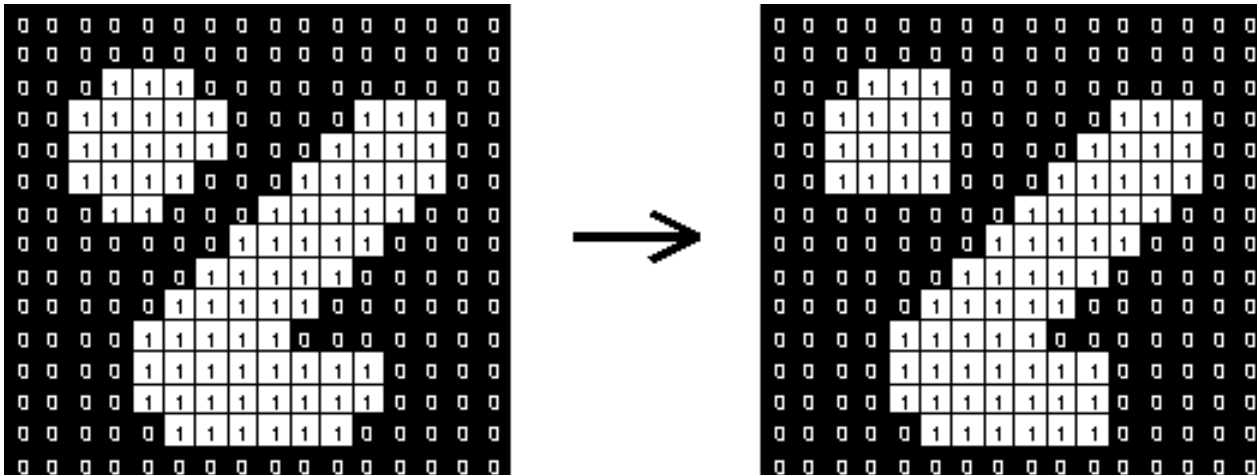


Structuring Element

Effect of opening

1	1	1
1	1	1
1	1	1

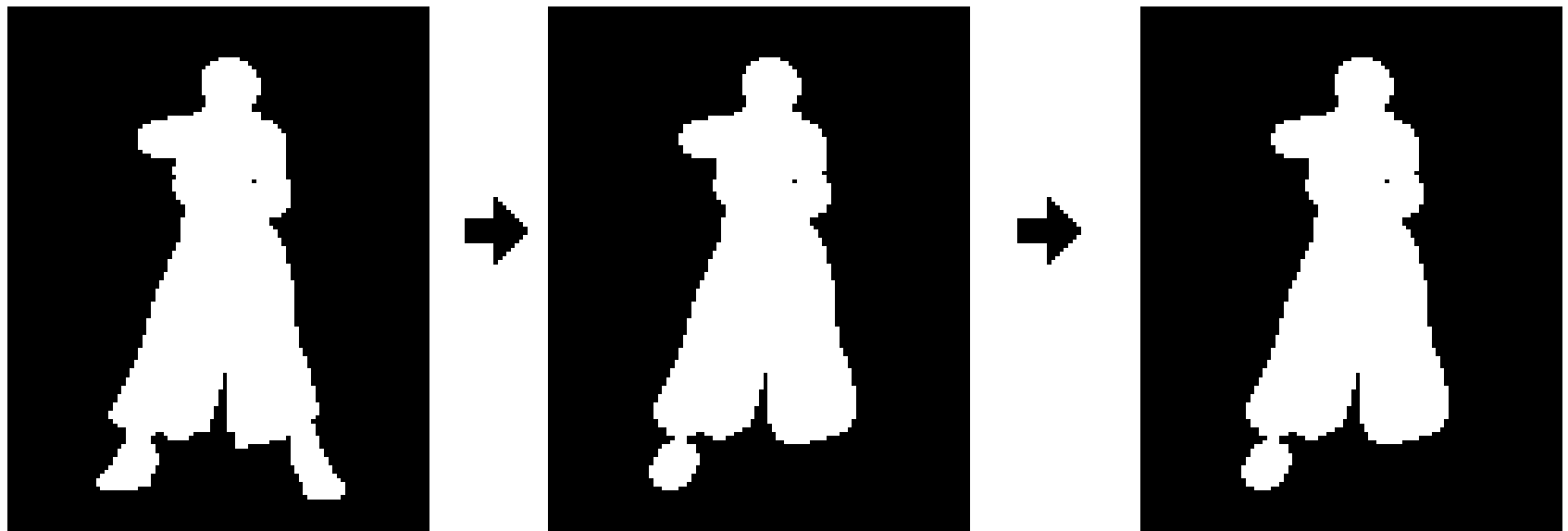
- Structuring element: 3x3 square



Effect of opening

- ❑ Smoothed the **outline**, by rounding off any **sharp points**, and
- ❑ Remove any parts that is smaller than the shape (SE) used.
- ❑ It will also disconnect or 'open' any thin bridges.
- ❑ It does not remove any 'holes', or gaps that may be present in the image.
- ❑ It does not make the basic 'core' size of the shape larger or smaller.

Note that performing an 'Open' on a shape that has already been opened, with the same kernel will result in no further change to the shape. For example...



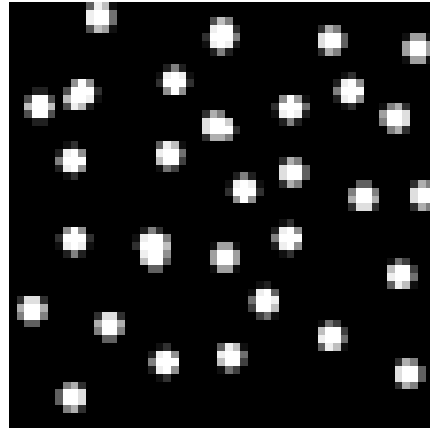
That is repeating a 'Open' operation, with the same kernel, has no effect on the result.

Effect of opening

Original Image

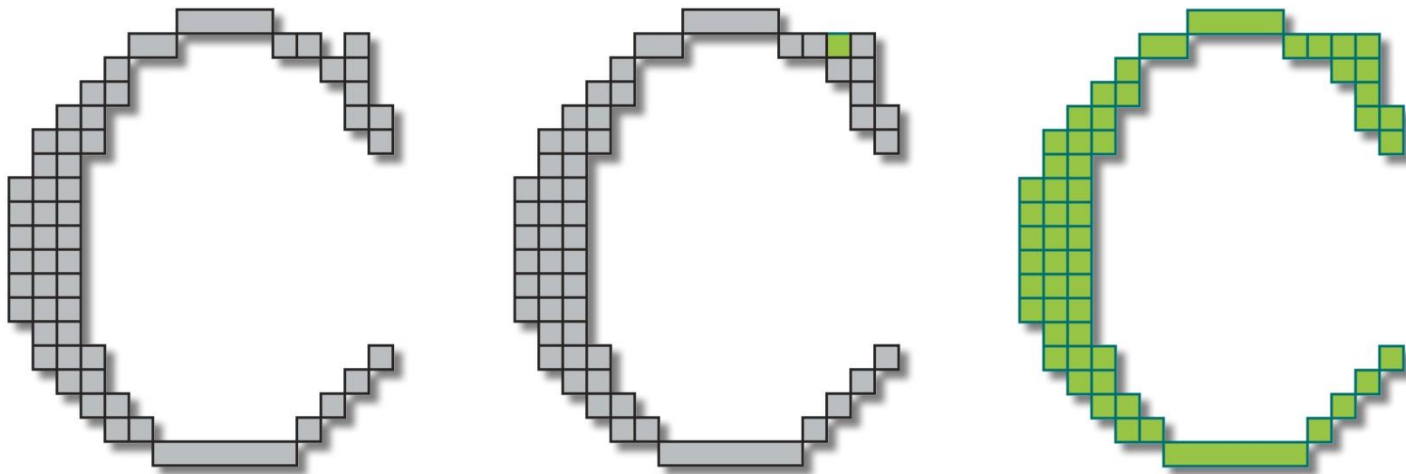


Processed Image



- ❑ Opening with a disk shaped structuring element 11 pixels in diameter gives the separation of circles from the lines.
- ❑ Some of the circles are slightly distorted, but
- ❑ In general, the lines have been almost completely removed while the circles remain almost completely unaffected.

Closing

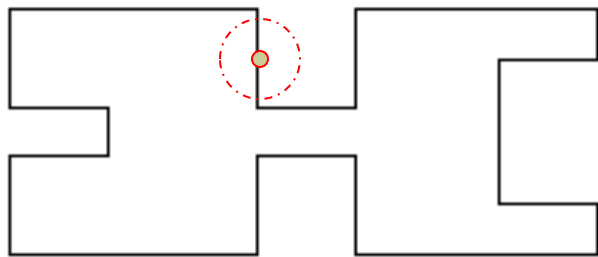


Dilation followed by an erosion

Closing

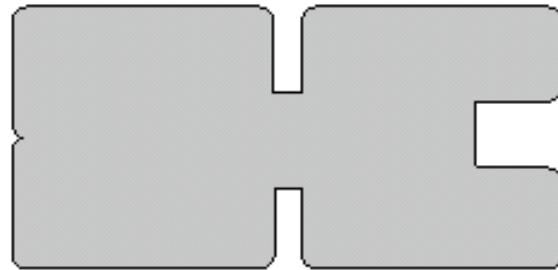
The closing of image f by structuring element s , denoted by $A \bullet B$ is simply a dilation followed by an erosion

$$A \bullet B = (A \oplus B) \ominus B$$



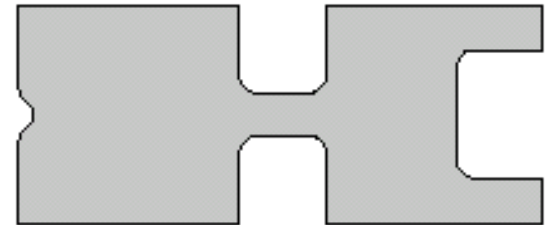
A

Original shape



$A \oplus B$

After dilation

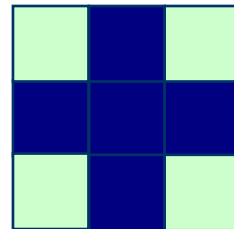
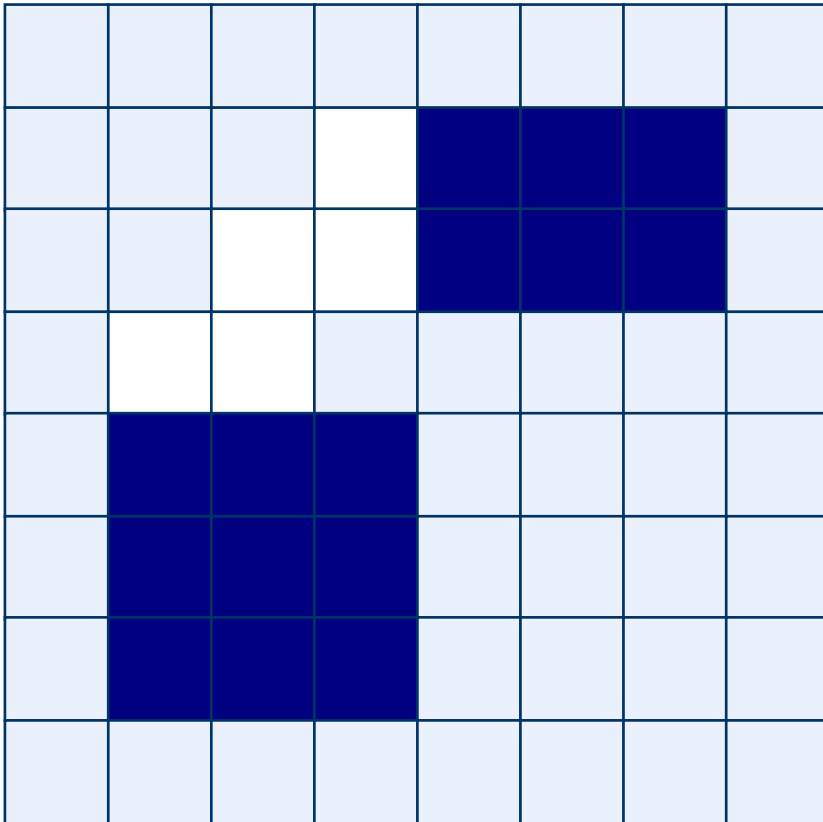


$A \bullet B = (A \oplus B) \ominus B$

After erosion
(closing)

Closing: Example

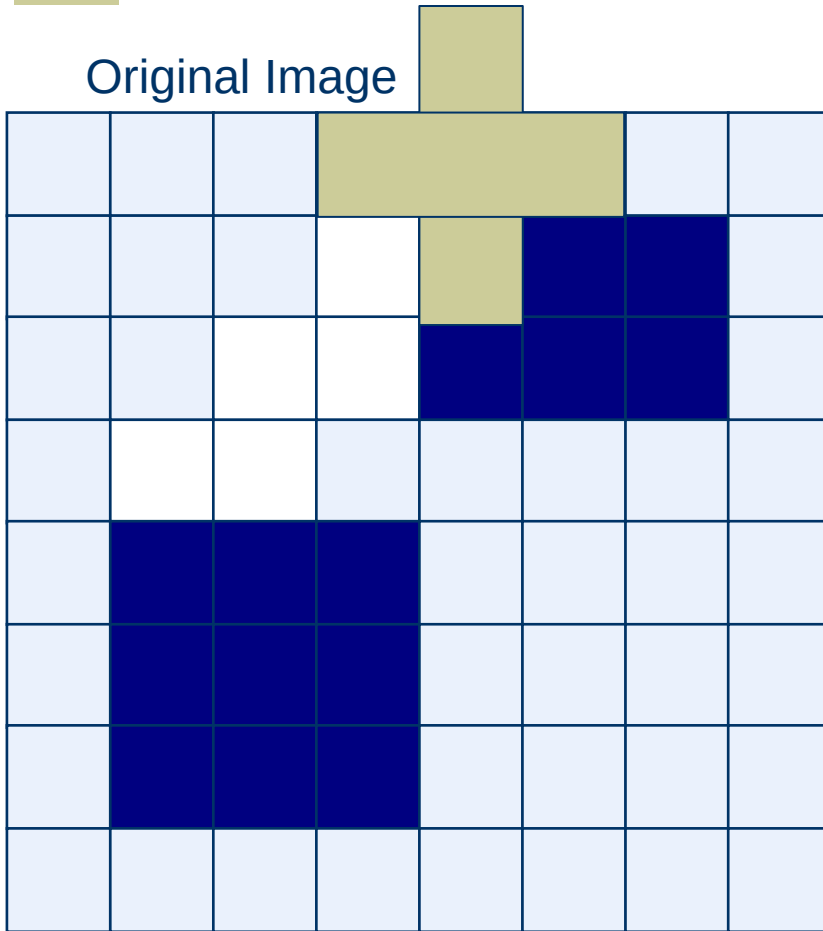
Original Image



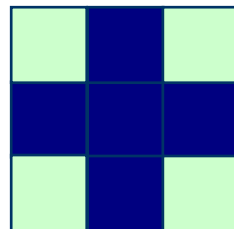
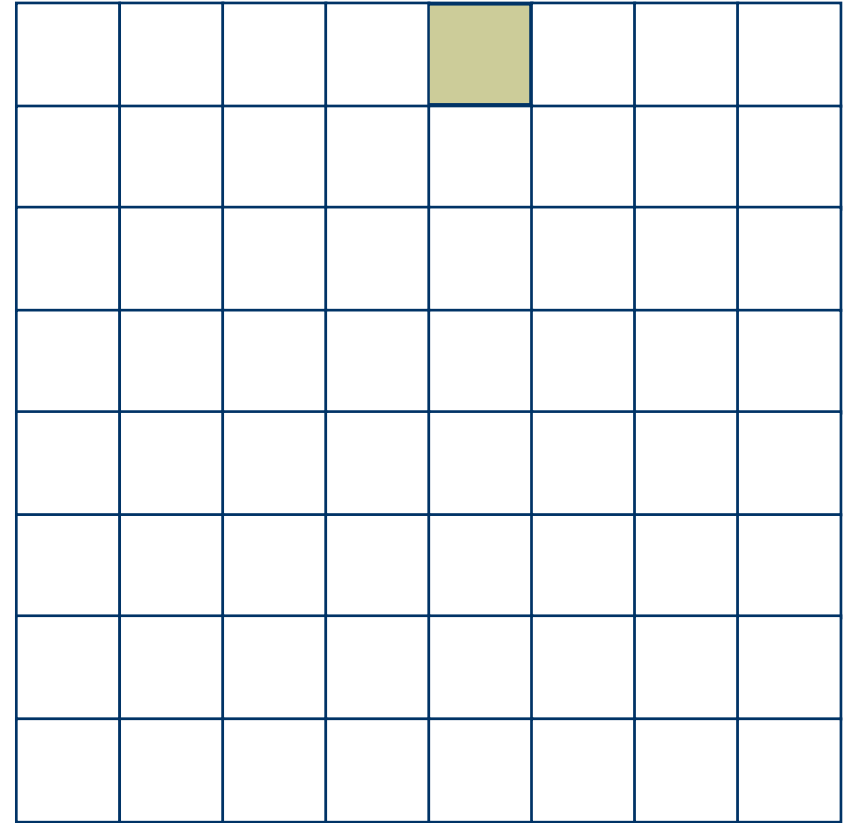
Structuring Element

Closing: Example (performing Dilation)

Original Image



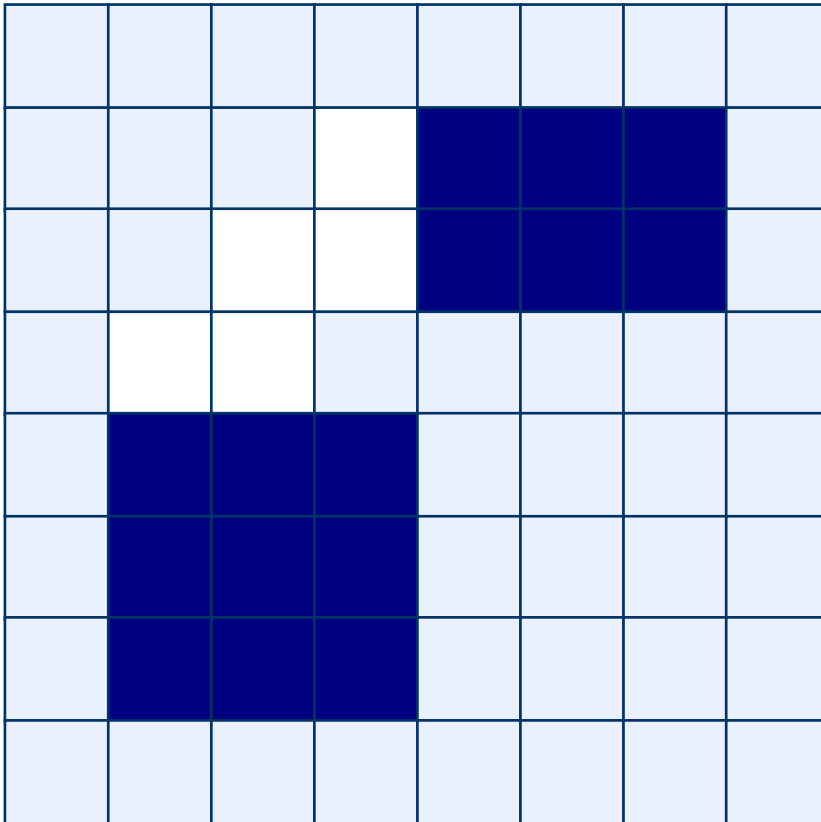
Processed Image



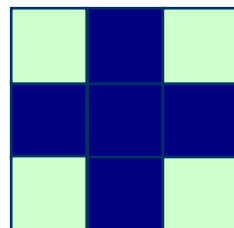
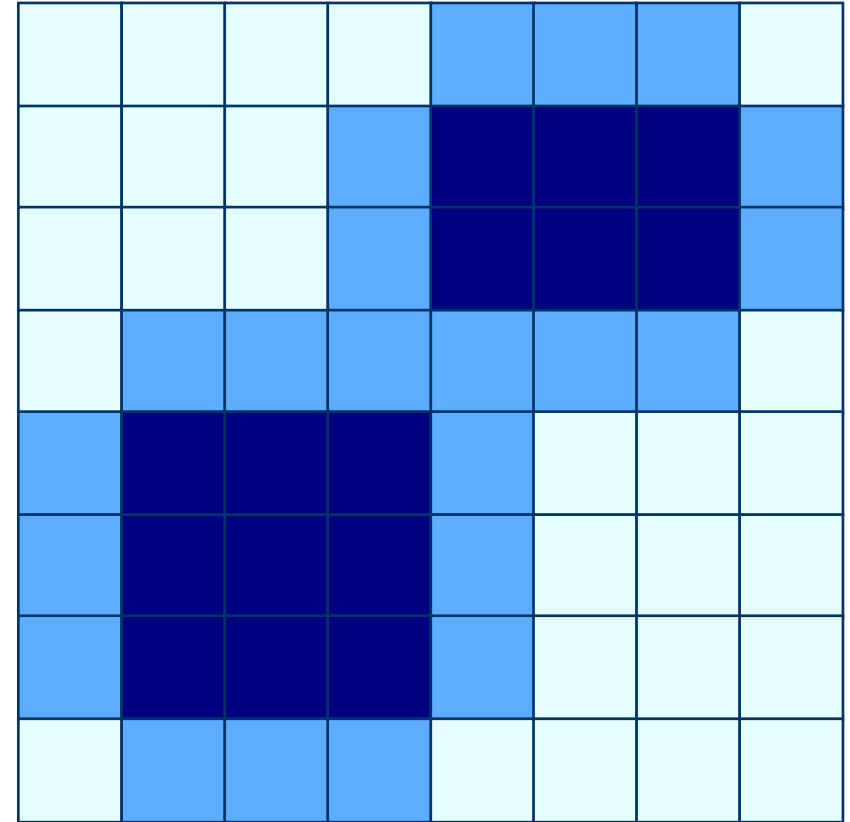
Structuring Element

Closing: Example (after Dilation)

Original Image



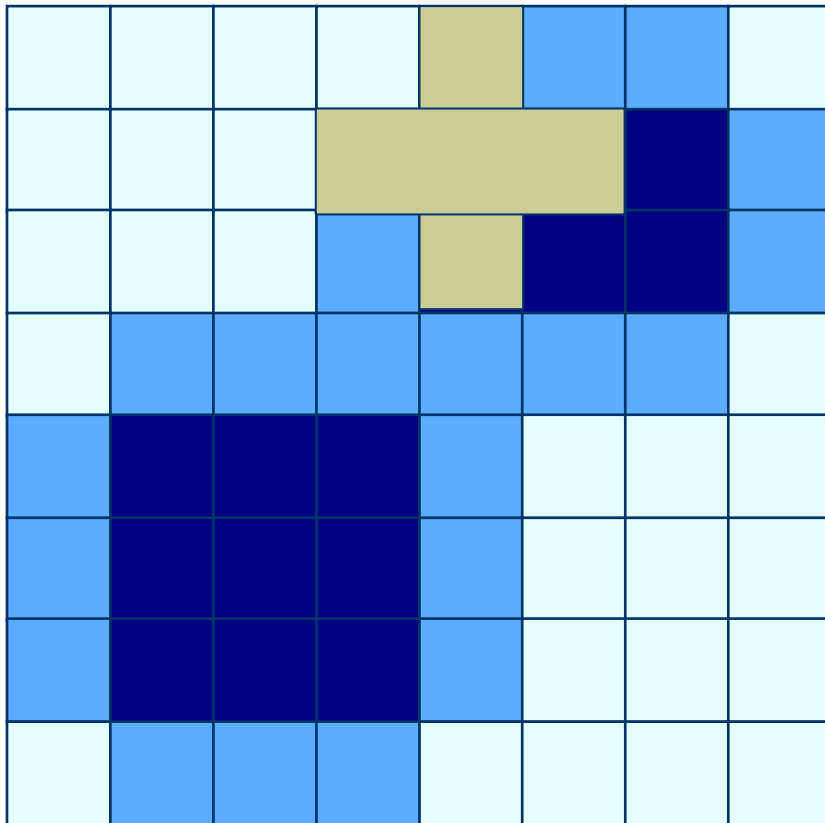
Processed Image



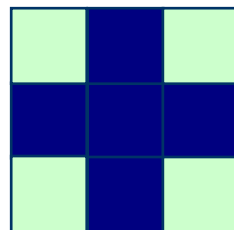
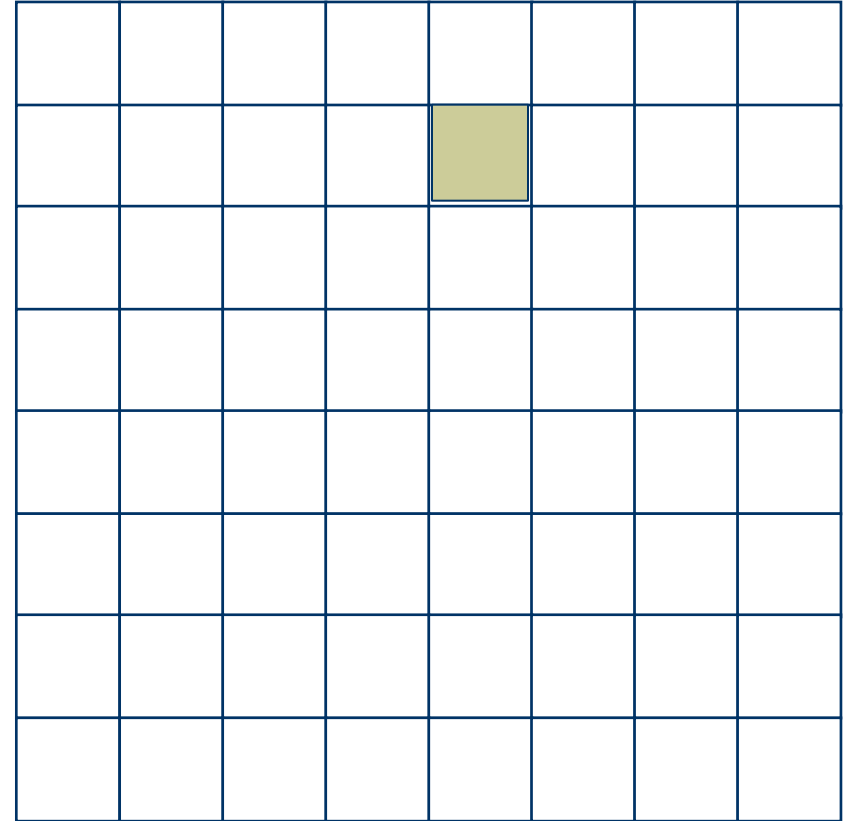
Structuring Element

Closing: Example(performing erosion on Dilated image)

Original Image



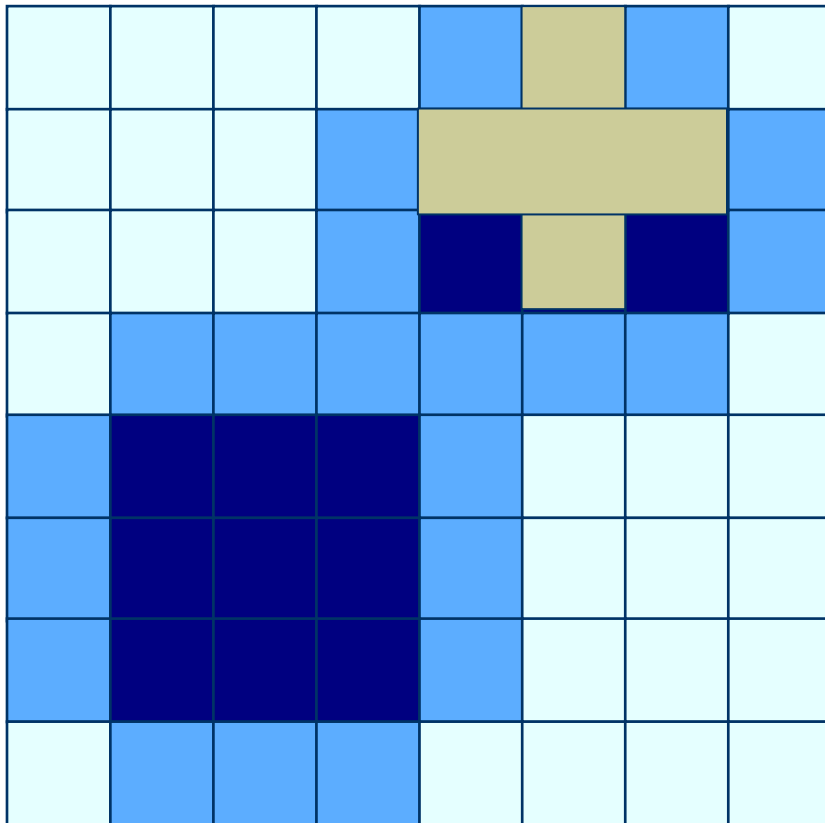
Processed Image



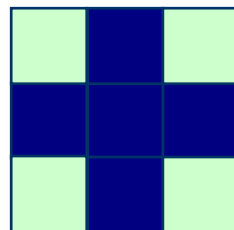
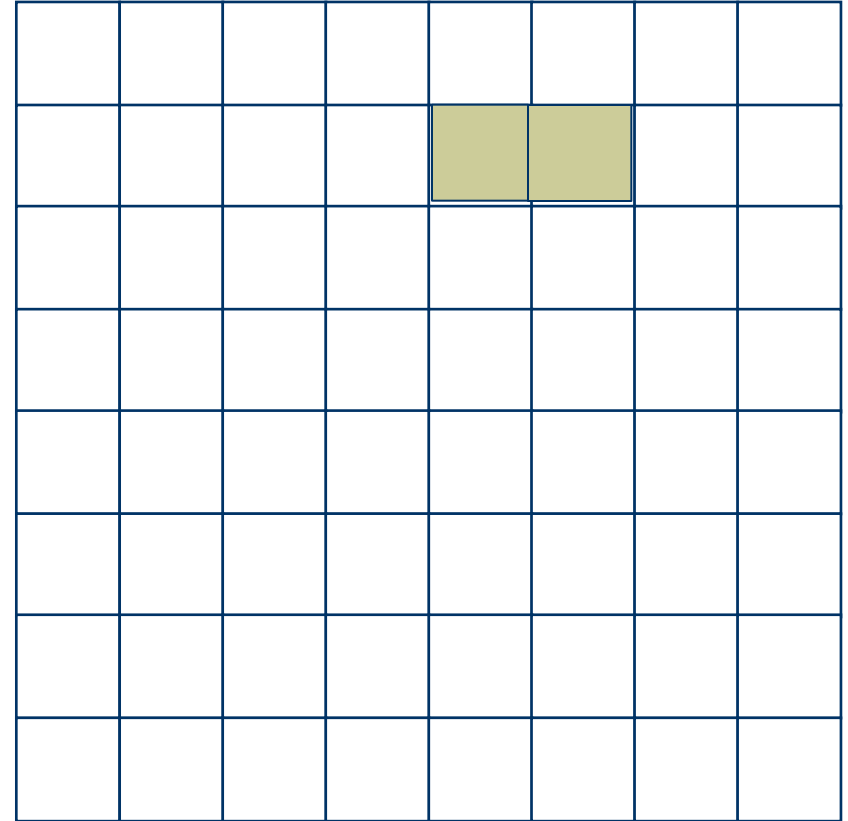
Structuring Element

Closing: Example(performing erosion on Dilated image)

Original Image



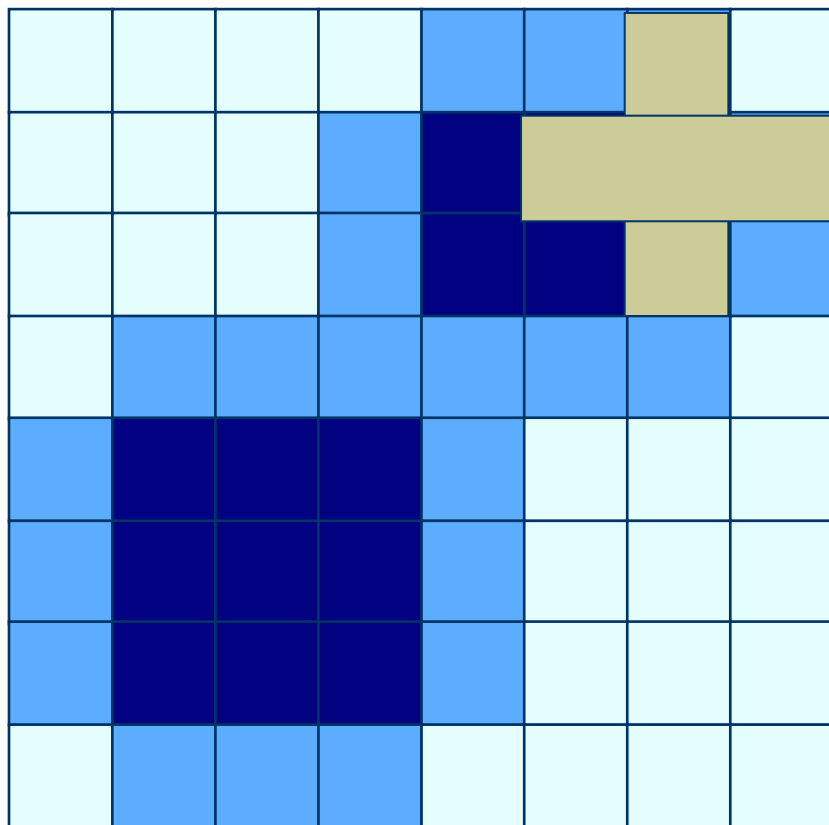
Processed Image



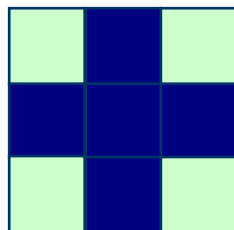
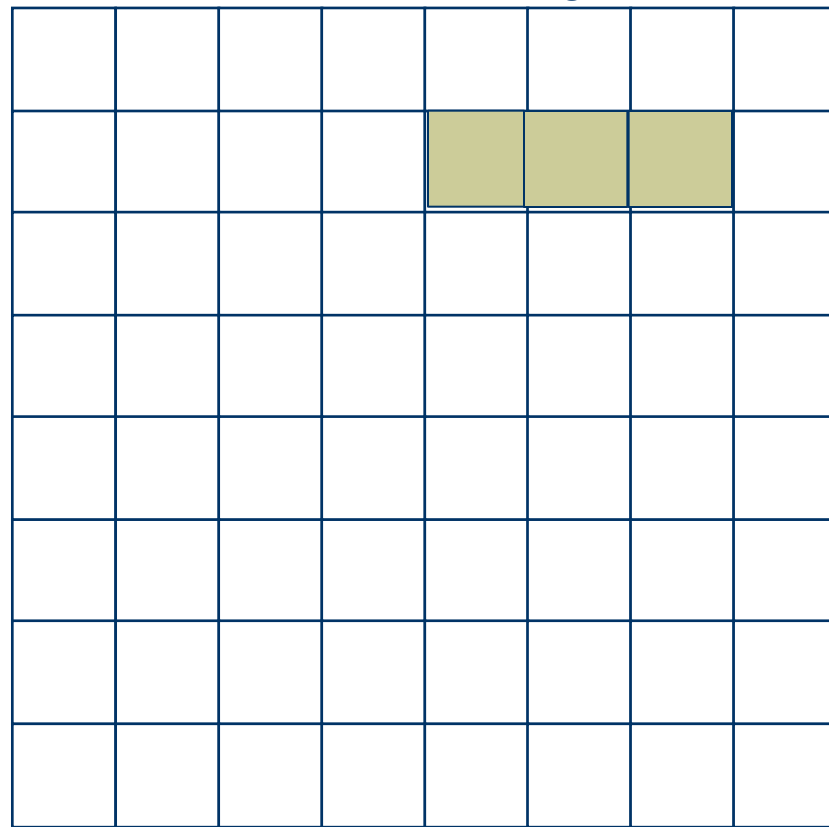
Structuring Element

Closing: Example(performing erosion on Dilated image)

Original Image



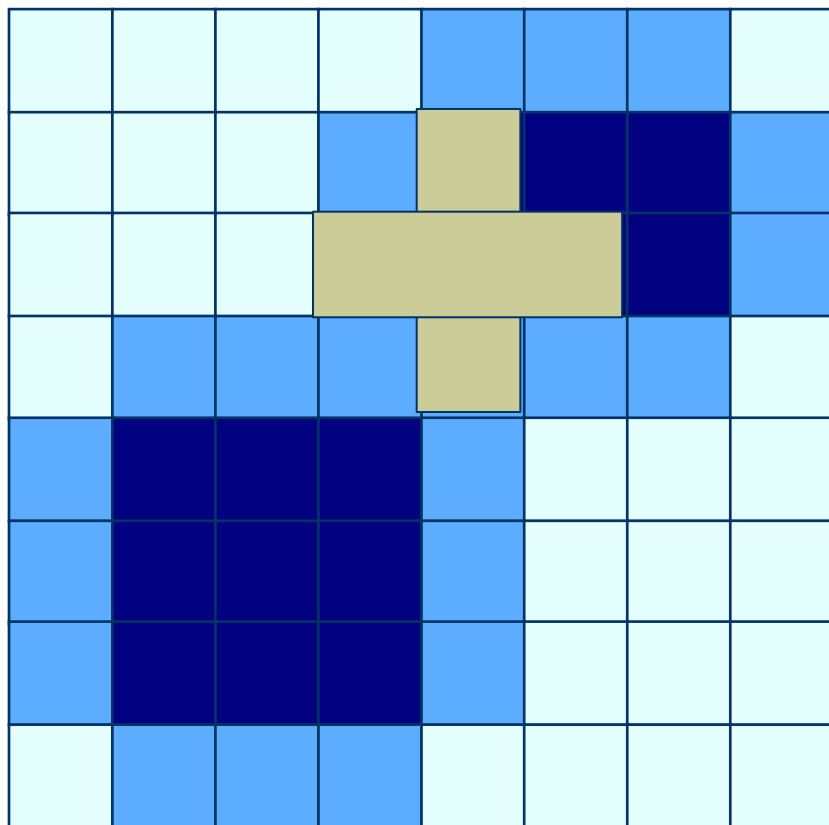
Processed Image



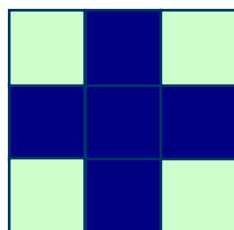
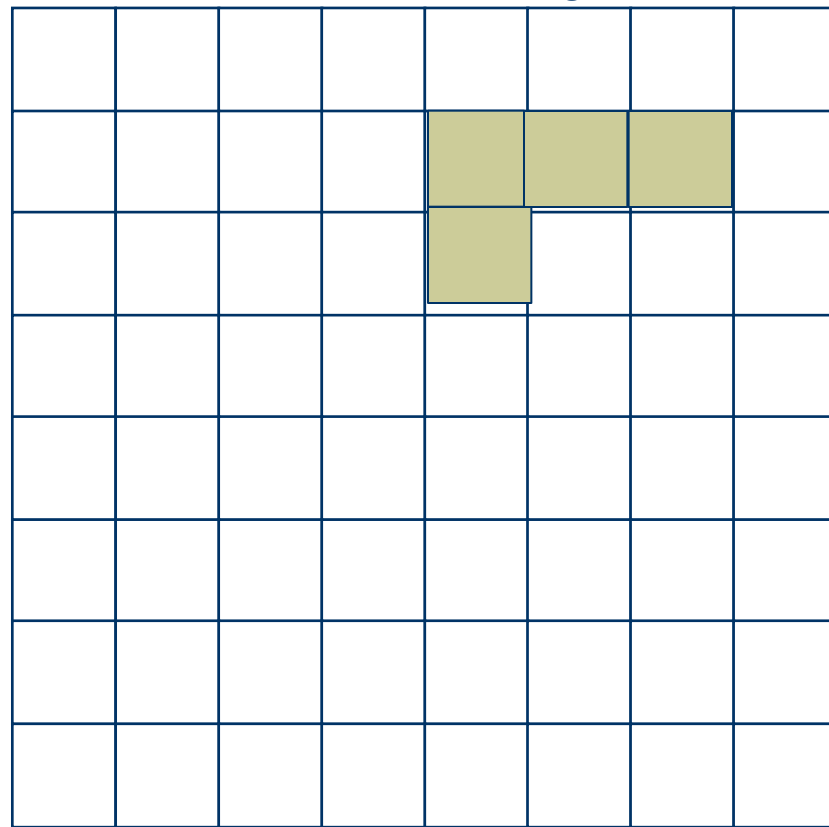
Structuring Element

Closing: Example(performing erosion on Dilated image)

Original Image



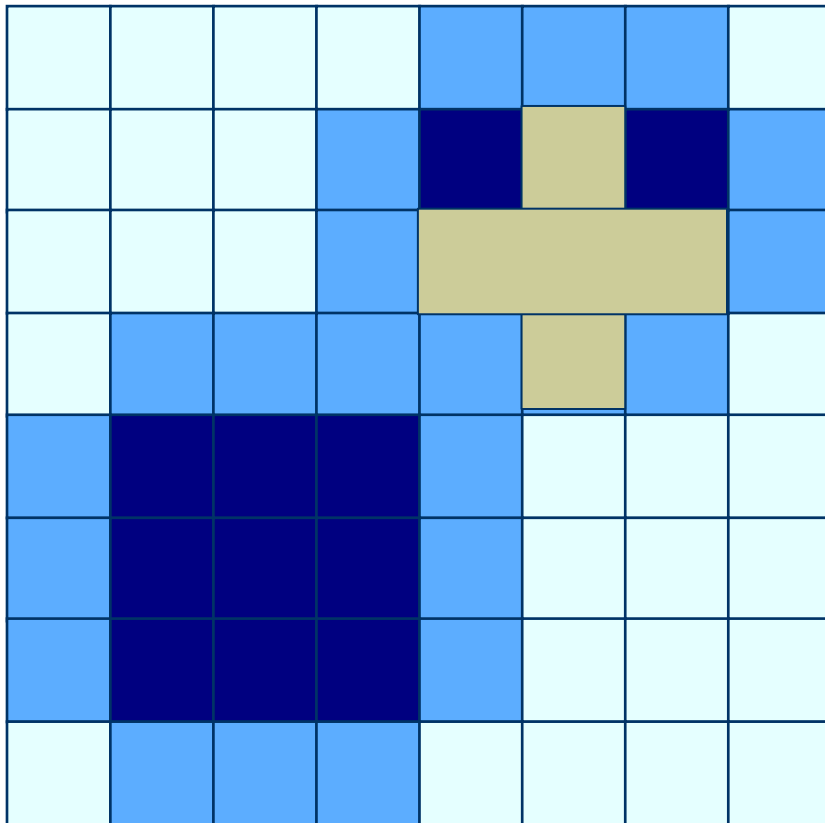
Processed Image



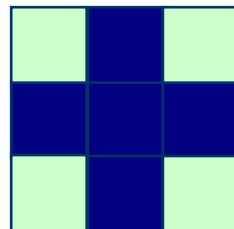
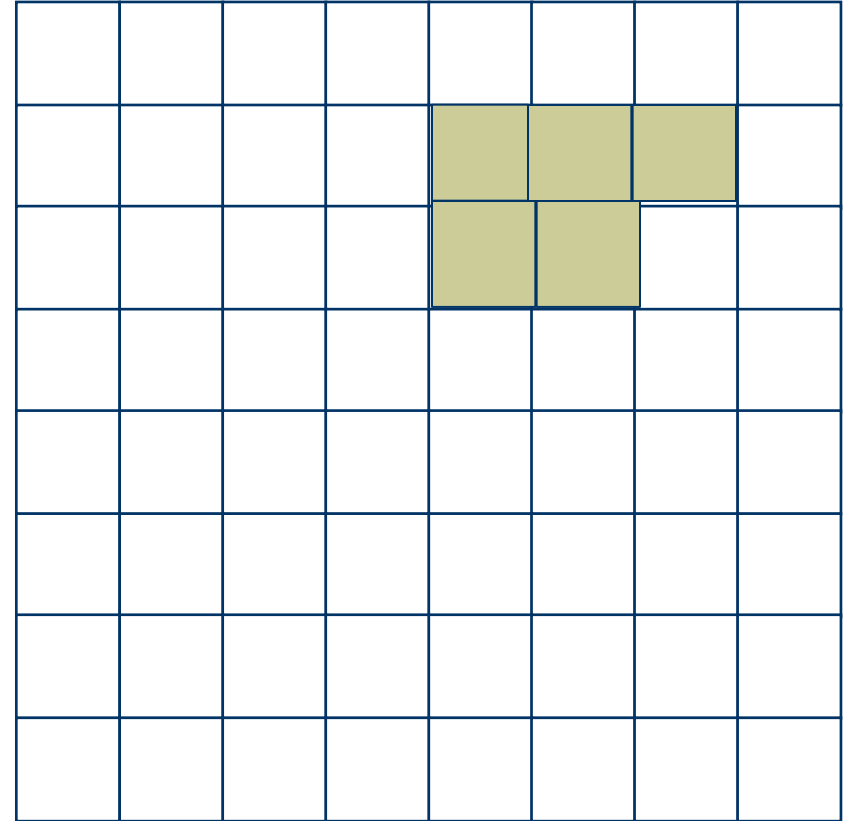
Structuring Element

Closing: Example(performing erosion on Dilated image)

Original Image



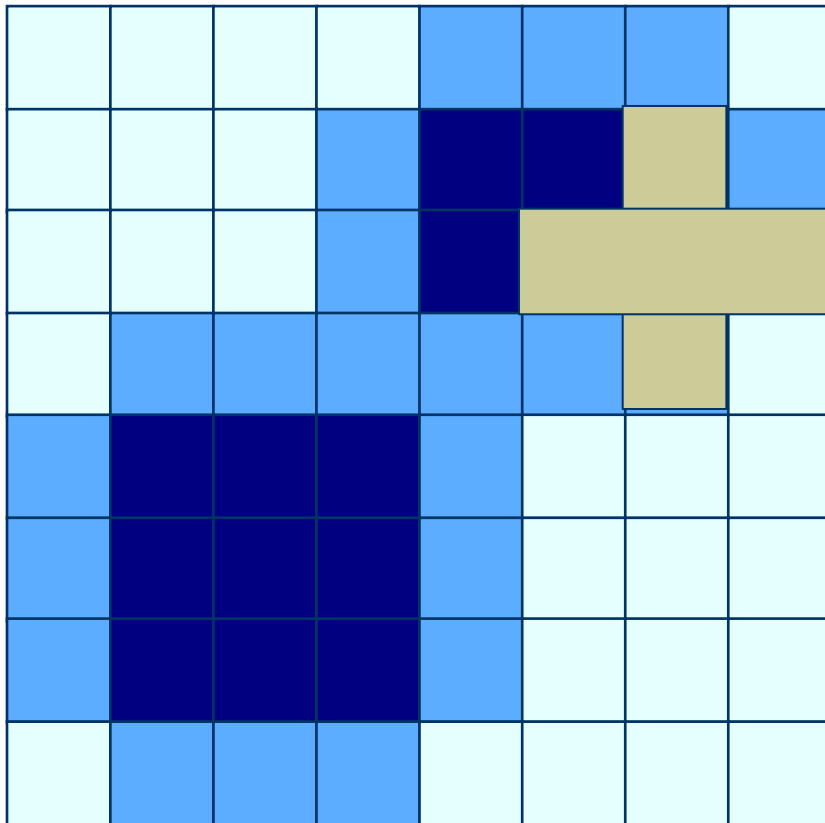
Processed Image



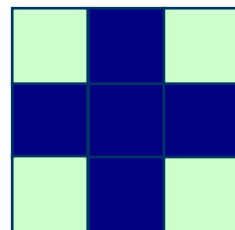
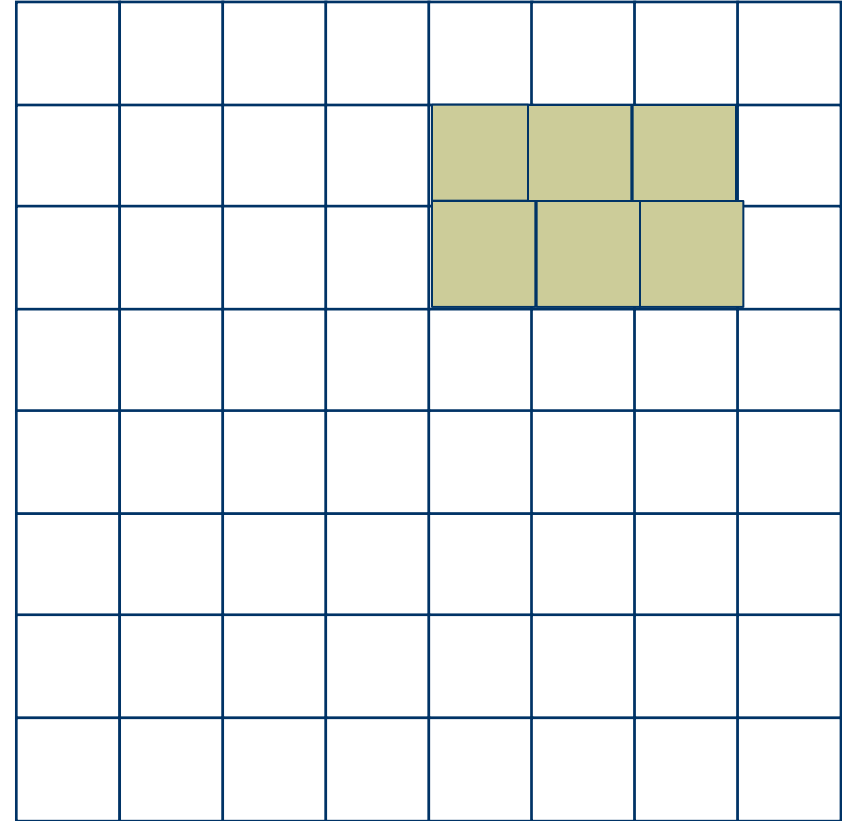
Structuring Element

Closing: Example(performing erosion on Dilated image)

Original Image



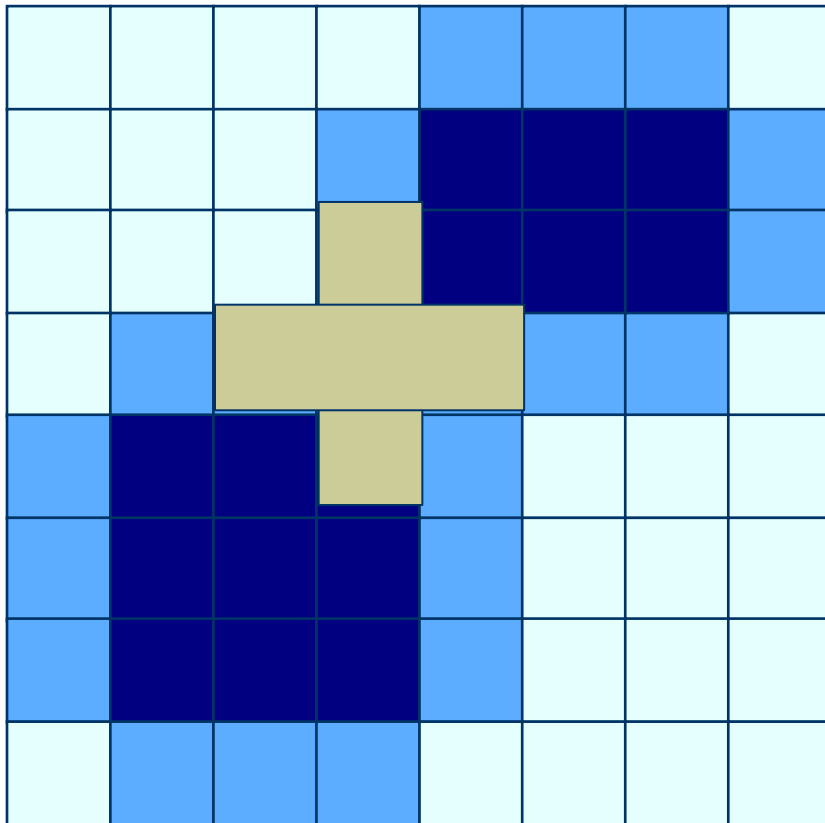
Processed Image



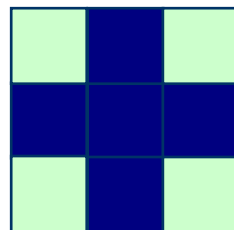
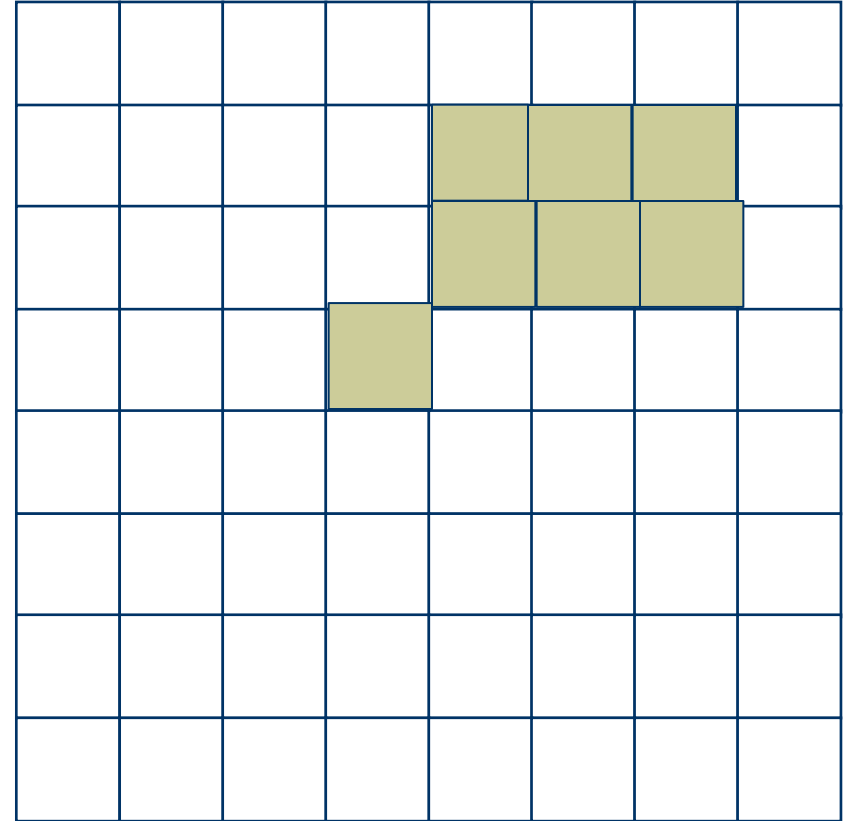
Structuring Element

Closing: Example(performing erosion on Dilated image)

Original Image



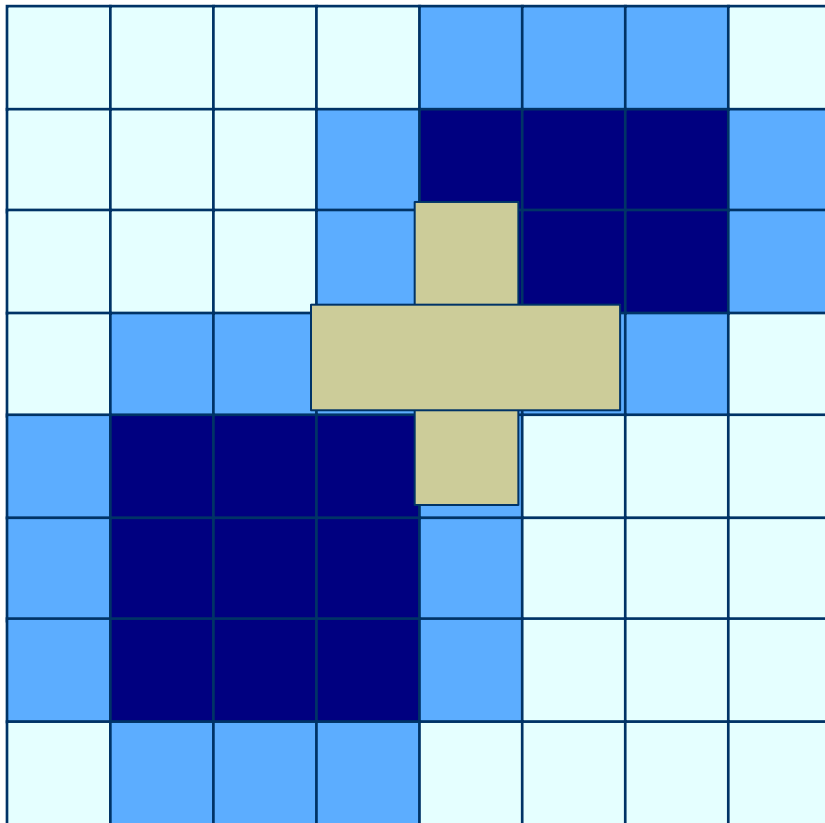
Processed Image



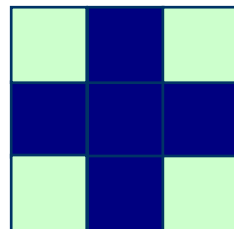
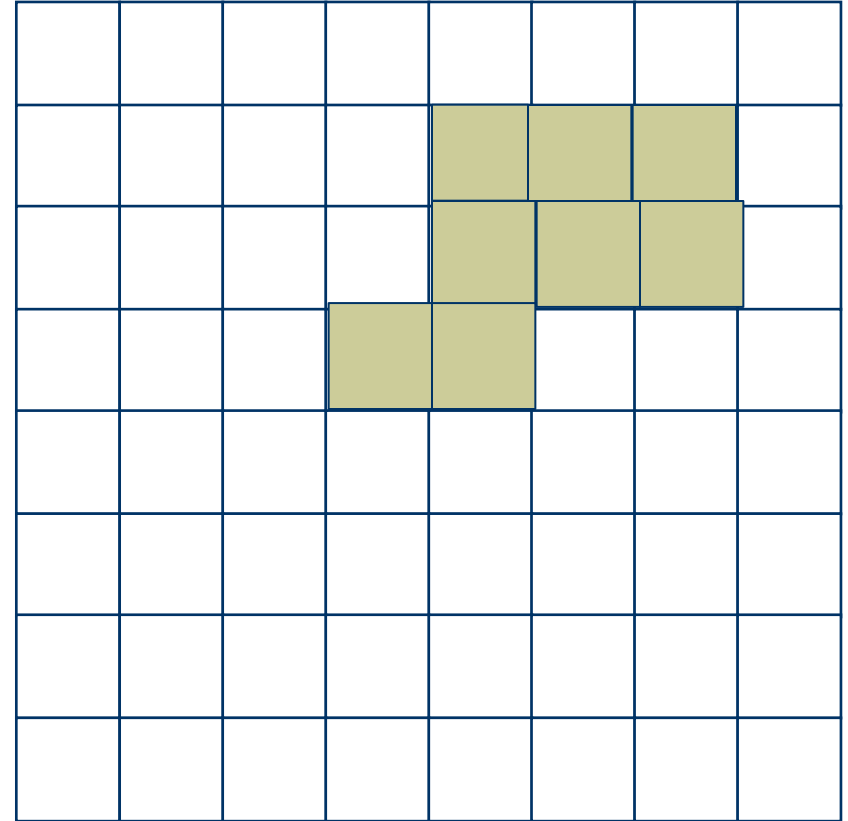
Structuring Element

Closing: Example(performing erosion on Dilated image)

Original Image



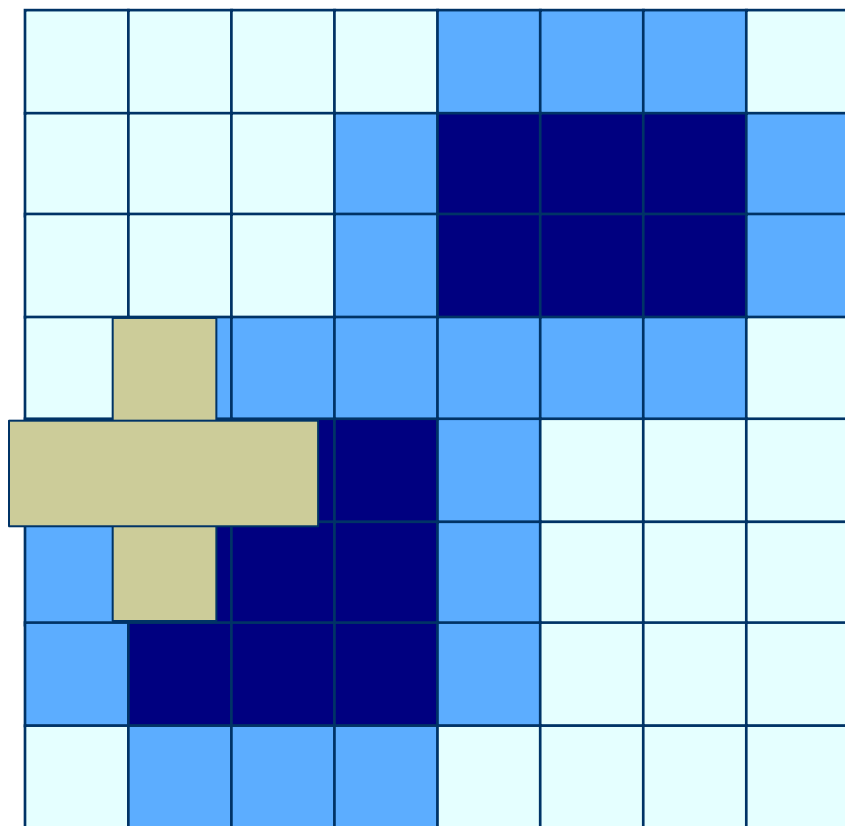
Processed Image



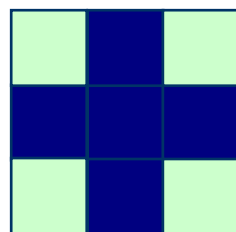
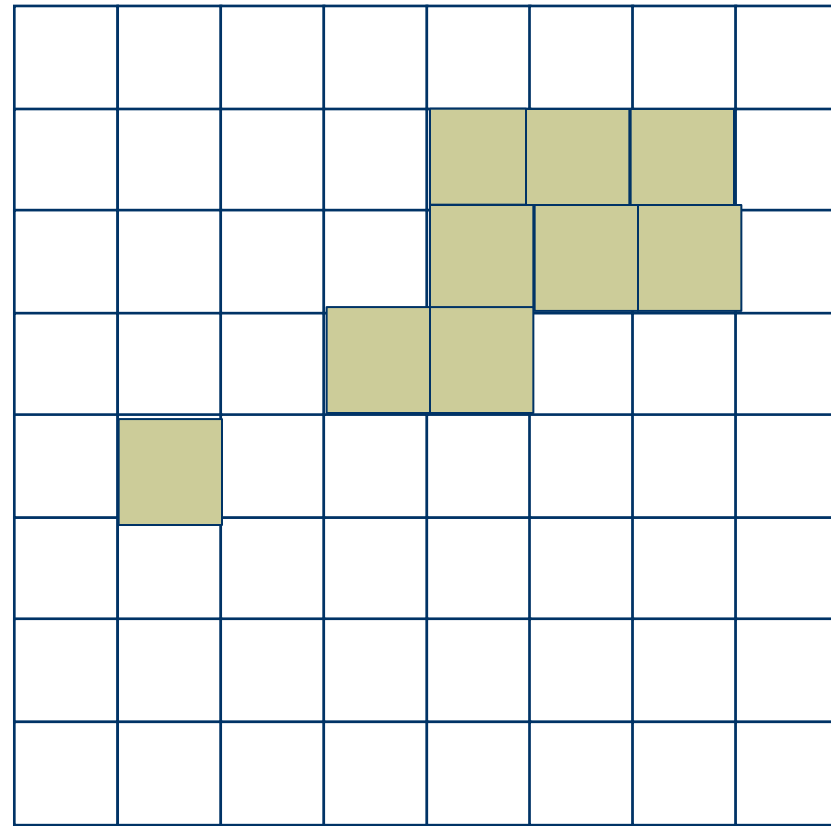
Structuring Element

Closing: Example(performing erosion on Dilated image)

Original Image



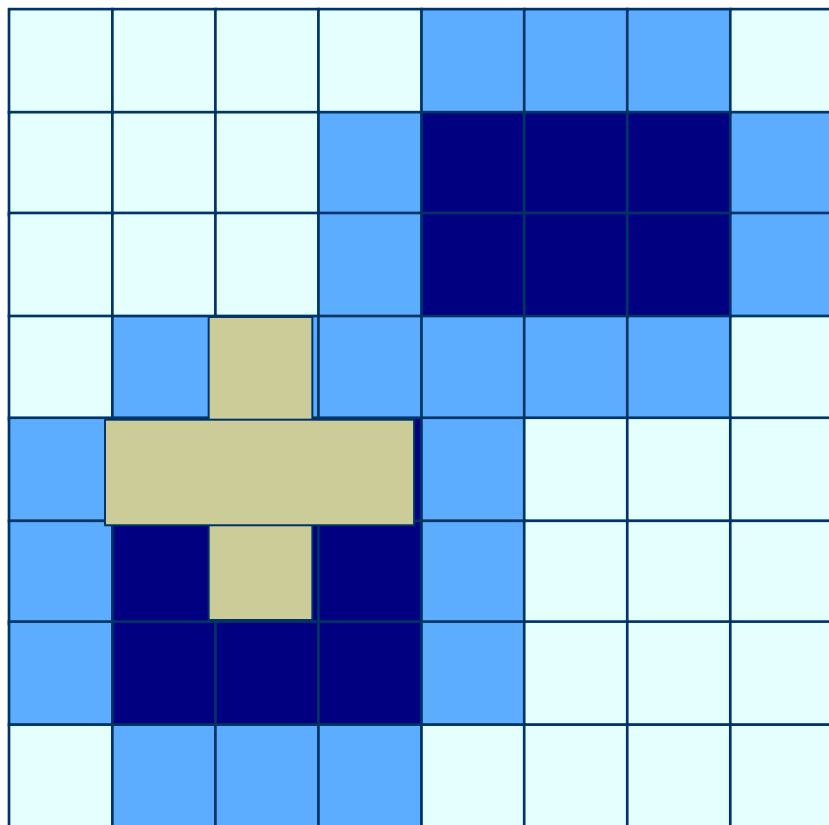
Processed Image



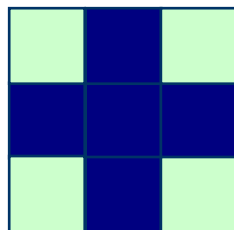
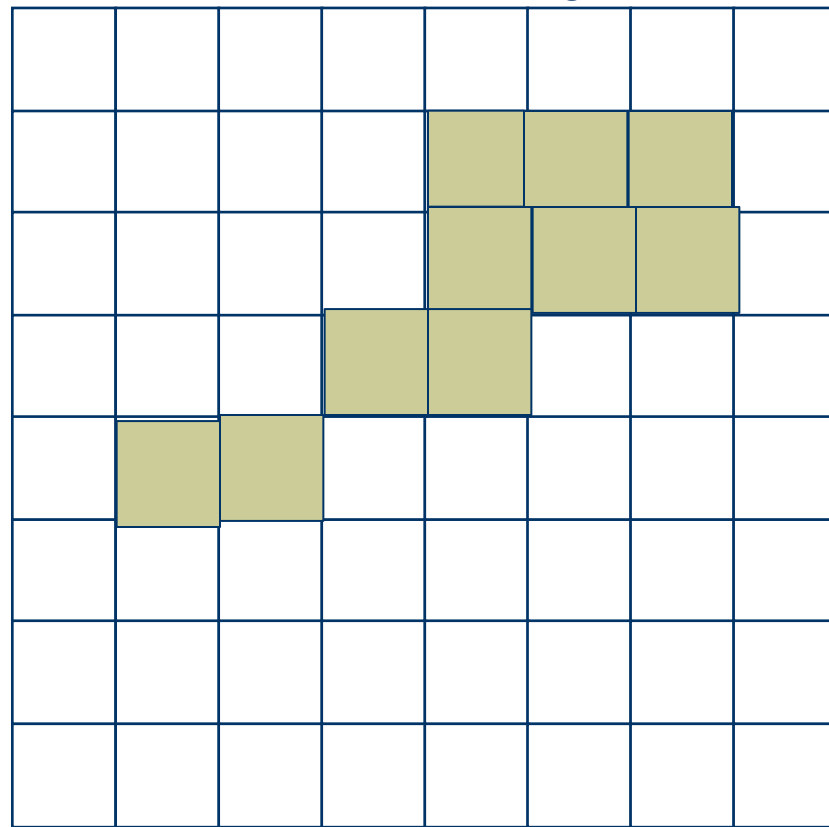
Structuring Element

Closing: Example(performing erosion on Dilated image)

Original Image



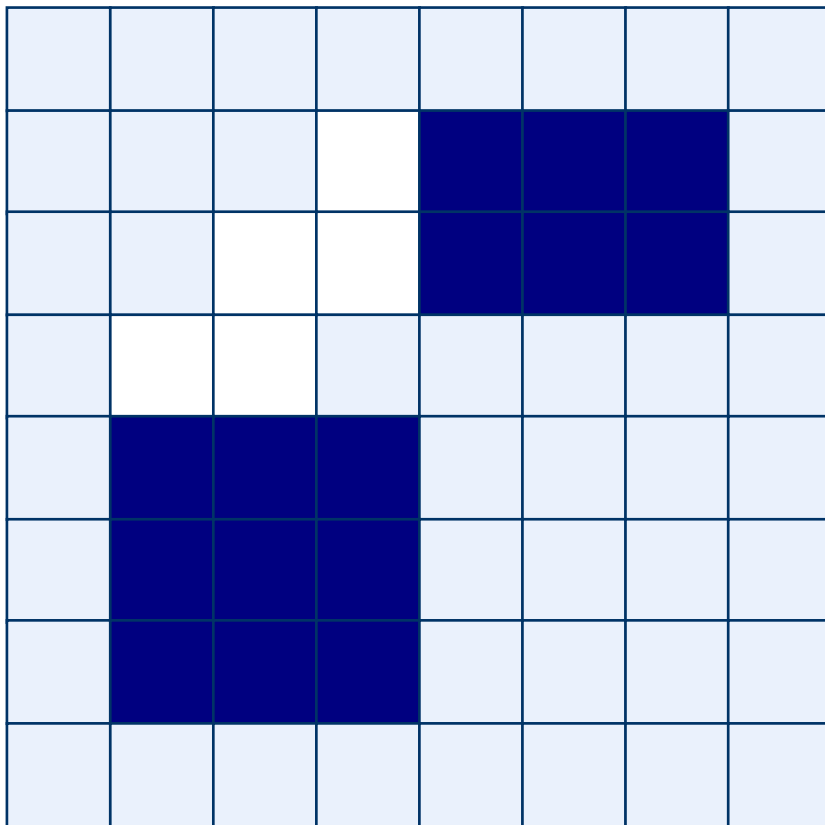
Processed Image



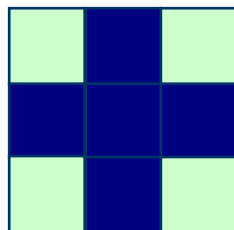
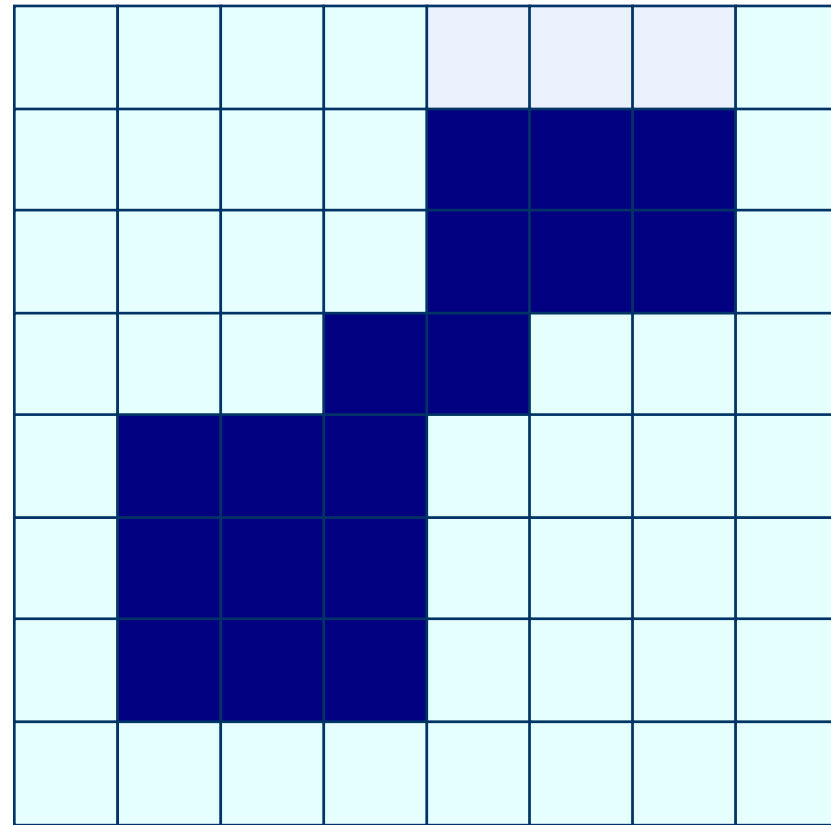
Structuring Element

Closing: Output

Original Image



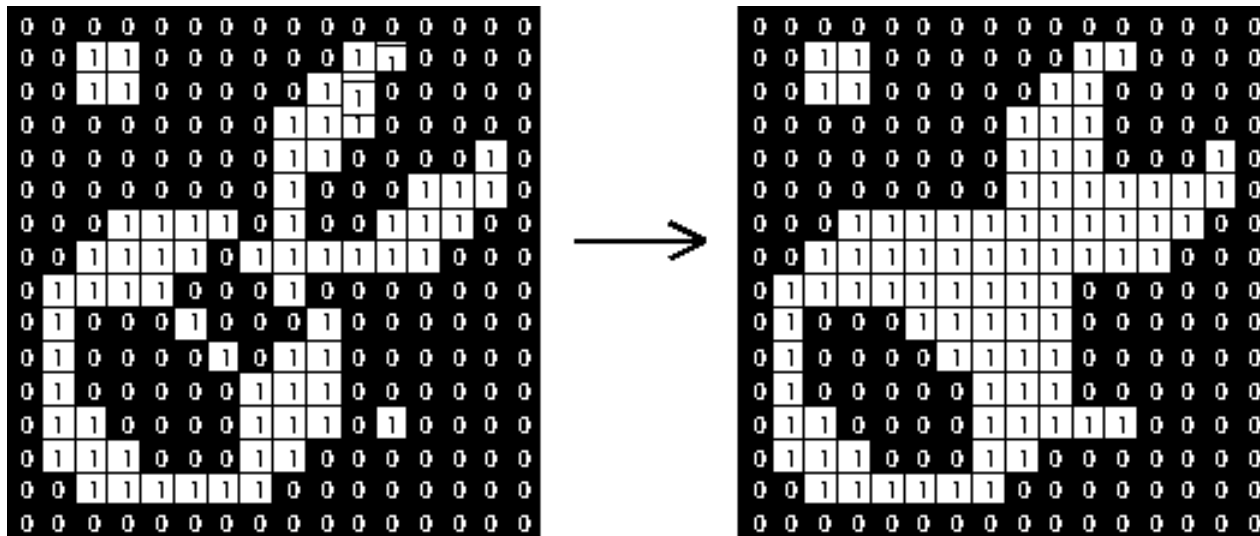
Processed Image



Structuring Element

Closing

- Structuring element: 3x3 square



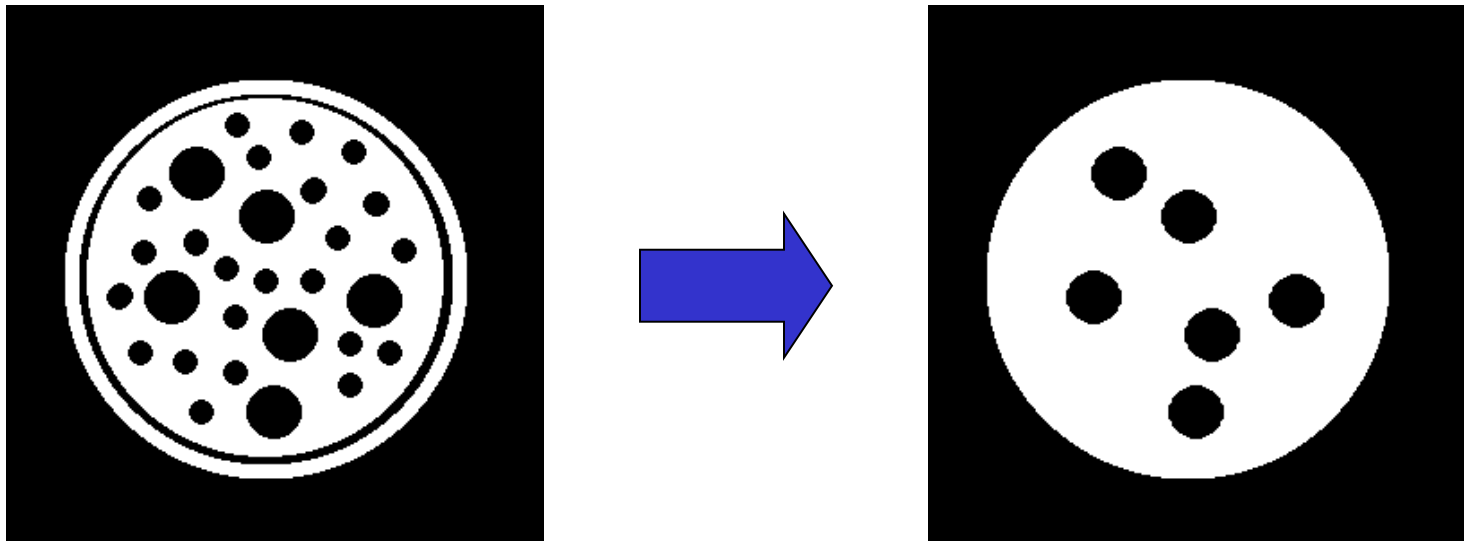
Effect of closing

- ❑ Smoothed the **outline**, by filling in (closing) any holes and indentations.
- ❑ It also will form connecting 'bridges' to other shapes that are close enough for the kernel (SE) to touch both simultaneously.
- ❑ But it does not make the basic 'core' size of the shape larger or smaller.

As with **Open**', repeating the '**Close**' method with the same kernel does not make any further changes to the image.

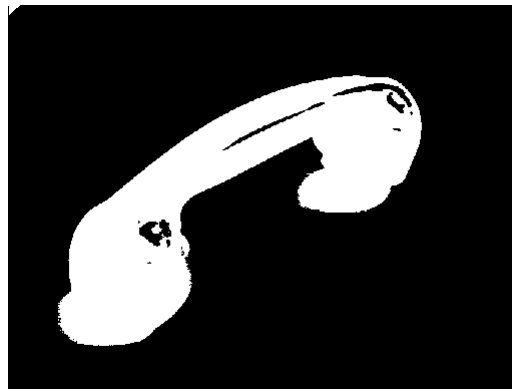
Closing Example

- Closing operation with a 22 pixel disc
- Closes small holes in the foreground



Closing Example 1

1. Threshold
2. Closing with disc of size 20



Thresholded closed

Opening & Closing

- Opening is the *dual* of closing
- *i.e.* opening the foreground pixels with a particular structuring element
- is equivalent to closing the background pixels with the same element.

***Home Work: Make an example with simulations.

Next class

Outlook:

**Hit-and-miss Transformation,
Thinning, and
Thickening**

Basic Set Theory

- ◆ The set space of binary image is Z^2
 - Each element of the set is a 2D vector whose coordinates are the (x,y) of a black (or white, depending on the convention) pixel in the image
- ◆ The set space of gray level image is Z^3
 - Each element of the set is a 3D vector: (x,y) and intensity level.

NOTE:

Set Theory

Logical operations

Basic Set Theory

- ◆ Let A be a set in $\mathbb{Z}^2$. if $a = (a_1, a_2)$ is an element of A , then we write

$$a \in A$$

- ◆ If a is not an element of A , we write

$$a \notin A$$

- ◆ Set representation

$$A = \{(a_1, a_2), (a_3, a_4)\}$$

- ◆ Empty or Null set

$$A = \emptyset$$

Basic Set Theory

- ◆ **Subset:** if every element of A is also an element of another set B, the A is said to be a subset of B

$$A \subseteq B$$

- ◆ **Union:** The set of all elements belonging either to A, B or both

$$C = A \cup B$$

- ◆ **Intersection:** The set of all elements belonging to both A and B

$$D = A \cap B$$

Basic Set Theory

- ◆ Two sets A and B are said to be **disjoint** or **mutually exclusive** if they have no common element

$$A \cap B = \emptyset$$

- ◆ **Complement:** The set of elements not contained in A

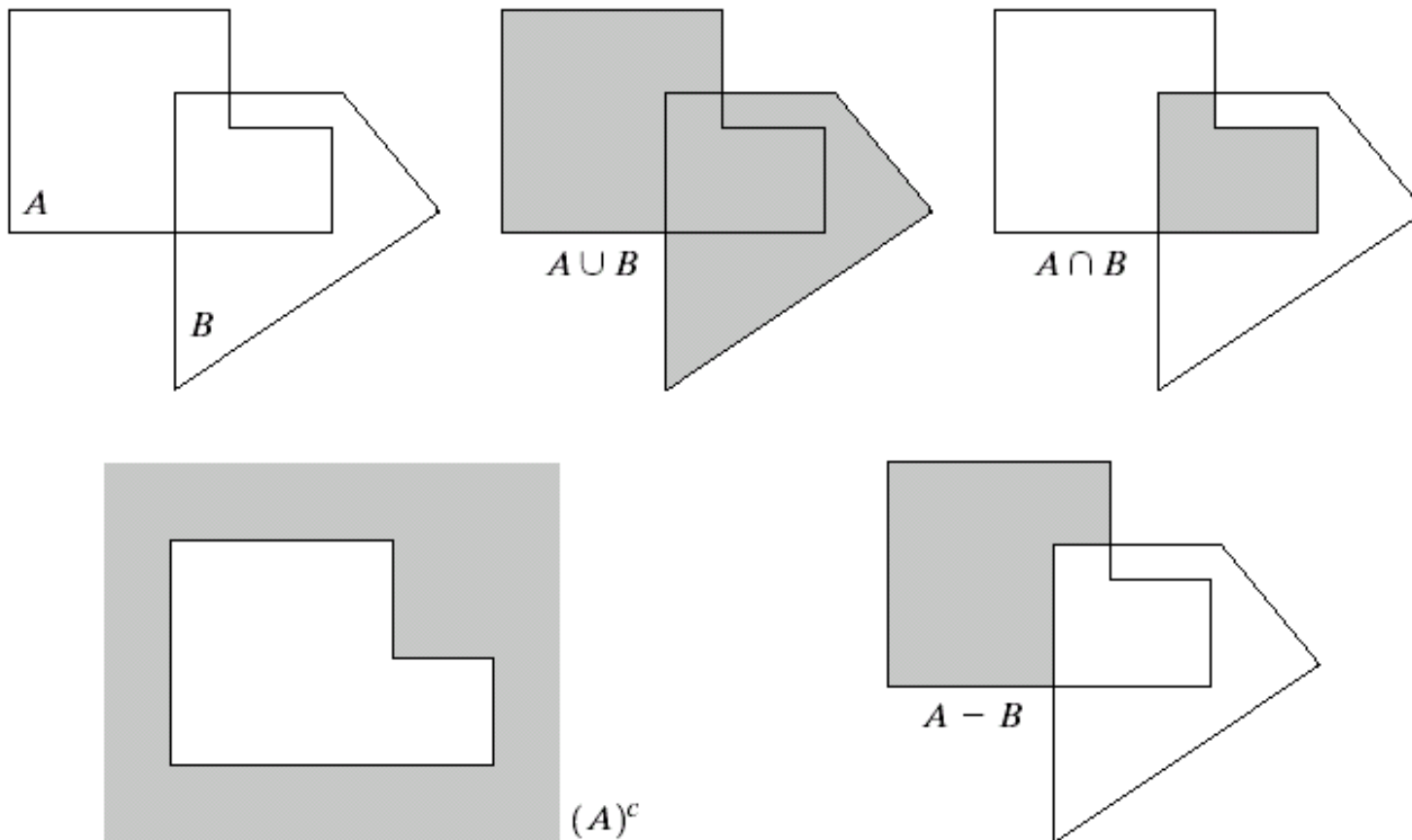
$$A^c = \{w \mid w \notin A\}$$

- ◆ **Difference** of two sets A and B, denoted by $A - B$, is defined as

$$A - B = \{w \mid w \in A, w \notin B\} = A \cap B^c$$

i.e. the set of elements that belong to A, but not to B

Basic Set Theory



a	b	c
d	e	

FIGURE 9.1
(a) Two sets A and B . (b) The union of A and B . (c) The intersection of A and B . (d) The complement of A . (e) The difference between A and B .

Example of some logic operations:

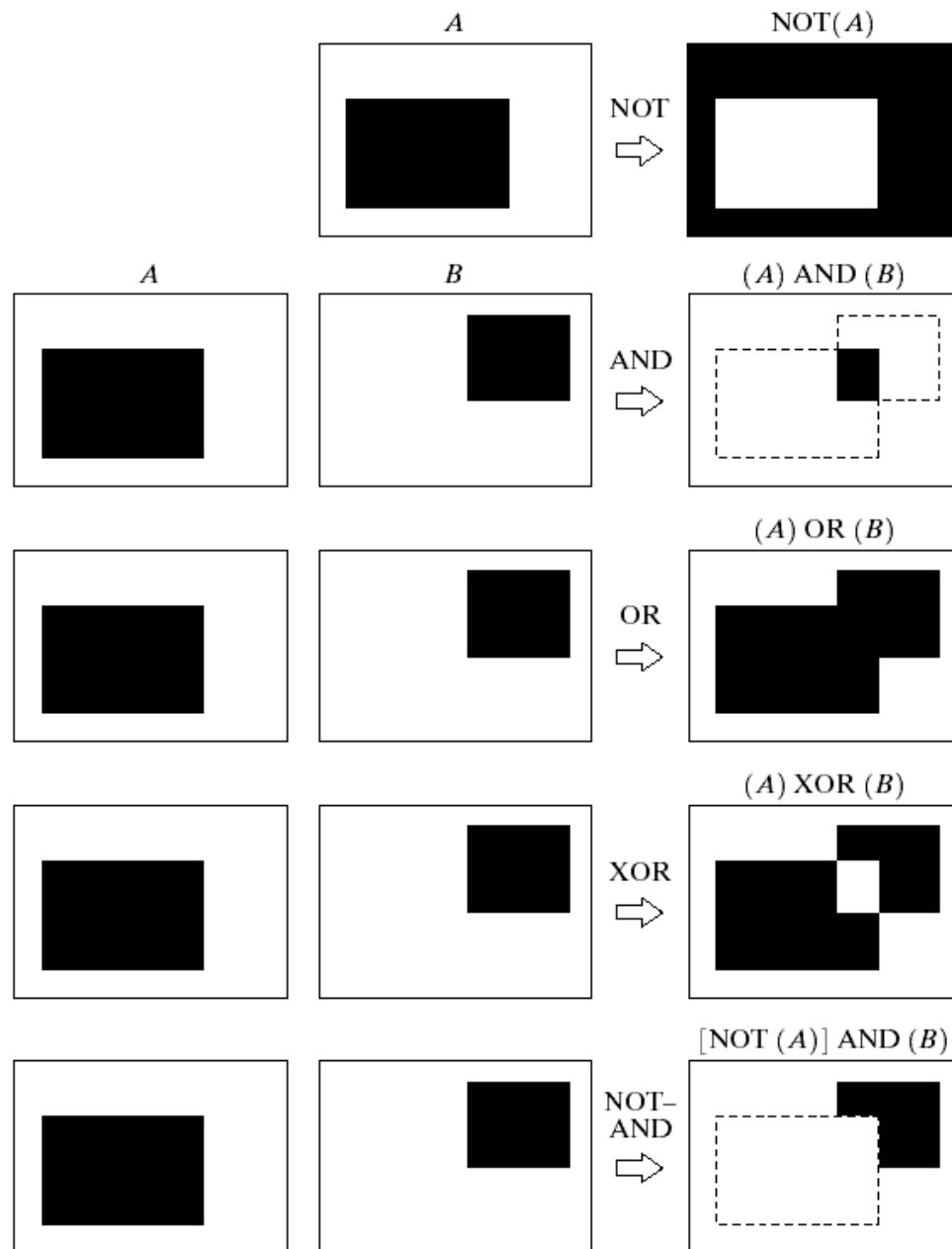


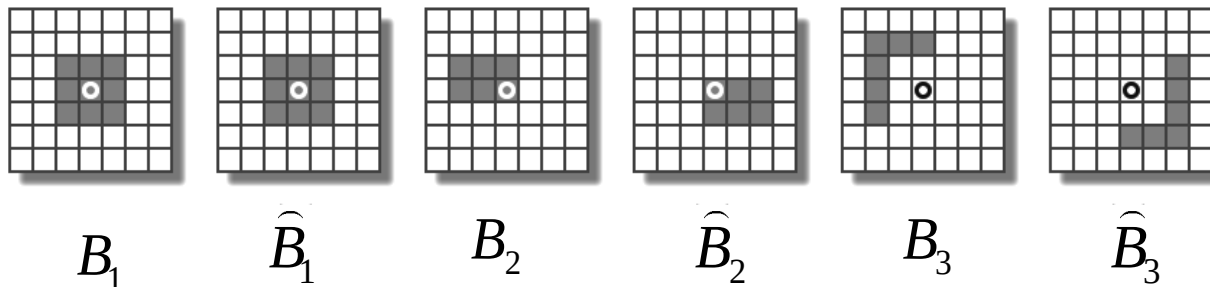
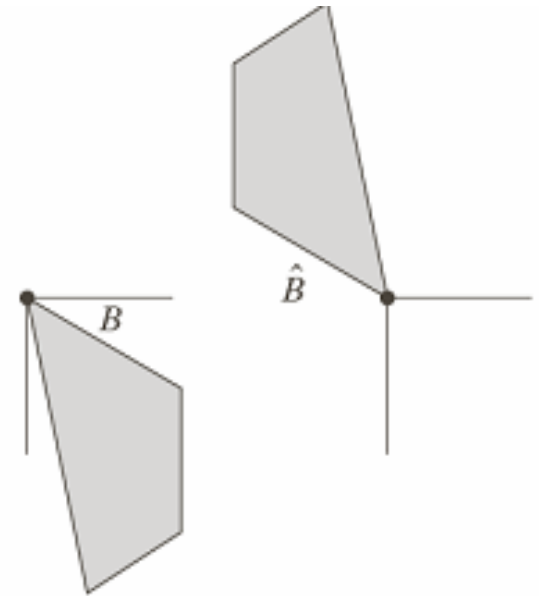
FIGURE 9.3 Some logic operations between binary images. Black represents binary 1s and white binary 0s in this example.

Reflection of set B

- ◆ Reflection of set B

$$B = \{w \mid w = -b, \text{ for } b \in B\}$$

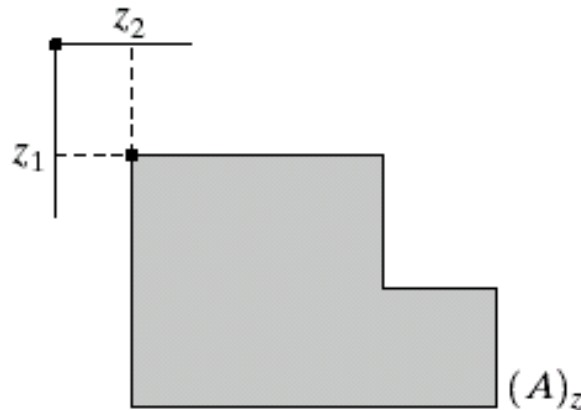
i.e. the set of element w , such that w is formed by multiplying each of two coordinates of all the elements of set B by -1



Translation of set A

- ◆ **Translation** of set A by point $z = (z_1, z_2)$, denoted $(A)_z$, is defined as

$$(A)_z = \{w \mid w = a + z, \text{ for } a \in A\}$$



Structuring Element: Translation

Let I be an image and B a SE.

$(B)_z$ means that B is moved so that its origin coincides with location z in image I .

$(B)_z$ is the *translate* of B to location z in I .

